UMC 0.11um LOGIC/MIXED-MODE Al Advanced Enhancement Native 2.5V MOSFET SPICE Model Document

Version 0.2 Phase1
2012/06/27

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2012/06/27	Eason Cheng	1. Original release.

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

2.5 V n-ch MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.24

For detailed descriptions of these compact model equations and parameters, please refer to this material.

BSIM3V3.24 MOSFET Model User Manual

In BSIM3V3.24, there are two types of models that can be used. One is a binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is a global model, which means one scalable parameter set is provided and it can cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for their many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the silicon data, which can be found in the following sections.

Note: Please make sure you would not design three kinds of 2.5V, 3.3V IO and 5V devices at same time. If any design requirement is necessary, please discuss in advance with UMC through CE department.

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.09
Cadence SPECTRE version 10.1.1.111.isr8 32bit
Mentor Graphics ELDO version 6.11_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below. Please note that 'fast' here means lower threshold voltage, higher leakage and driving current, and 'slow' means higher threshold voltage, lower leakage and driving current.

TT: Typical n-ch MOSFET and Typical p-ch MOSFET
FF: Fast n-ch MOSFET and Fast p-ch MOSFET
SS: Slow n-ch MOSFET and Slow p-ch MOSFET
FNSP: Fast n-ch MOSFET and Slow p-ch MOSFET
SNFP: Slow n-ch MOSFET and Fast p-ch MOSFET

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE
- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9
.lib "./ModelFileName.lib" CaseName
, where CaseName can be TT, FF, SS, FNSP, or SNFP.
- Define device condition by 'Mxx' statement;

Example 1:
M1 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1

For Cadence Spectre
- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

SetOption1 options scale=0.9
include "./ModelFileName.lib.scs" section=CaseName
, where CaseName can be TT, FF, SS, FNSP, or SNFP.
- Define device condition by 'Mxx' statement;

Example 1:
m1 (Vdd Vgg Vss Vbb) DeviceName W=W1 L=L1

For Mentor Graphics Eldo
- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation
directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9
.lib "./ModelFileName.lib.eldo" CaseName
, where CaseName can be TT, FF, SS, FNSP, or SNFP.
- Define device condition by 'Mxx' statement;

Example 1:
M1 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1

ModelFileName for 2.5 V MOSFET: l110ae_nvt_25_v021
DeviceName for 2.5 V n-ch MOSFET: n_nvt_25_l110ae

As stated above, one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range (After 90% shrink):

For 2.5 V n-ch MOSFET:

1.62 um <=L_{DES} <= 10 um
1 um <= W_{DES} <= 10 um

Note: It is recommended to adopt multi-finger structures if W_{DES} > 10 um.

Adopt multi-finger structures if WDES > 10 um.

Voltage range:

For 2.5 V n-ch MOSFET:

0 V <= V_{GS} <= 2.5 V(*1.1)
0 V <= V_{DS} <= 2.5 V(*1.1)
-2.5 V<= V_{BS} <= 0V

Temperature range:

-40 C ~ +125 C

Feature\ GDSII	Drawing in shrinkable L130AE GDS database (unit : um)	L110AE GDSII =0.9*(L130AE GDS) Dimension (unit : um)	L110AE SPICE Dimension (unit : um)
Core (W/L)	10/0.12	9/0.108	HS(L + 8 nm): 9/0.116 LL/SP(L +12 nm): 9/0.12
2.5V I/O (W/L)	10/0.24	9/0.216	(L +12 nm): 9/0.228
3.3V I/O (W/L)	10/0.34	9/0.306	(L +12 nm): 9/0.318
5V I/O (W/L)	10/0.542	9/0.488	(L +12 nm): 9/0.5
HG 3.3V I/O (W/L)	HG IO-PMOS : 10/0.32 HG IO-NMOS : 10/0.34	HG IO-PMOS : 9/0.228 HG IO-NMOS : 9/0.306	HG IO-PMOS (L +12 nm): 9/0.3 HG IO-NMOS (L +12 nm): 9/0.318
Salicide and Non-salicide resistor	(Width/Length)	0.9*(Width/Length)	0.9*(Width/Length)
Mixed-Mode Device	(Width/Length)	0.9*(Width/Length)	0.9*(Width/Length)
RF Device	(Width/Length)	0.9*(Width/Length)	0.9*(Width/Length)
Contact size	0.16*0.16	0.144*0.144	0.144*0.144
Via size(1X;2X)	0.2*0.2; 0.4*0.4	0.18*0.18; 0.36*0.36	MVA1(W -0.02um) : 0.16*0.2 Other 1X MVA : 0.18*0.18 2X MVA (W -0.04um): 0.28*0.44
Metal size	Width/Space	0.9*(Width/Space)	0.9*(Width/Space)
Sloting inserting	Metal line width>5.556um	Metal line width>5um	

Note:

1. Because process bias make drawn and physical dimension are different, please refer to interconnect capacitance model (G-04 document in individual DSM) for precise interconnection-capacitance extraction.

2. Please refer to SPICE model (G-05 document in individual DSM) for precise device characteristics.

2 2.5 V N-ch MOSFET

2.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW1RA	BSW1RA	BSW1RA	BSW1RA
Lot ID	D6HTK	D6HTK	D6HTK	D6HTK
Wafer Number	#7	#7	#7	#7

2.2 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at ID =300nA*WDES*0.9/(LDES*0.9+12nm) for NMOS

All parameters are given for T=25 C, V_{BS}=0 V unless specified otherwise.
(The dimensions are after 90% shrink and 12nm sizing channel length back.)

Table 2-2 Electrical parameters for NMOS

All parameters are given for T=25 ^{o}C, V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
Long channel device (W/L= 10 µm / 10 µm)						
V_{TLIN}	V_{DS} = 0.1 V	V	-0.179	-0.177	-0.178	-0.179
I_{ON}	V_{DS}=V_{GS}= 2.5 V	µA/µm	74.81	74.36	74.21	74.81
Wide and short channel device (W/L= 10 µm / 1.632 µm)						
V_{TLIN}	V_{DS} = 0.1 V	V	-0.207	-0.205	-0.206	-0.207
V_{TSAT}	V_{DS} = 2.5 V	V	-0.271	-0.269	-0.276	-0.271
Body Effect V_T shift	V_{DS} = 2.5 V V_{BS}= 0 ~ -2.5 V	V	--	0.068	0.070	0.070
DIBL	V_{DS}= 0.1~2.5 V	V	0.064	0.064	0.070	0.064
I_{ON}	V_{DS}=V_{GS}= 2.5 V	µA/µm	366.4	365.62	364.32	366.40
I_{OFF}	V_{DS}= 2.5 V, V_{GS}=0 V	A/µm	--	6.7E-06	7.5E-06	7.3E-06
Narrow and long channel device (W/L= 1 µm / 10 µm)						
V_{TLIN}	V_{DS} = 0.1 V	V	-0.113	-0.114	-0.115	-0.113
V_{TSAT}	V_{DS}= 2.5 V	V	-0.116	-0.117	-0.118	-0.116
I_{ON}	V_{DS}=V_{GS}= 2.5 V	µA/µm	63.74	63.10	62.88	63.74
Narrow and short channel device (W/L= 1 µm / 1.632 µm)						
V_{TLIN}	V_{DS} = 0.1 V	V	-0.129	-0.128	-0.125	-0.129
V_{TSAT}	V_{DS}= 2.5 V	V	-0.142	-0.144	-0.139	-0.142
I_{ON}	V_{DS}=V_{GS}= 2.5 V	µA/µm	332.5	331.94	329.03	332.50
I_{OFF}	V_{DS}= 2.5 V, V_{GS}=0 V	A/µm	--	2.3E-06	2.1E-06	2.2E-06
Capacitance						
CJ	V_J = 0 V	F/m^2	--	1.13E-04	1.14E-04	1.14E-04
CJSW	V_J = 0 V	F/m	--	3.13E-10	3.09E-10	3.09E-10
CJSWG	V_J = 0 V	F/m	--	8.65E-11	9.93E-11	9.93E-11
C_{OVL}	V_{GS} = -0.5 V	F/m	--	1.00E-09	4.50E-10	4.50E-10

2.3 MOSFET DC Model

2.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 2-3 Geometry of NMOS devices measured for DC parameter extraction

W\L	10	5	3	2.5	2	1.8	1.632
10	X	X	X	X	X	X	X
5	X	X	X		X	X	X
2.5	X	X	X	X	X	X	X
2	X		X		X	X	X
1.6	X	X	X	X	X	X	X
1.4	X		X	X		X	X
1	X	X	X	X	X	X	X

2.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 2-4 Test conditions for IV curves of NMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	0.1	2.5	2.4
V_{GS} (V)	-2	2.75	0.05
V_{BS} (V)	0	-2.5	-0.625

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	2.75	0.05
V_{GS} (V)	0	2.5	0.25
V_{BS} (V)	0	-2.5	-2.5

2.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (LDES *0.9+12nm and WDES *0.9) respectively.

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

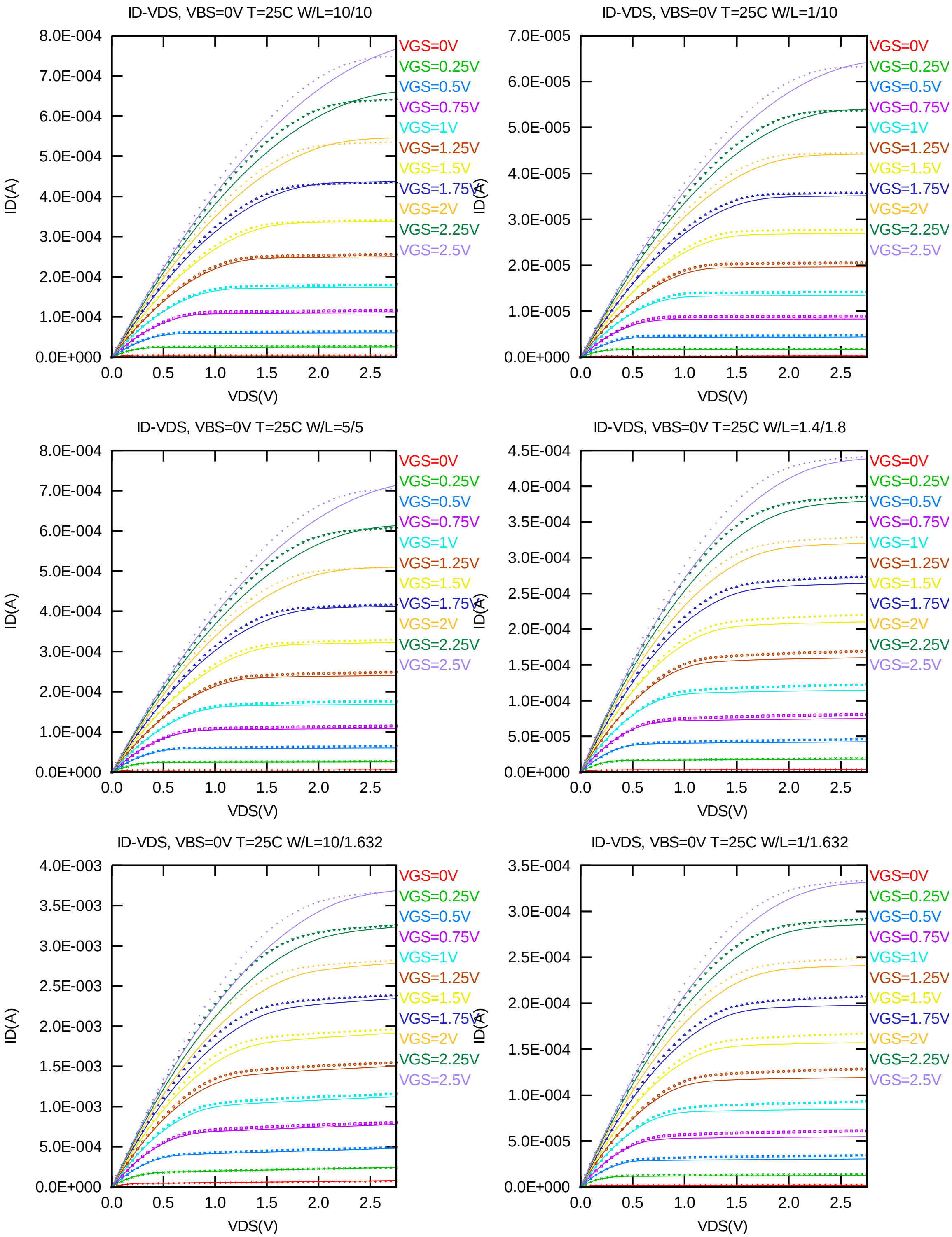

Figure 2-1 I_D vs. V_{DS} at V_{BS}=0 V for NMOS

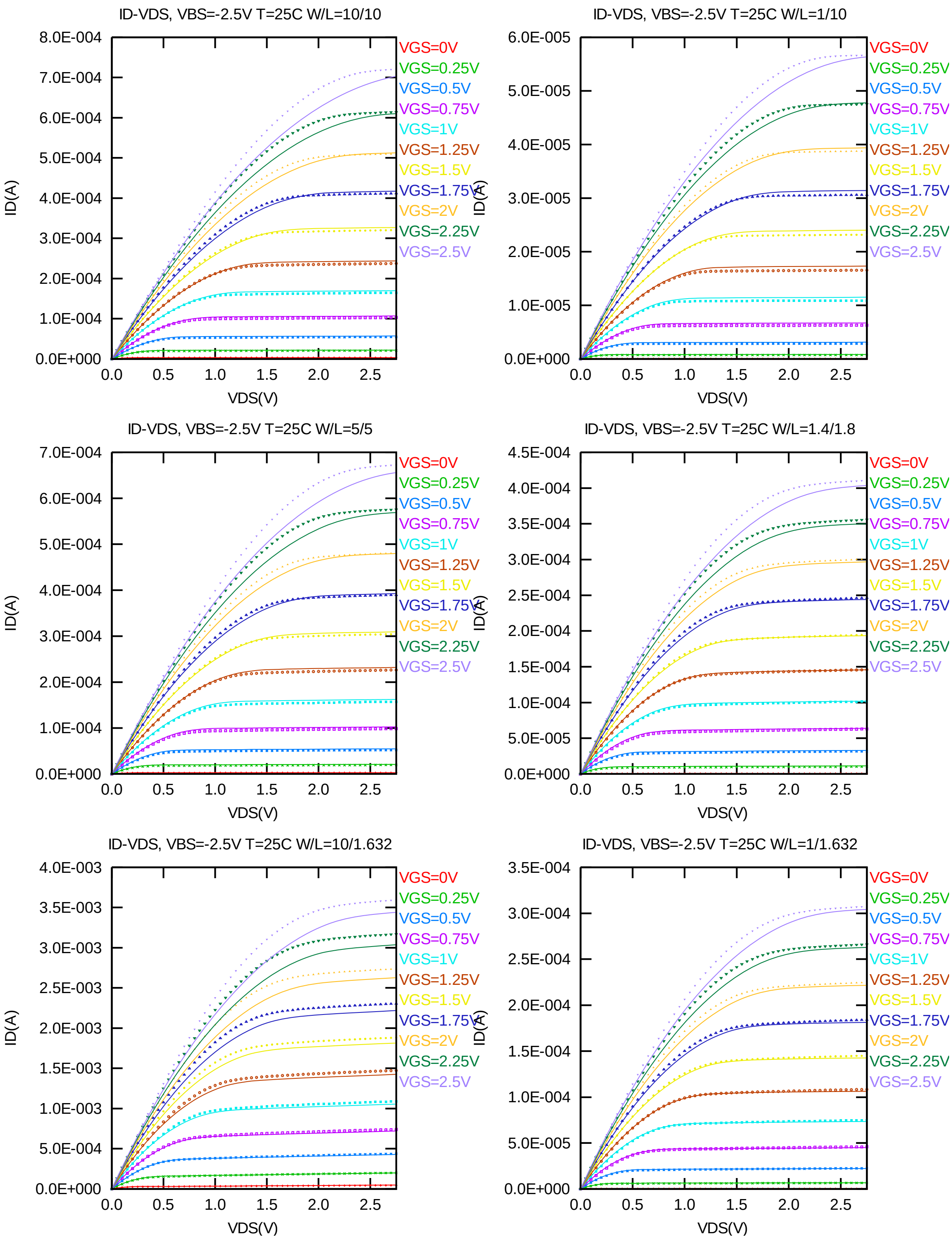

Figure 2-2 I_D vs. V_{DS} at V_{BS}=-2.5 V for NMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

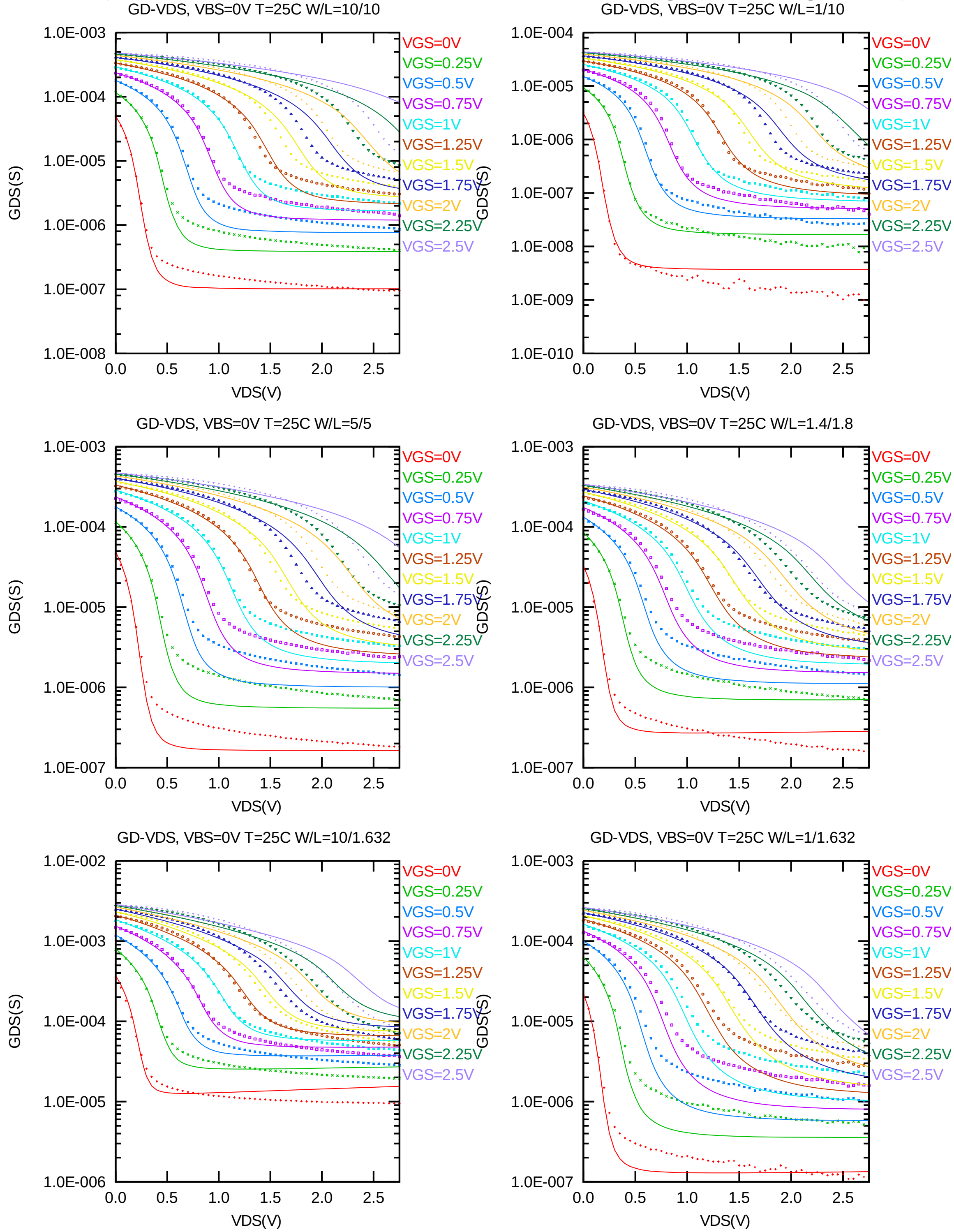

Figure 2-3 G_{DS} vs. V_{DS} at V_{BS}=0 V for NMOS

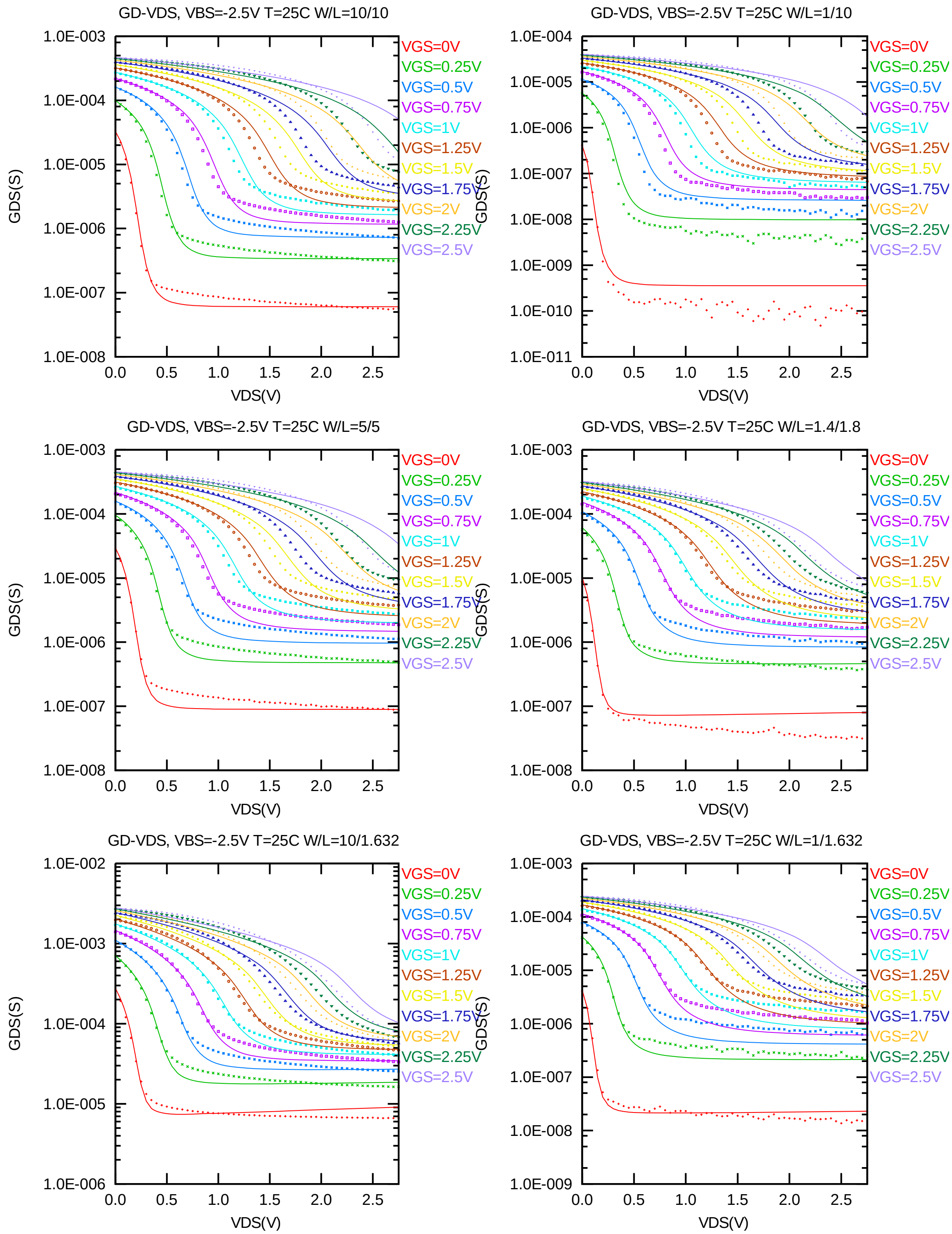

Figure 2-4 G_{DS} vs. V_{DS} at V_{BS}=-2.5 V for NMOS

I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

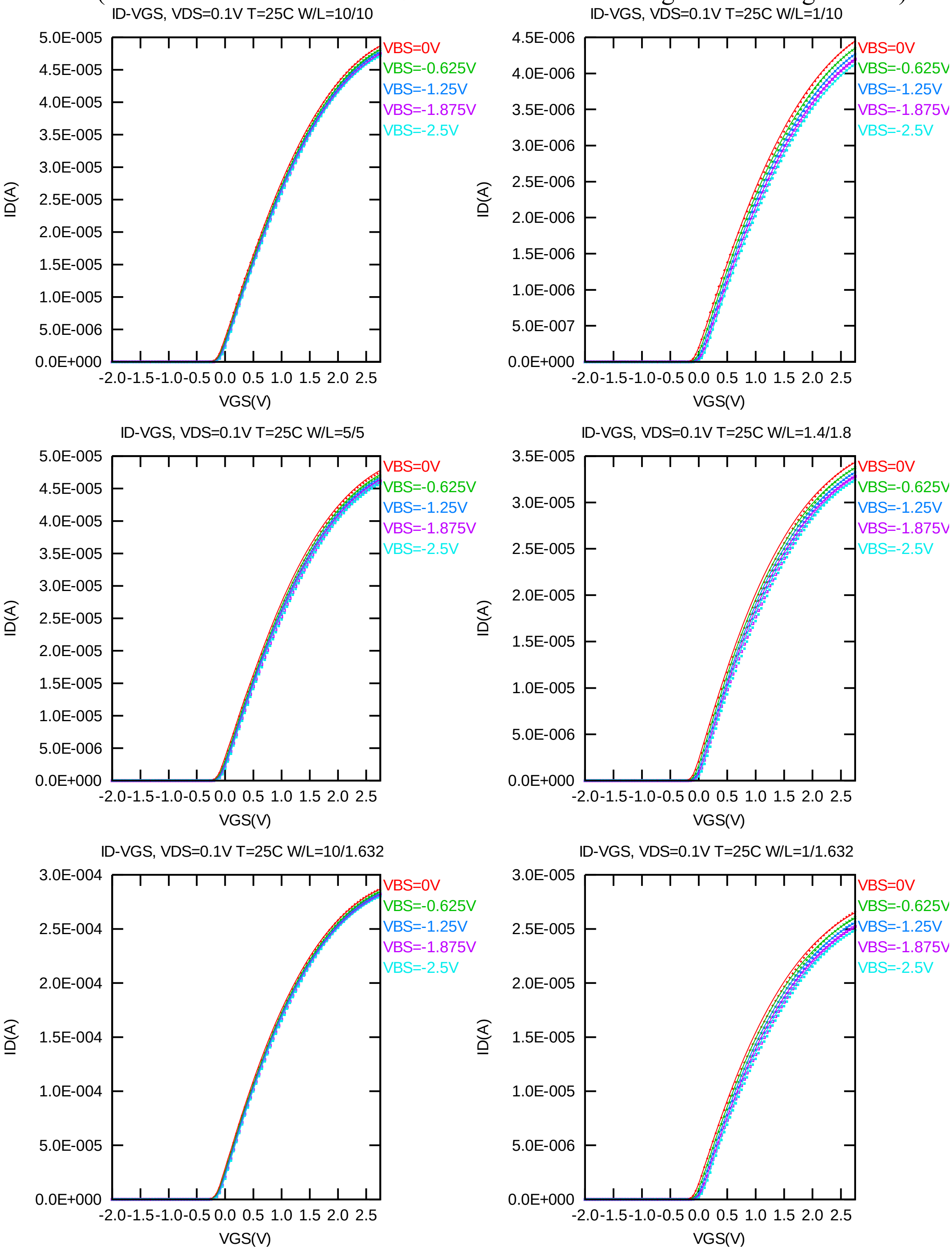

Figure 2-5 I_D vs. V_{GS} at V_{DS}=0.1 V for NMOS

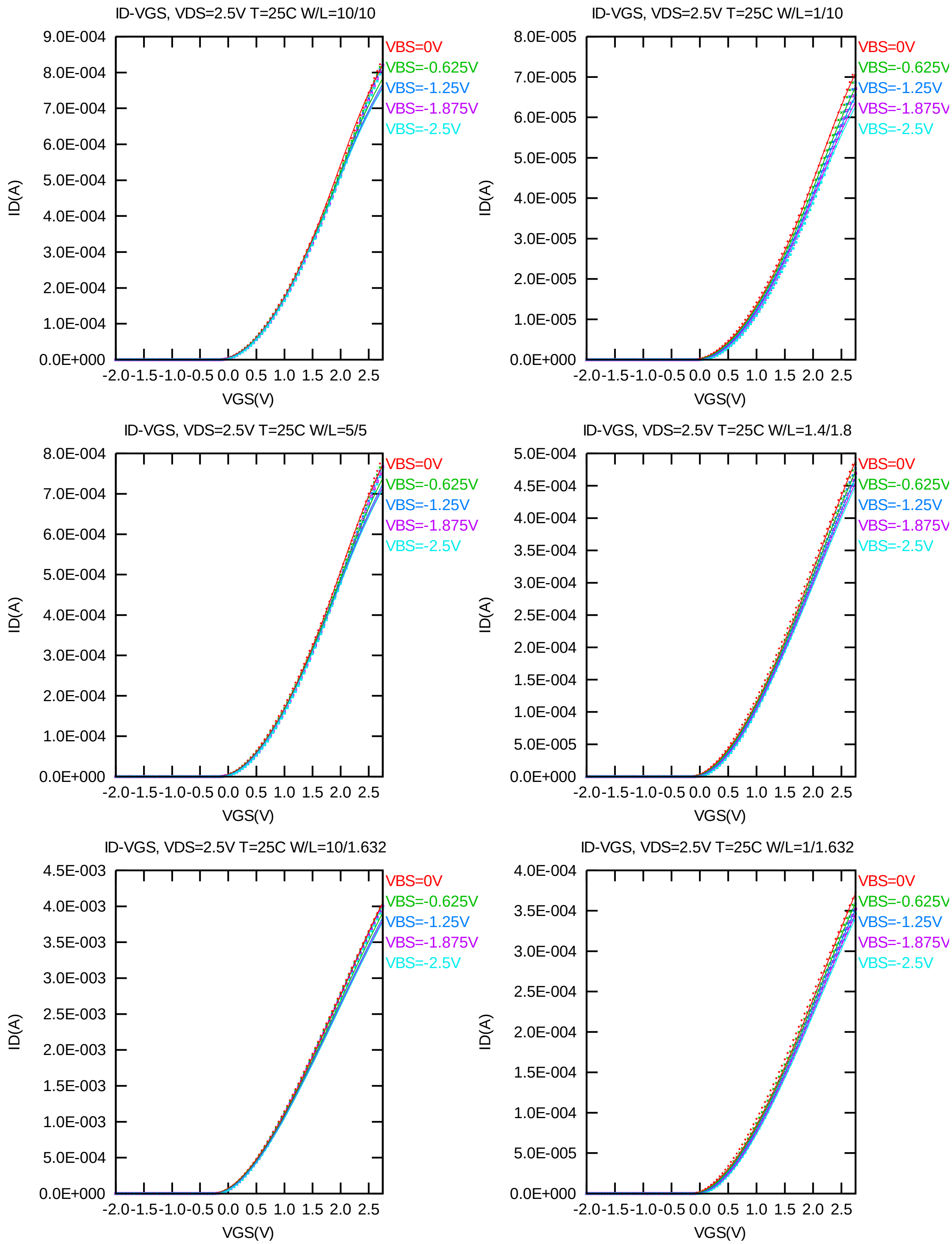

Figure 2-6 I_D vs. V_{GS} at V_{DS}=2.5 V for NMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

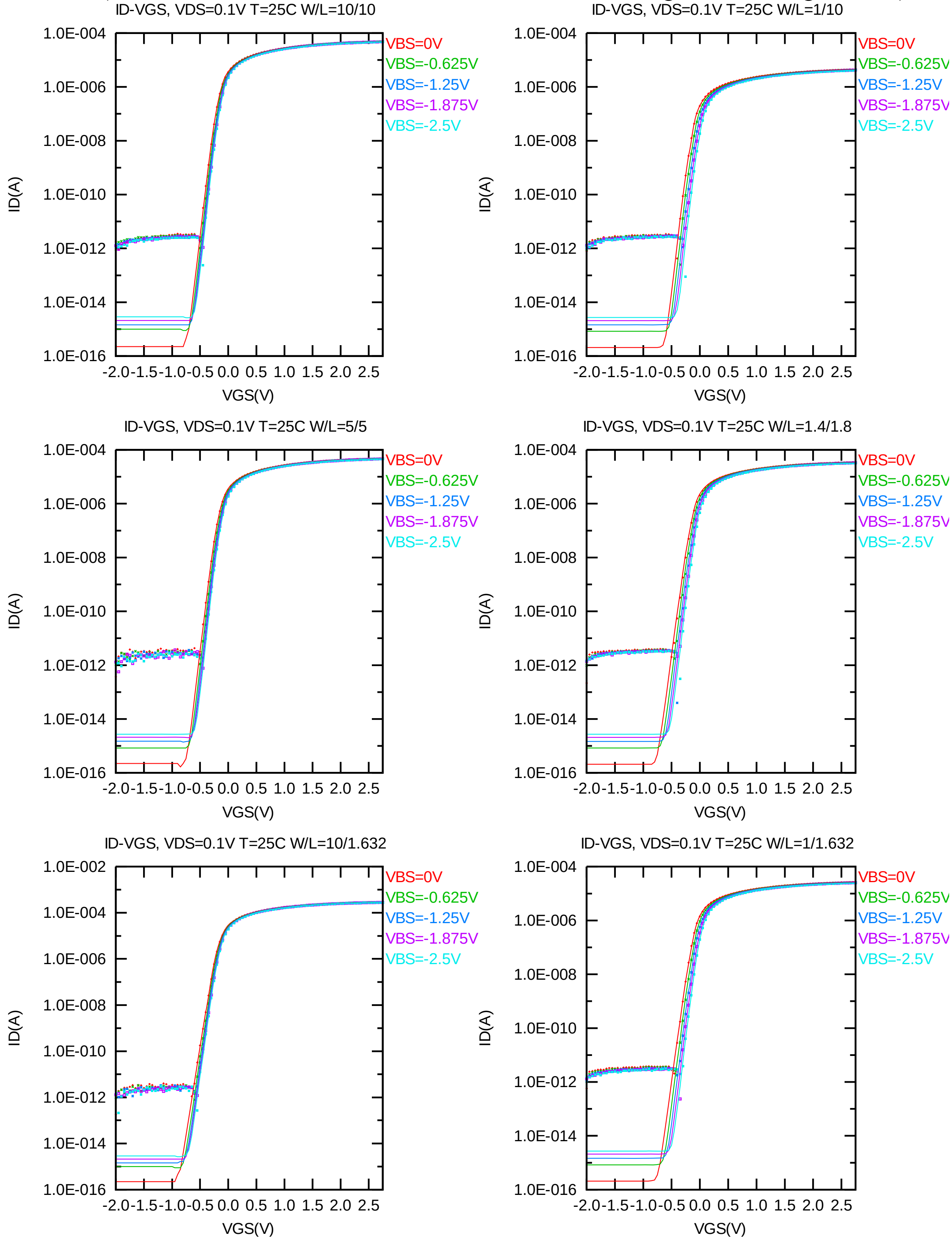

Figure 2-7 Log scaled I_D vs. V_{GS} at V_{DS}=0.1 V for NMOS

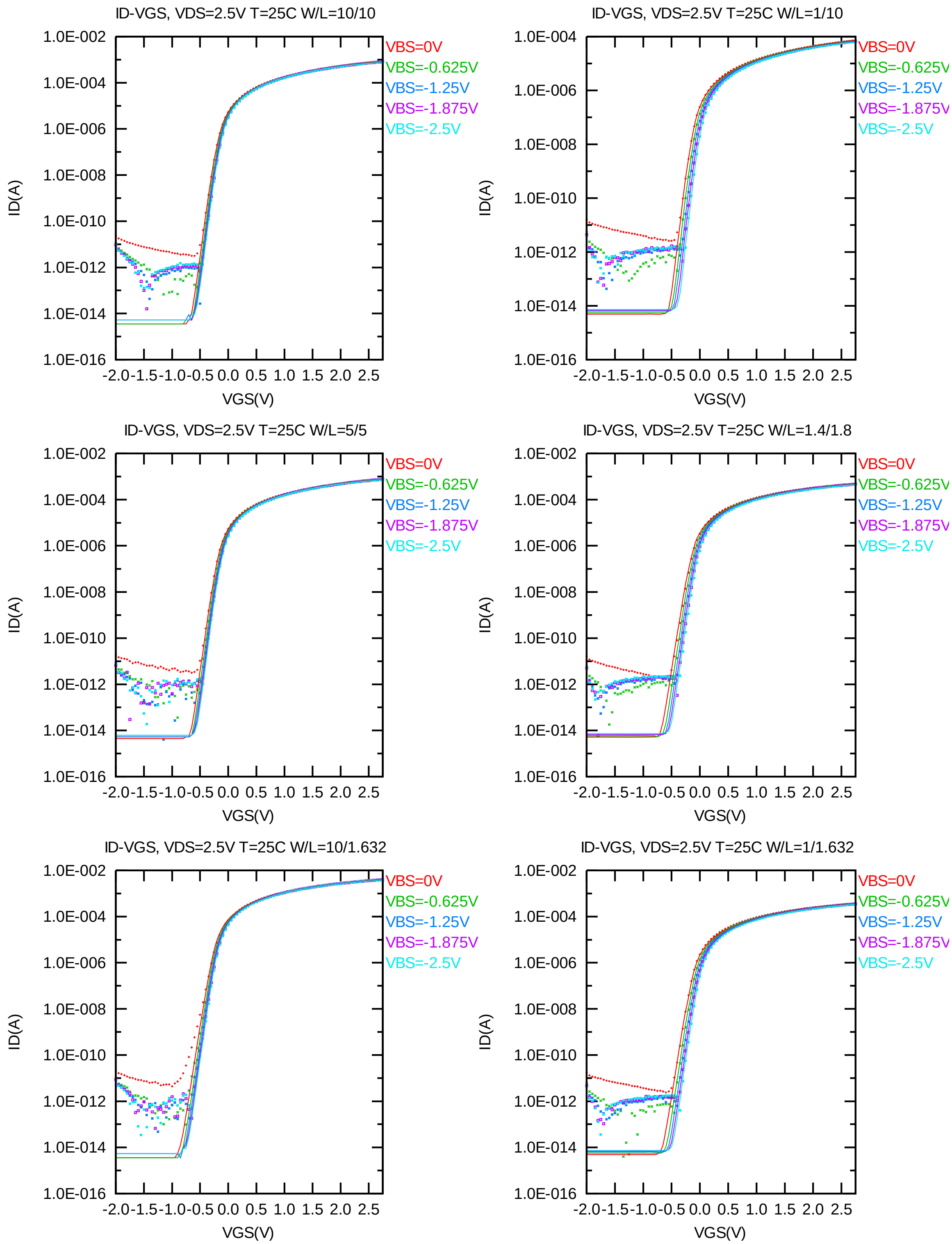

Figure 2-8 Log scaled I_D vs. V_{GS} at V_{DS}=2.5 V for NMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

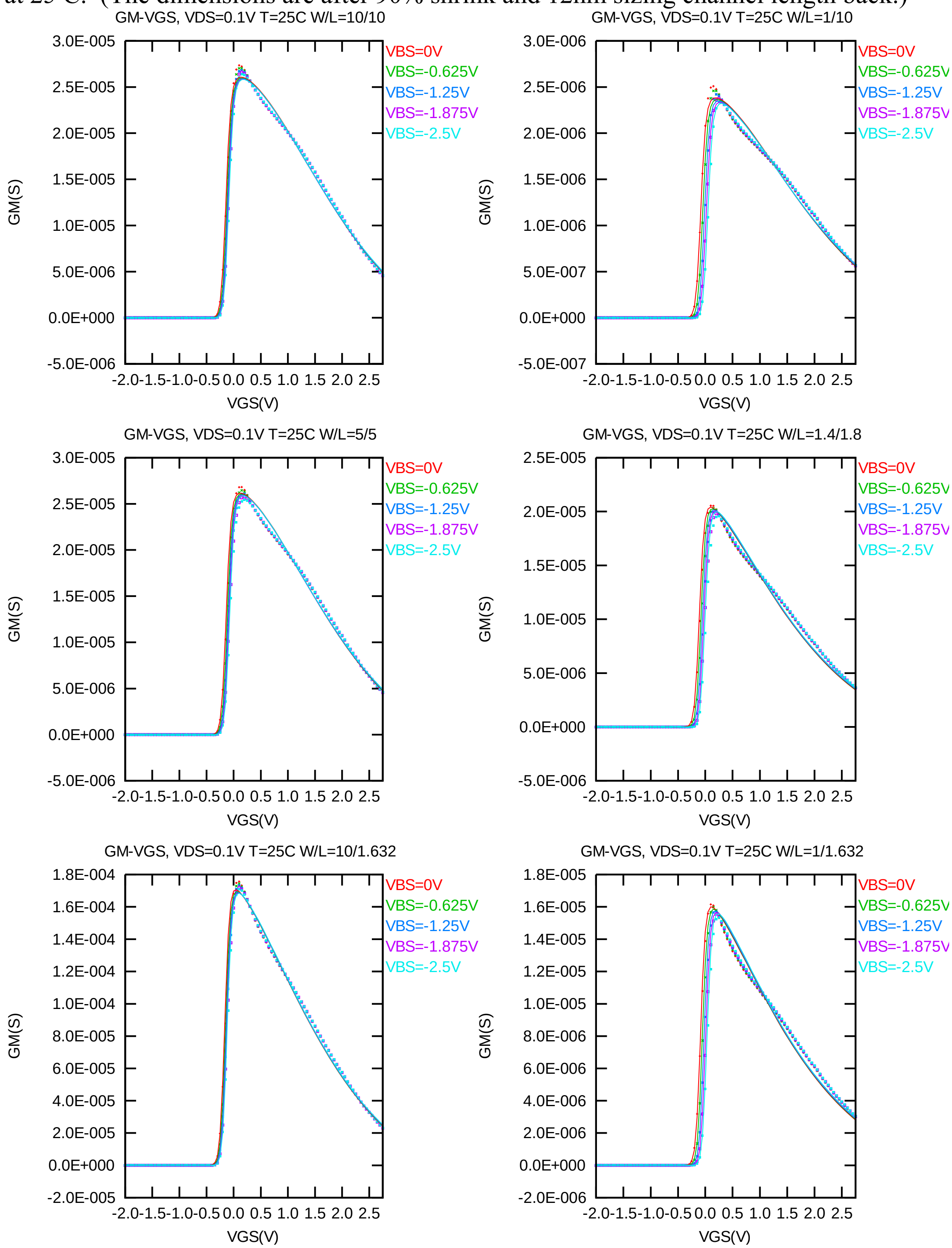

Figure 2-9 G_M vs. V_{GS} at V_{DS}=0.1 V for NMOS

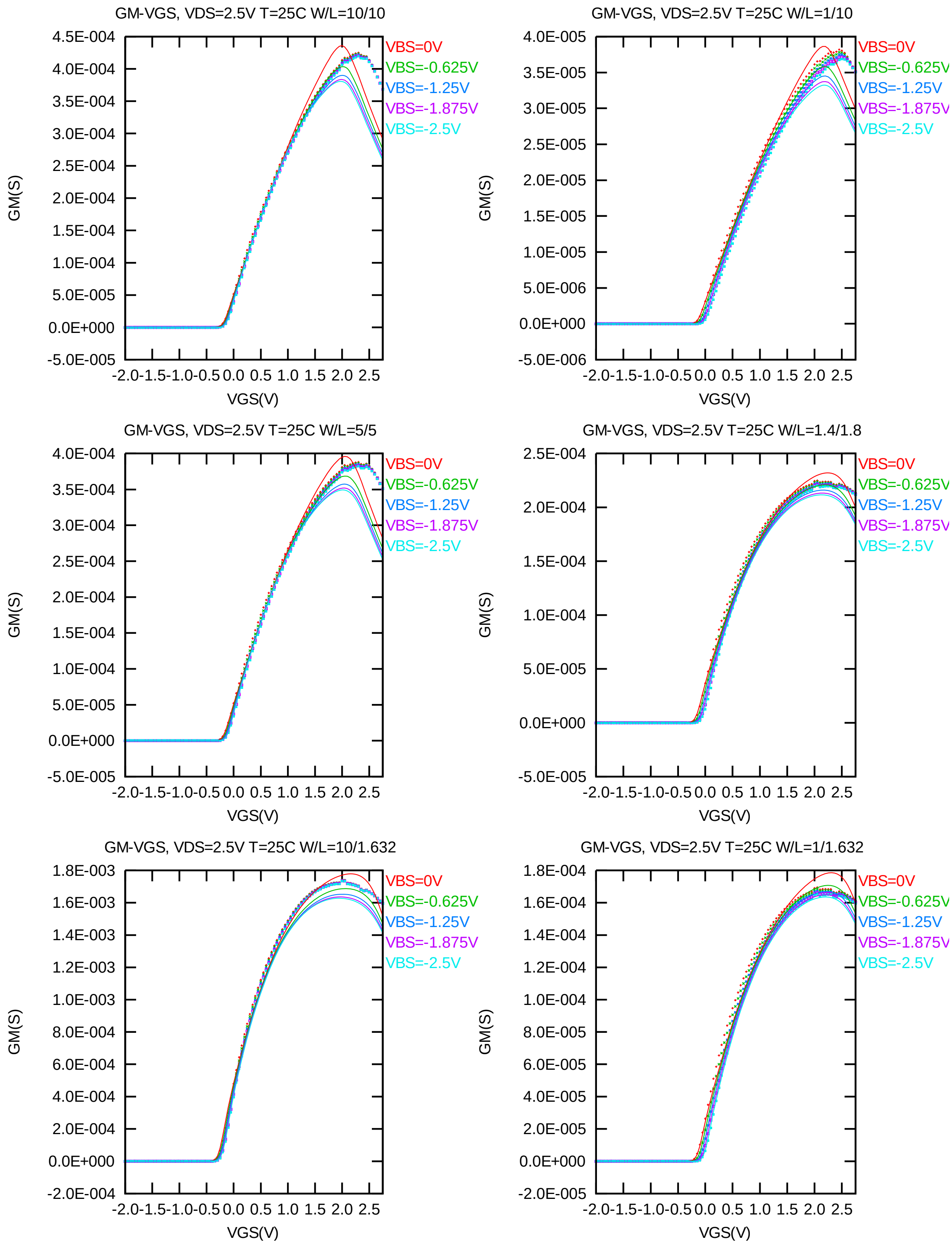

Figure 2-10 G_M vs. V_{GS} at V_{DS}=2.5 V for NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at ID =300nA*(WDES*0.9)/(LDES*0.9+12nm) at 25 ℃. The plots show the physical dimension. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

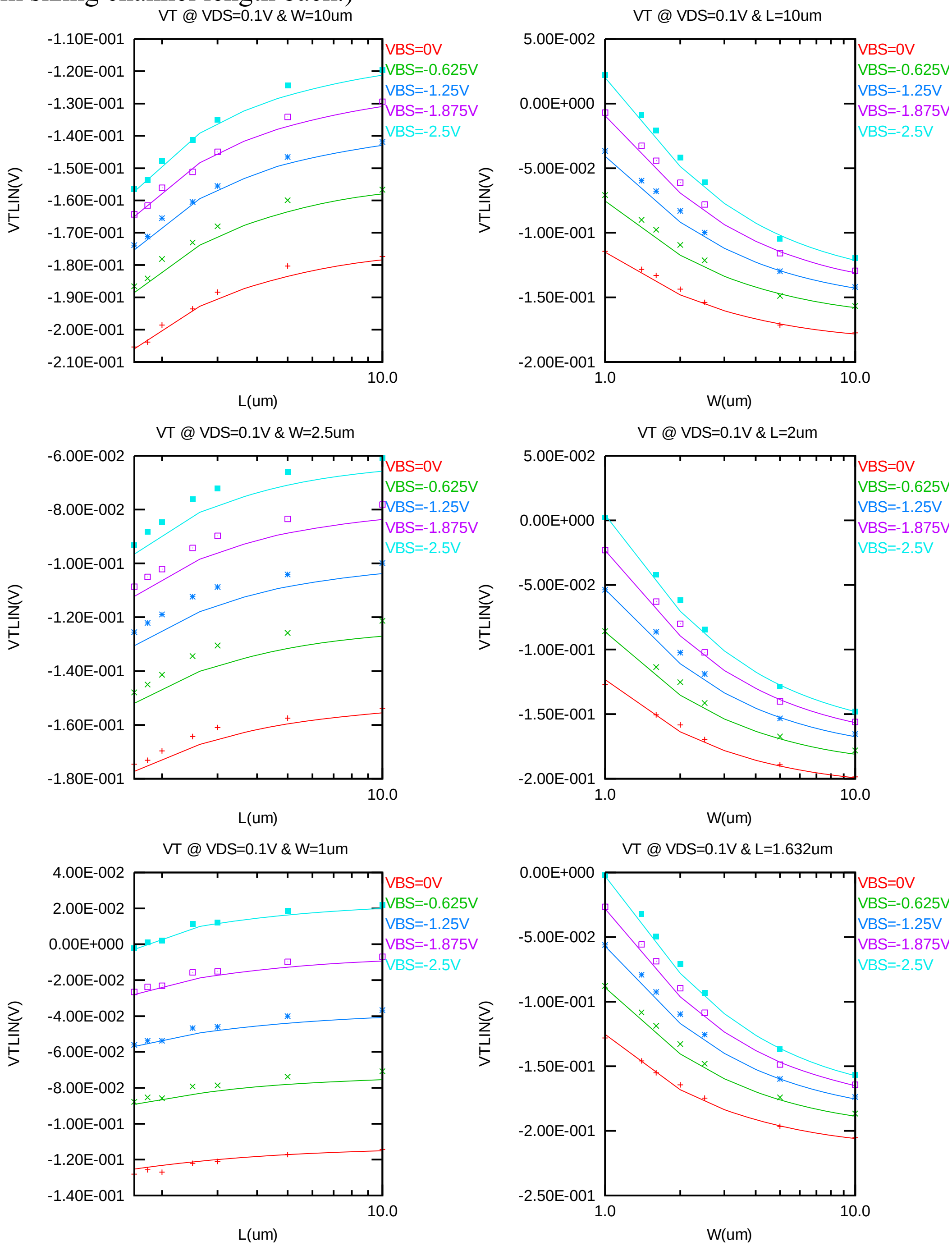

Figure 2-11 V_{TLIN} vs. channel length & channel width for low drain bias (V_{DS}=0.1 V) for NMOS

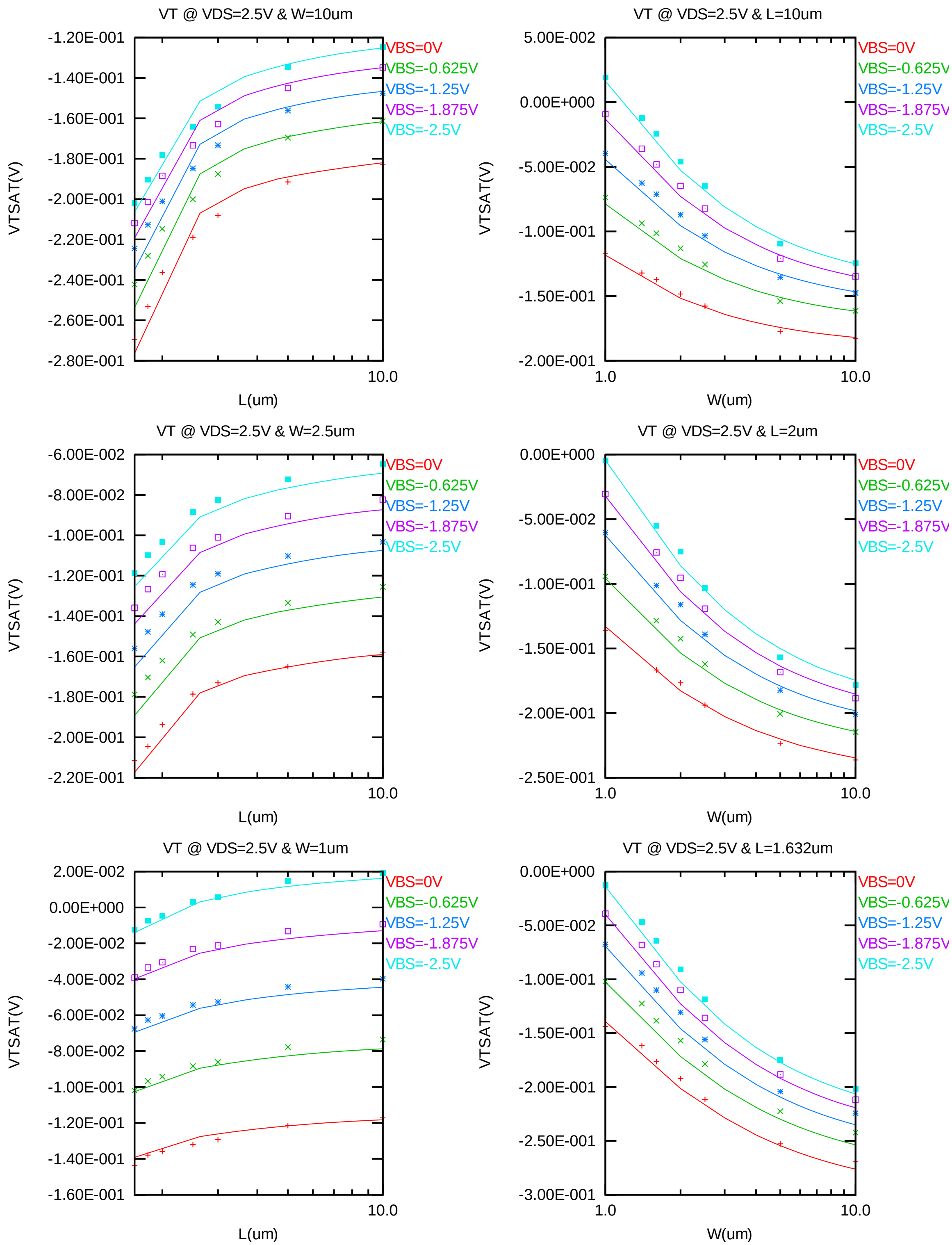

Figure 2-12 V_{TSAT} vs. channel length & channel width for high drain bias (V_{DS}=2.5 V) for NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following plots show the channel length and width dependence of I_{ON} extracted at V_{DS}=V_{GS}=2.5 V for V_{BS}=0 V and V_{BS}=-2.5 V at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

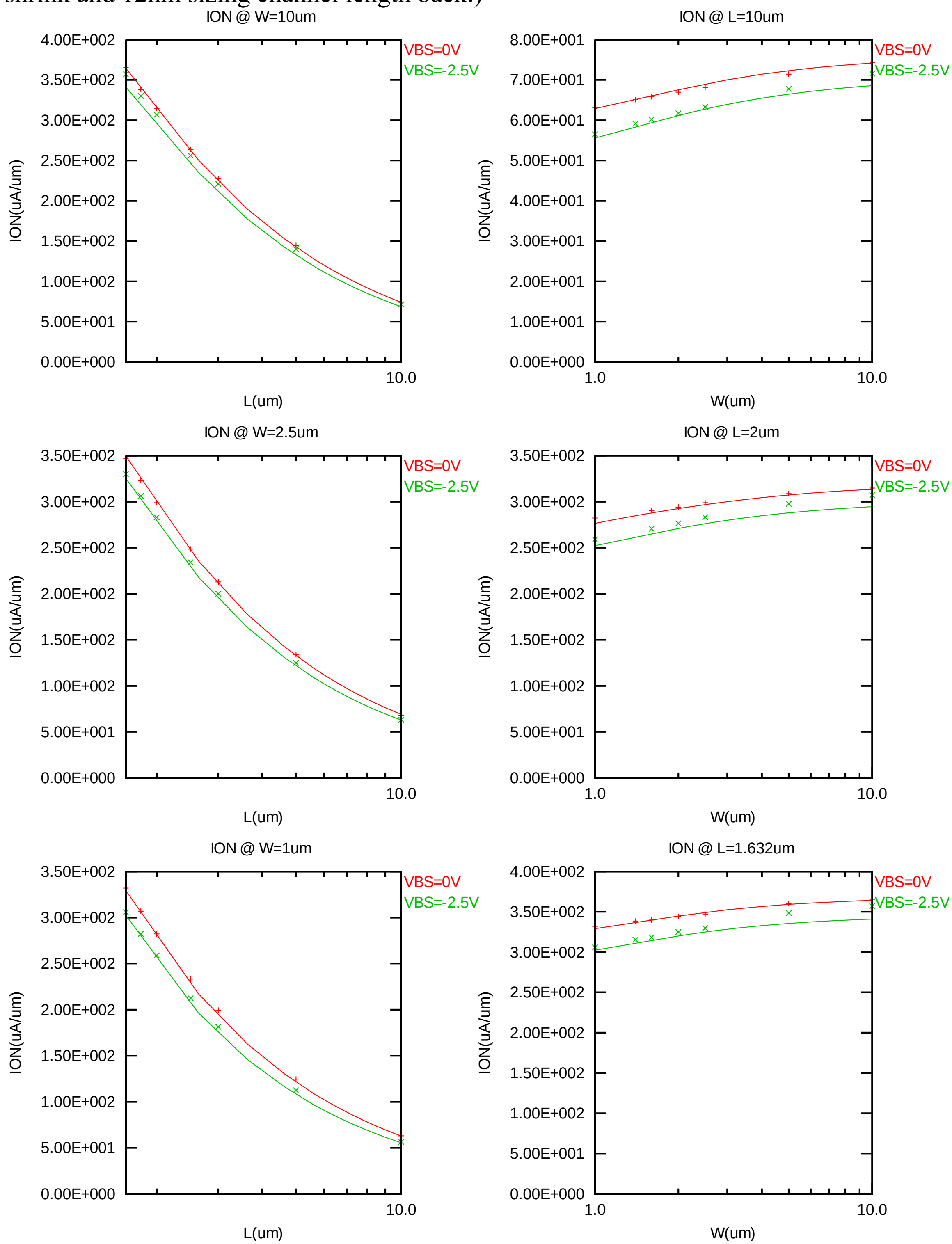

Figure 2-13 Saturation current I_{ON} vs. channel length and channel width for NMOS

The L and W in the plots refer to LDES and WDES respectively.

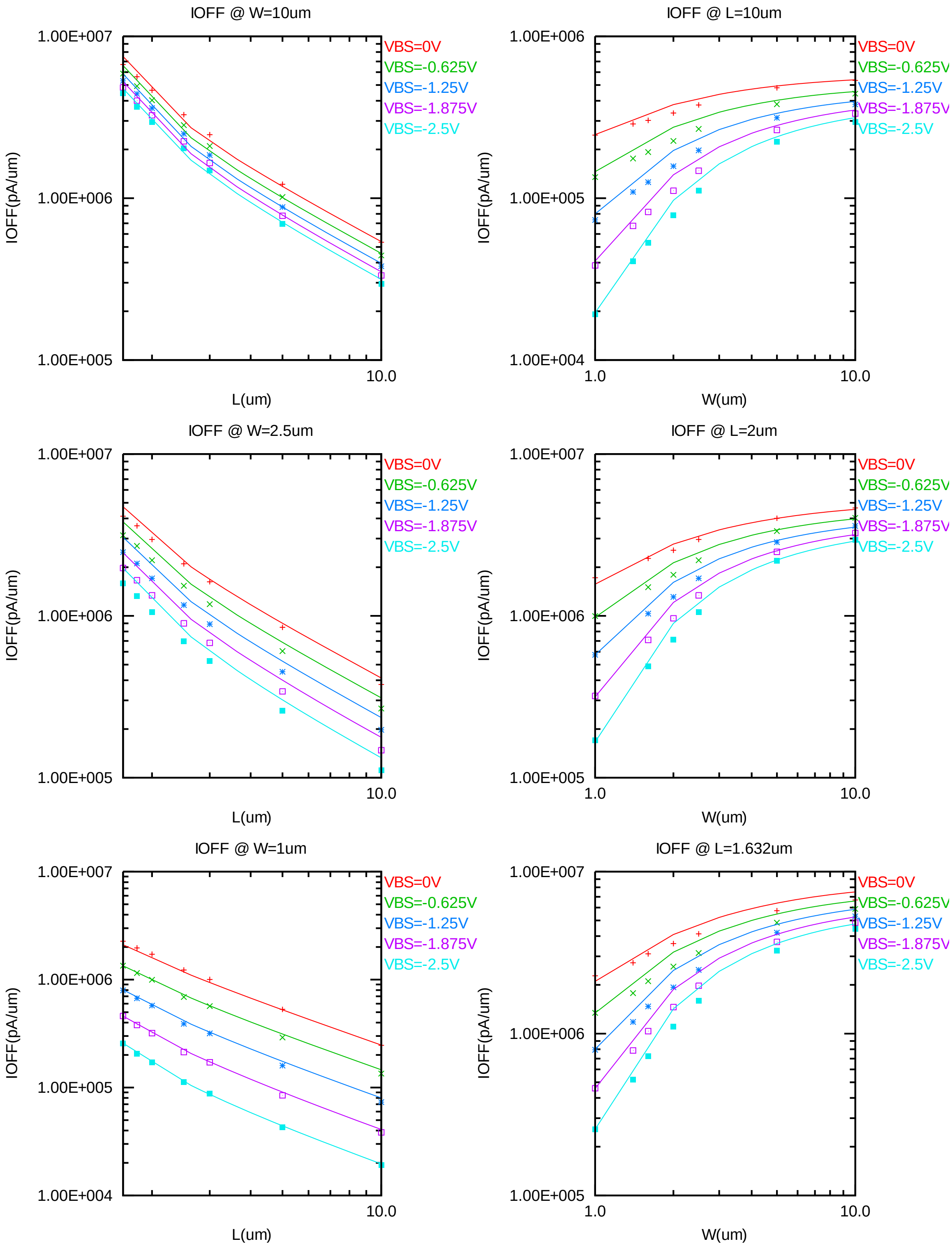

Figure 2-14 I_{OFF} vs. channel length and channel width for NMOS

V_{TLIN} vs. temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at ID=300nA*(WDES*0.9)/(LDES*0.9+12 μm). (The dimensions are after 90% shrink and 12nm sizing channel length back.)
The L and W in the plots refer to LDES and WDES respectively.

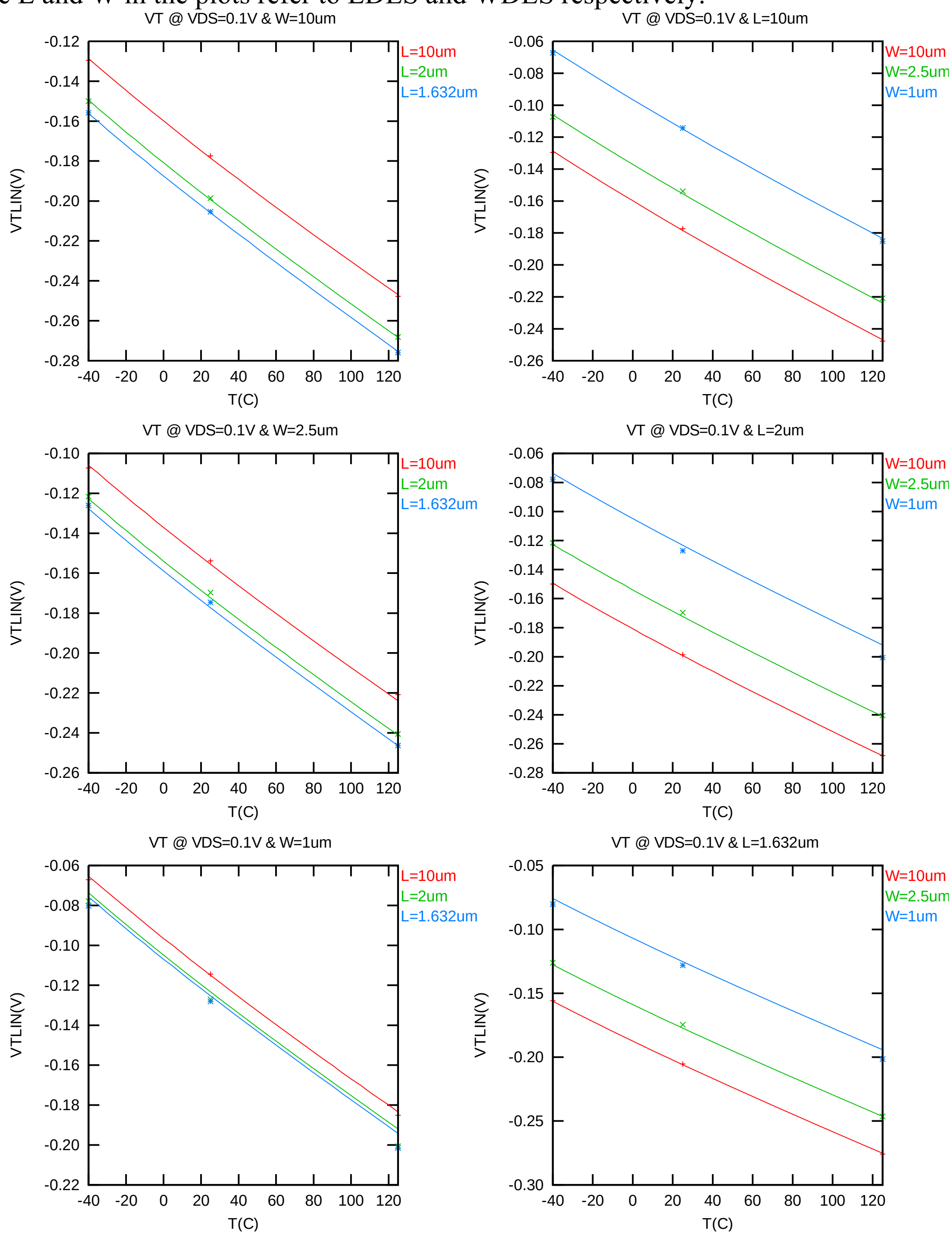

Figure 2-15 V_{TLIN} vs. temperature for NMOS

V_{TSAT} vs. temperature

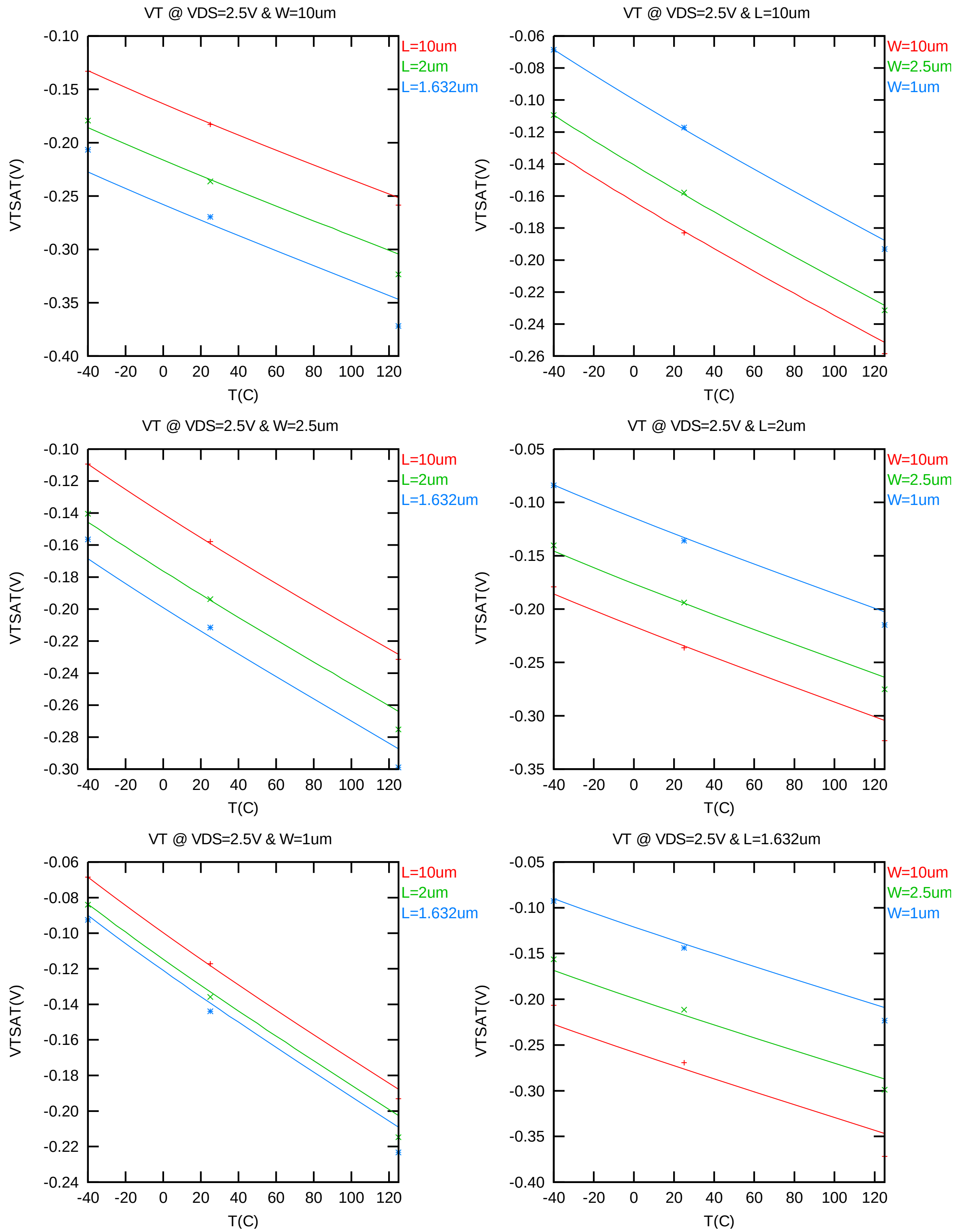

Figure 2-16 V_{TSAT} vs. temperature for NMOS

I_{ON} vs. temperature

The following plots show the temperature dependence of I_{ON} extracted at V_{GS}=V_{DS}=2.5 V, V_{BS}=0 V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

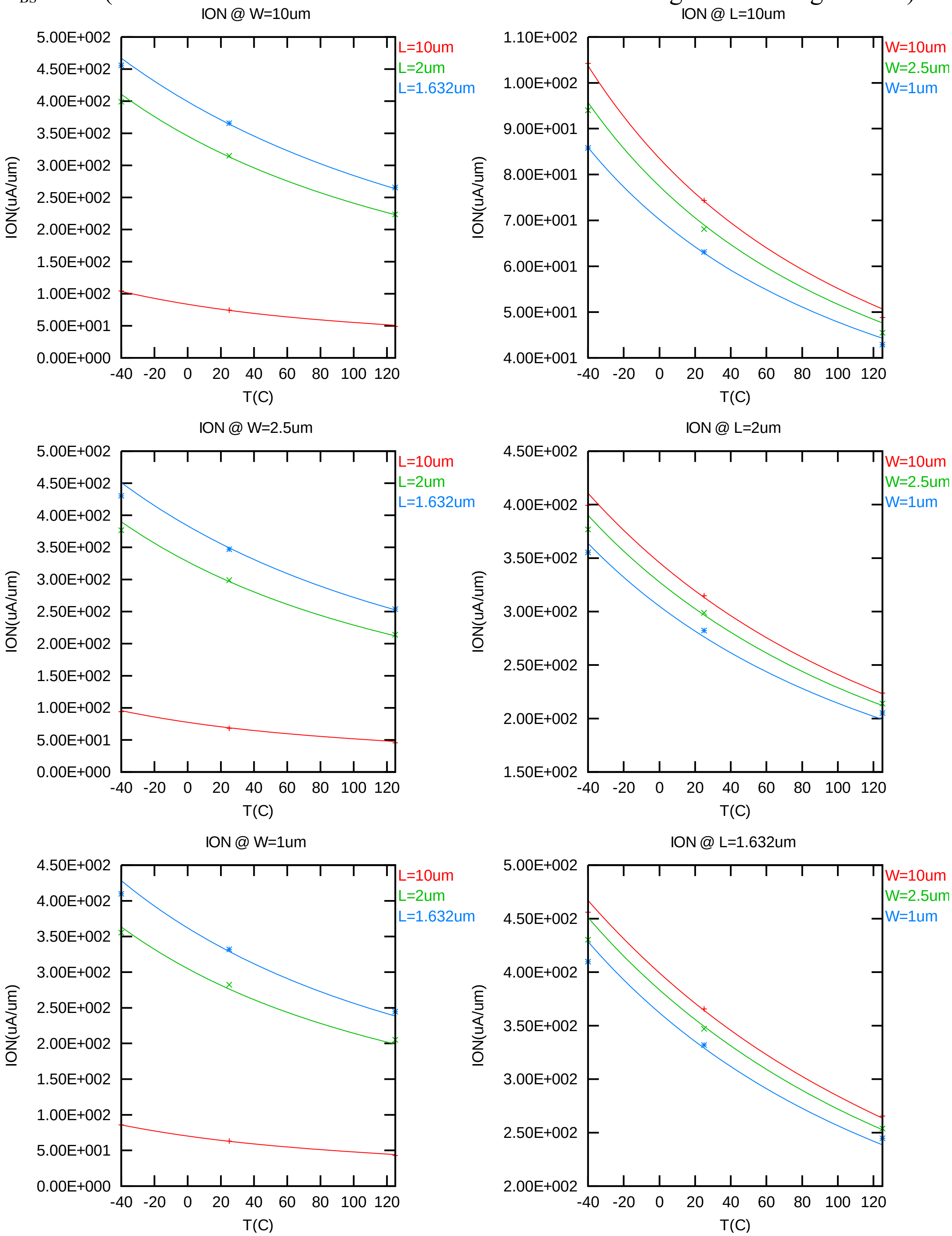

Figure 2-17 Saturation current vs. temperature for NMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at V_{DS}=2.5 V, V_{GS}=V_{BS}=0 V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

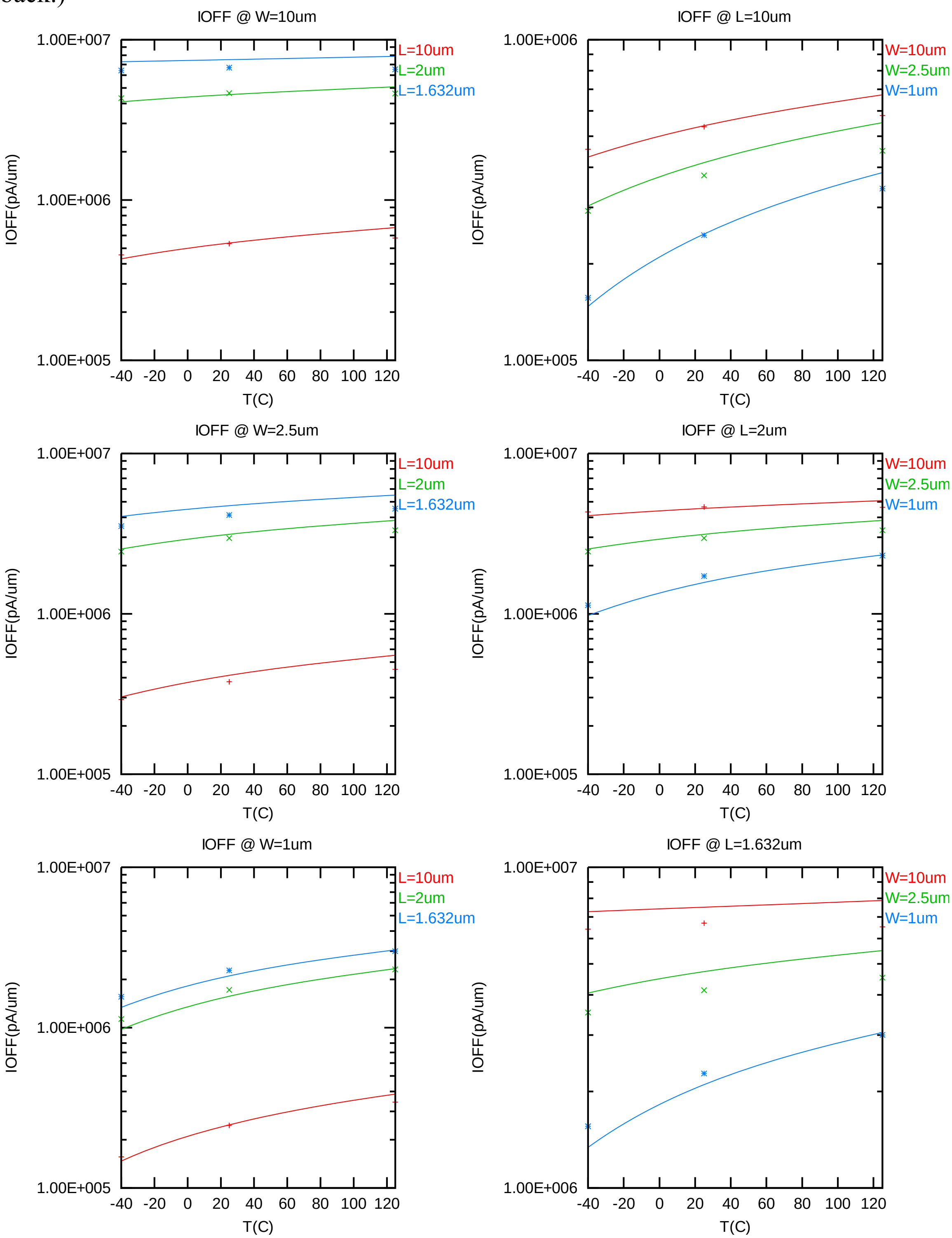

Figure 2-18 I_{OFF} vs. temperature for NMOS

2.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D, G_M and G_{DS} is quantified using the following equation:

error %=100 * | measurement - simulation | / [(measurement + simulation) / 2]

Different error criteria are defined for different parameters (I_D, G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

a. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
b. I_D in moderate inversion region, i.e. $V_T \leqq V_{GS} < V_T + V_X$
c. I_D in strong inversion region, i.e. $V_T + V_X \leqq V_{GS} \leqq V_{CC}$

, where V_X=(V_{CC} - V_T) / 2

2) G_M and G_{DS} accuracy criteria

d. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
e. G_M and G_{DS} in inversion region, i.e. $V_T \leqq V_{GS} \leqq V_{CC}$

Table 2-5 Requirements for model fit quality

Target of Quality Check	$V_{GS}< V_T$ (sub-threshold)	$V_T\leq V_{GS} < V_T+V_X$ (moderate inversion)	$V_T+V_X \leq V_{GS}\leq V_{CC}$ (strong inversion)
I_D	≤75%	≤25%	≤7%
G_M	≤50%	≤20%	≤20%
G_{DS}	≤100%	≤50%	≤50%

I_D accuracy

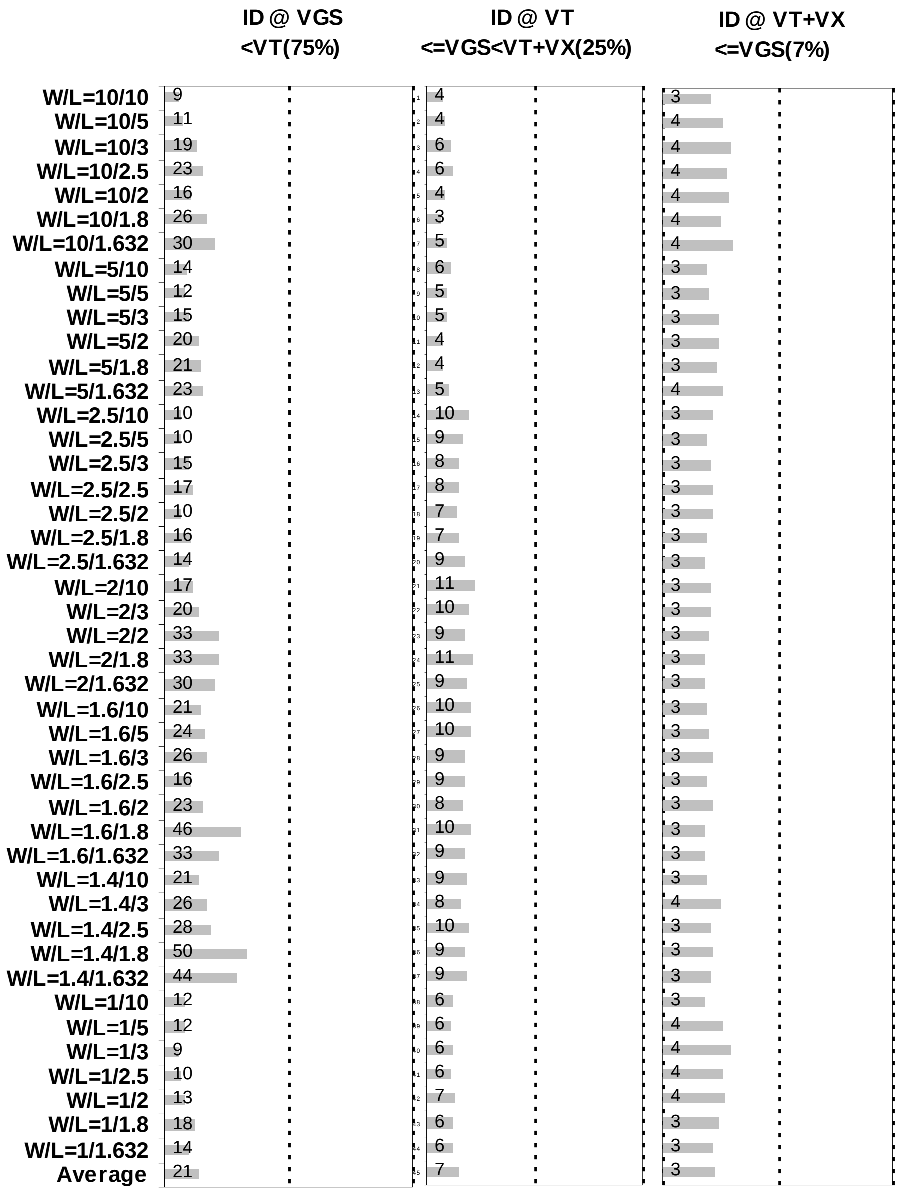

Figure 2-19 I_D accuracy at 25 C for NMOS

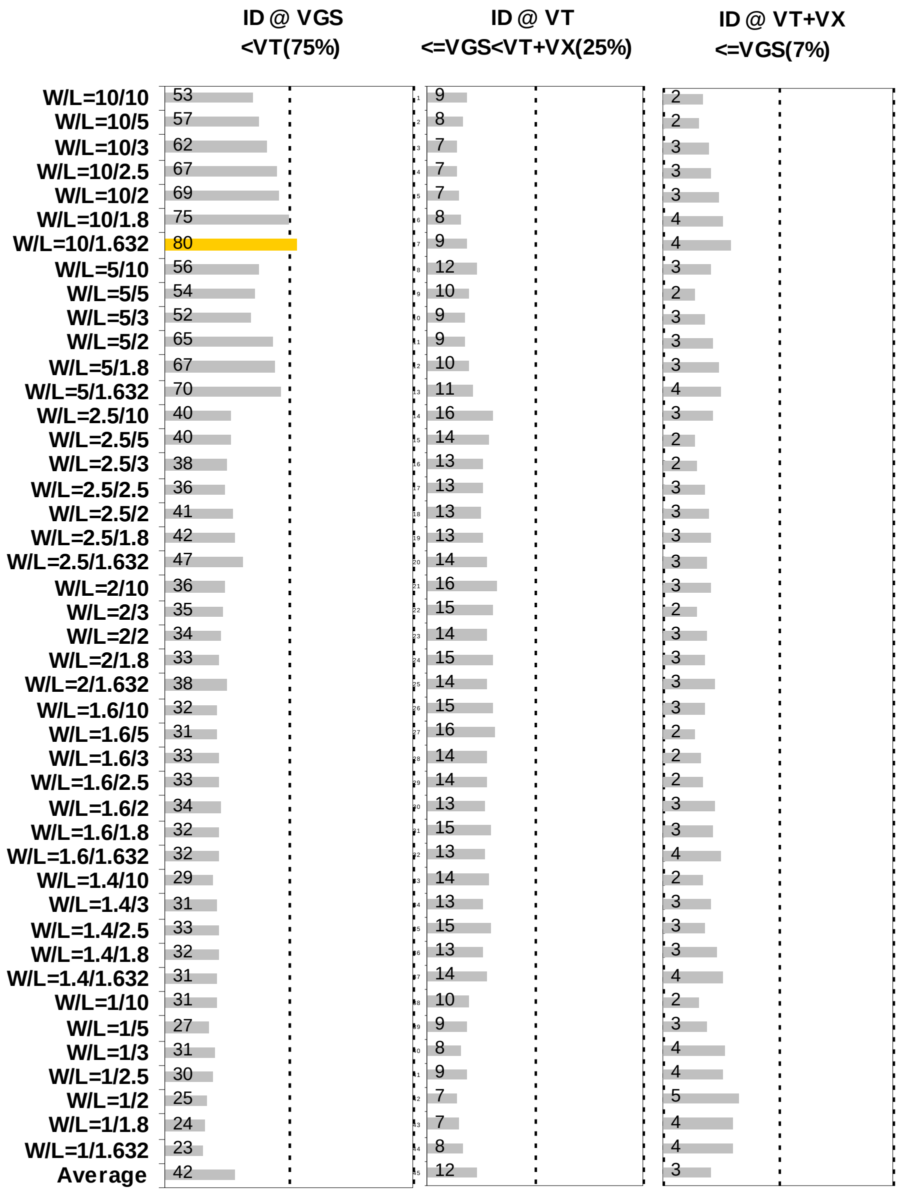

Figure 2-20 I_D accuracy at 125 C for NMOS

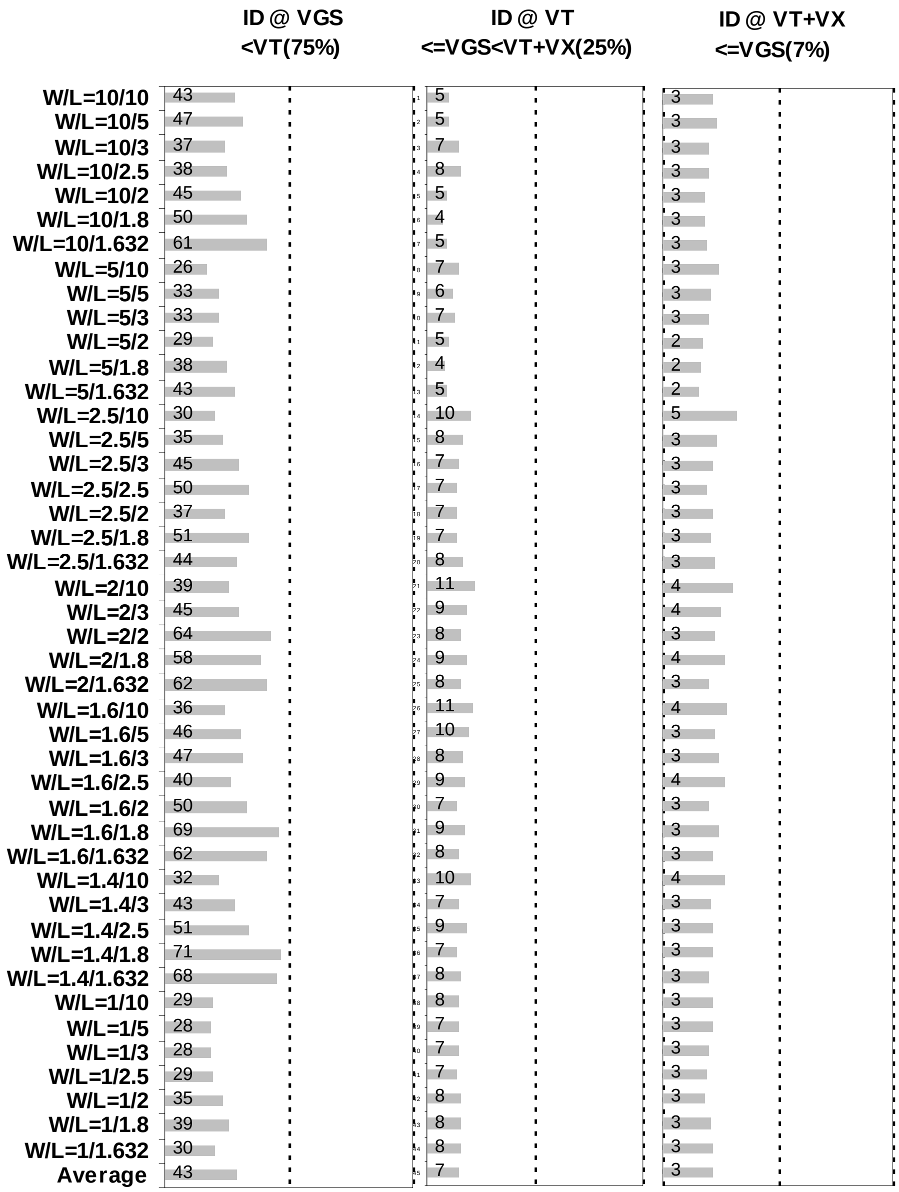

Figure 2-21 I_D accuracy at -40 C for NMOS

G_M and G_{DS} accuracy

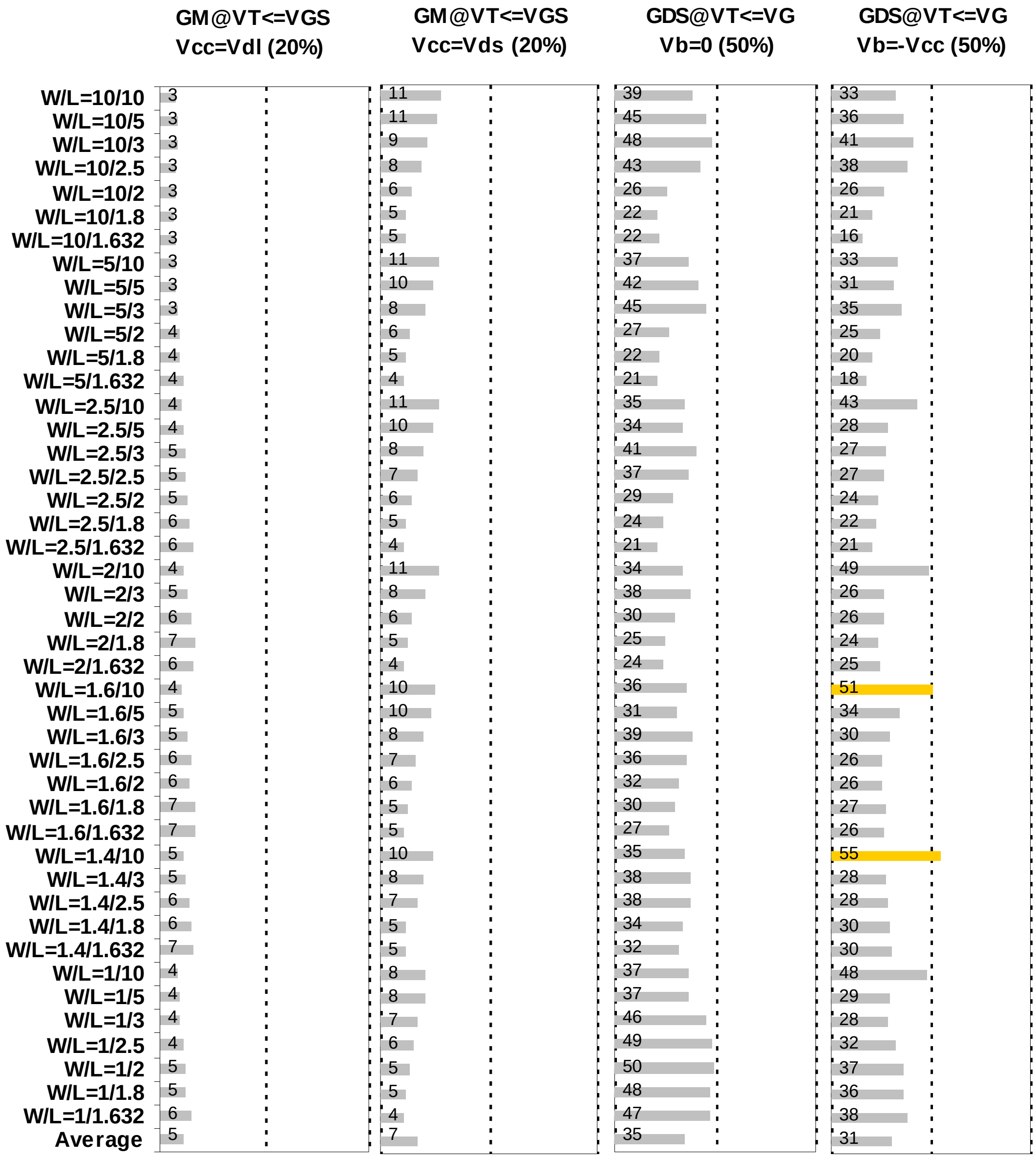

Figure 2-22 G_M and G_{DS} accuracy at 25 C for NMOS

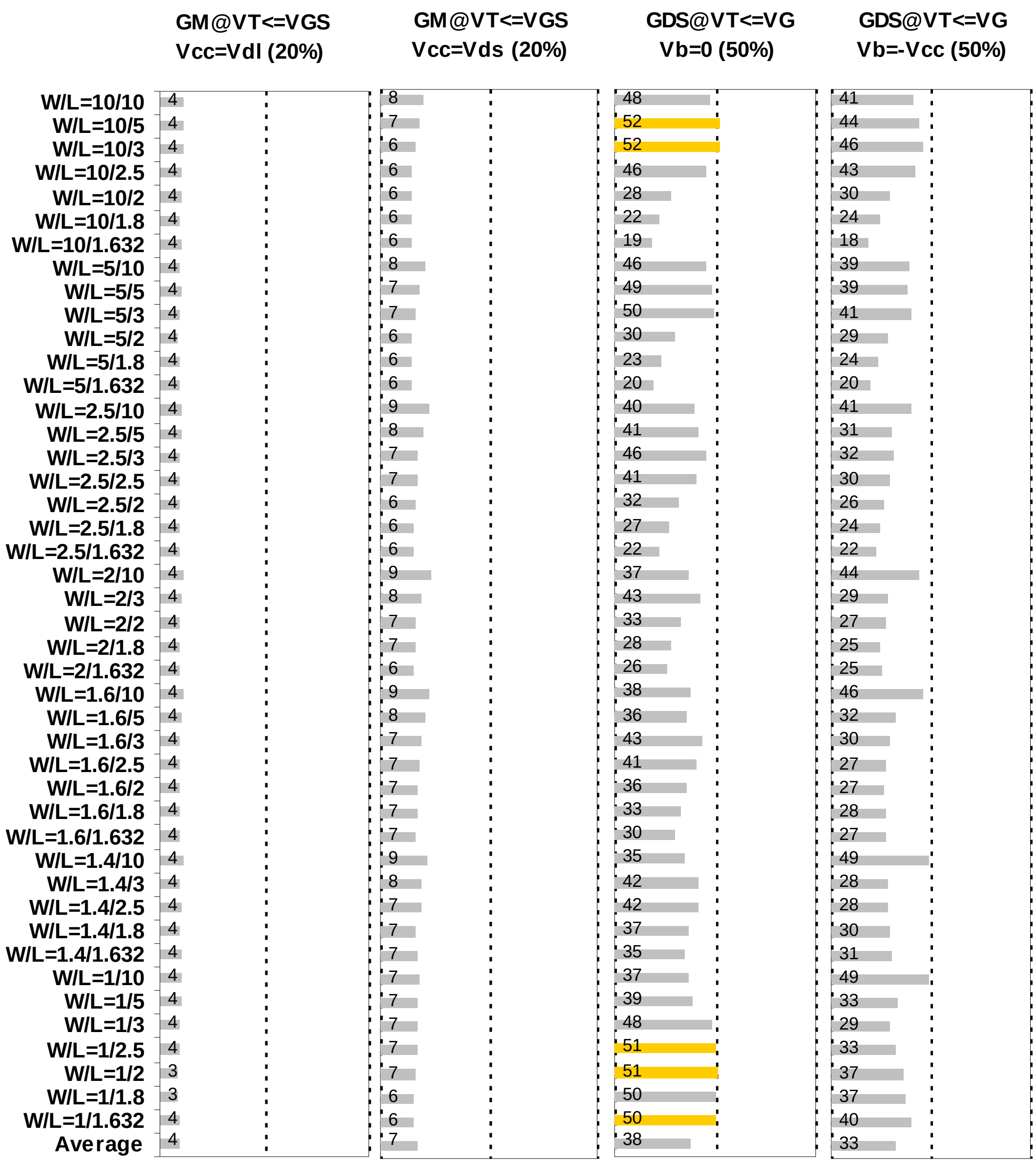

Figure 2-23 G_M and G_{DS} accuracy at 125 C for NMOS

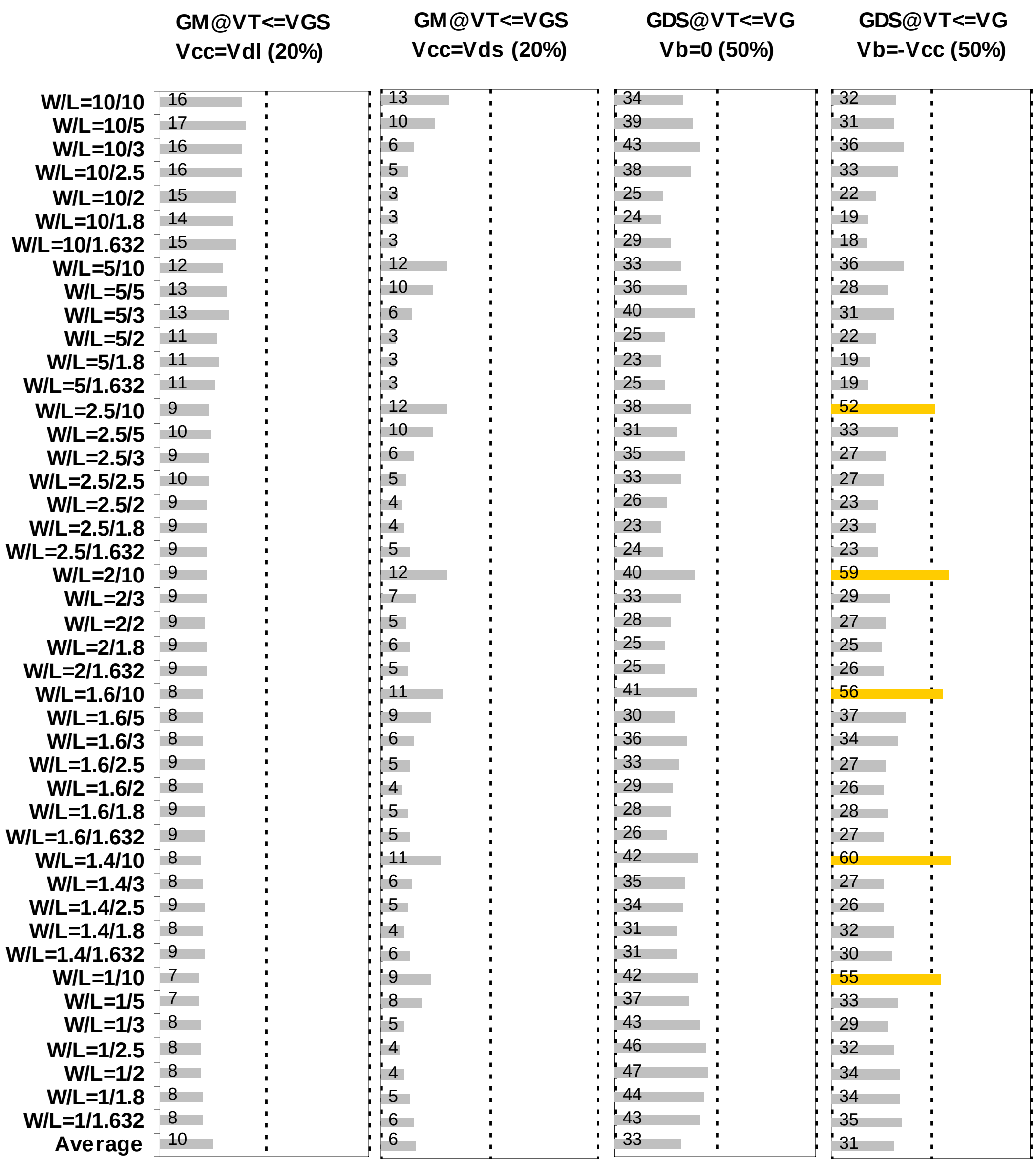

Figure 2-24 G_M and G_{DS} accuracy at -40 C for NMOS

2.4 MOSFET Capacitance Model

2.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-6 Measurement structure for NMOS Cgg capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components, C_{GDS}	80	80

Table 2-7 Large area measurement structure for NMOS Cgc capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	2	180
	2	100

2.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized in Table 2-8 below.

Table 2-8 Measurement conditions for NMOS capacitance model

V_{BIAS} [V]	From -3.63 to 3.63 V in 0.066 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

2.4.3 T_{OX} extraction

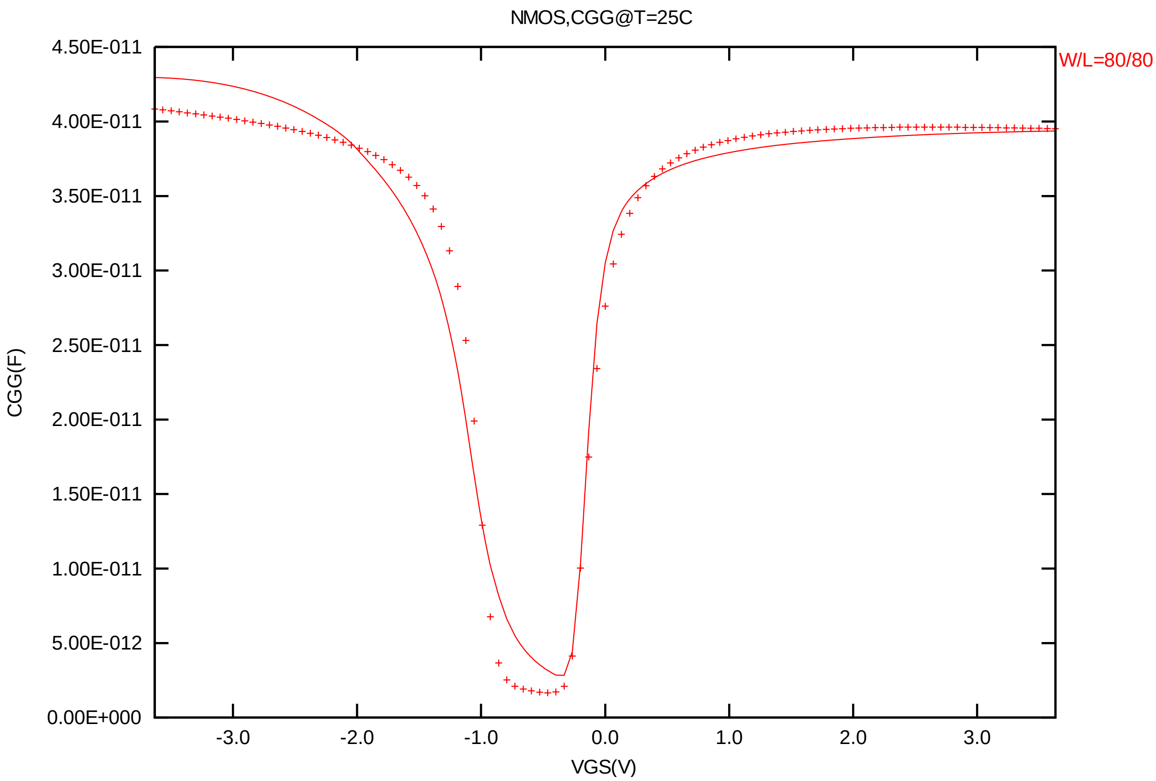

Figure 2-25 Measured and simulated C_{GG} curves

2.4.4 Width and length AC model accuracy

Width and length AC model accuracy

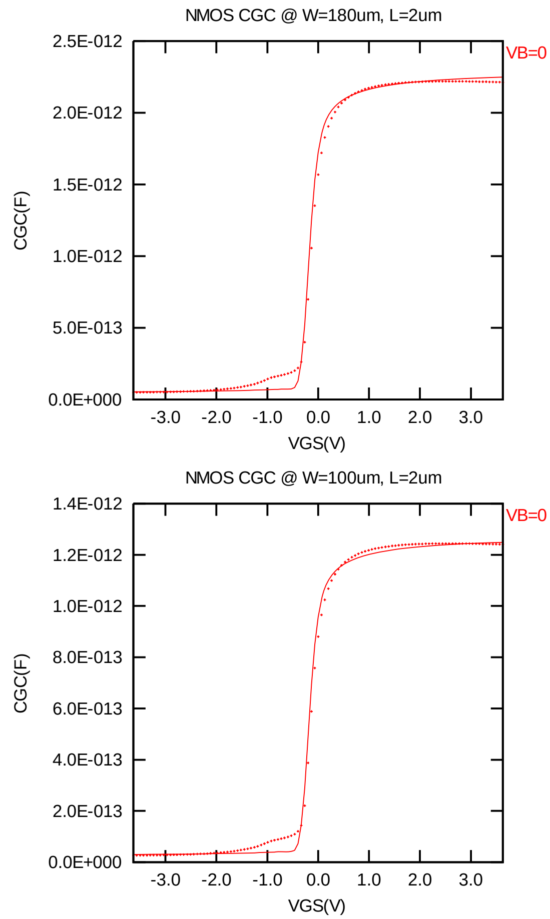

Figure 2-26 C_{GC} plot for NMOS

2.5 Parasitic Source / Drain Junction Capacitance Model

2.5.1 Test devices

The following test structures were used for parameter extraction.

Table 2-9 Junction capacitor structures for NMOS

Structures	Area[m^2]	Perimeter[m]	Description
Bottom area junction capacitance	5.00E-08	2.00E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	2.70E-08	1.83E-02	Finger diode
Inner gate sidewall junction	2.70E-08	1.83E-02	Poly finger structure

2.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-10 Junction capacitor measurement conditions for NMOS

V_{BIAS} [V]	VH: pwell & VL: n+ From 0 to 3.63 V in 0.033 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-40,25,125

2.5.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from silicon data.

Table 2-11 Extracted parameters from junction capacitor measurement for NMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	Unit	Value	Parameter	Unit	Value	Parameter	Unit	Value
CJ	F/m^2	1.14E-04	CJSW	F/m	3.09E-10	CJSWG	F/m	9.93E-11
PB	V	4.41E-01	PBSW	V	8.36E-01	PBSWG	V	3.48E-01
MJ	--	3.04E-01	MJSW	--	1.23E-01	MJSWG	--	3.56E-01
TCJ	1/K	2.69E-03	TCJSW	1/K	1.00E-03	TCJSWG	1/K	6.00E-03
TPB	V/K	1.89E-03	TPBSW	V/K	3.81E-03	TPBSWG	V/K	3.03E-04

CJ, CJSW vs. Vbias plots

The following plots show the variation of bottom area junction and STI bounded sidewall at different bias voltages and temperatures.

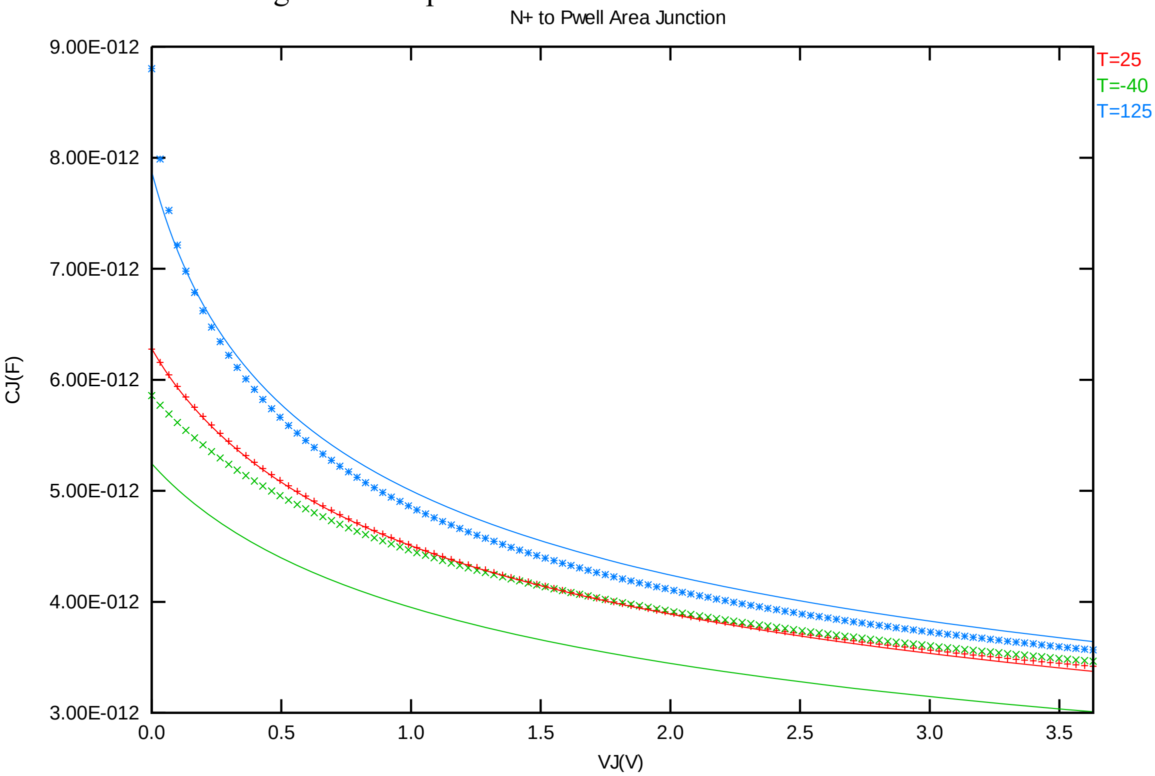

Figure 2-27 Bottom area junction capacitance at -40 C, 25 C and 125 C for NMOS

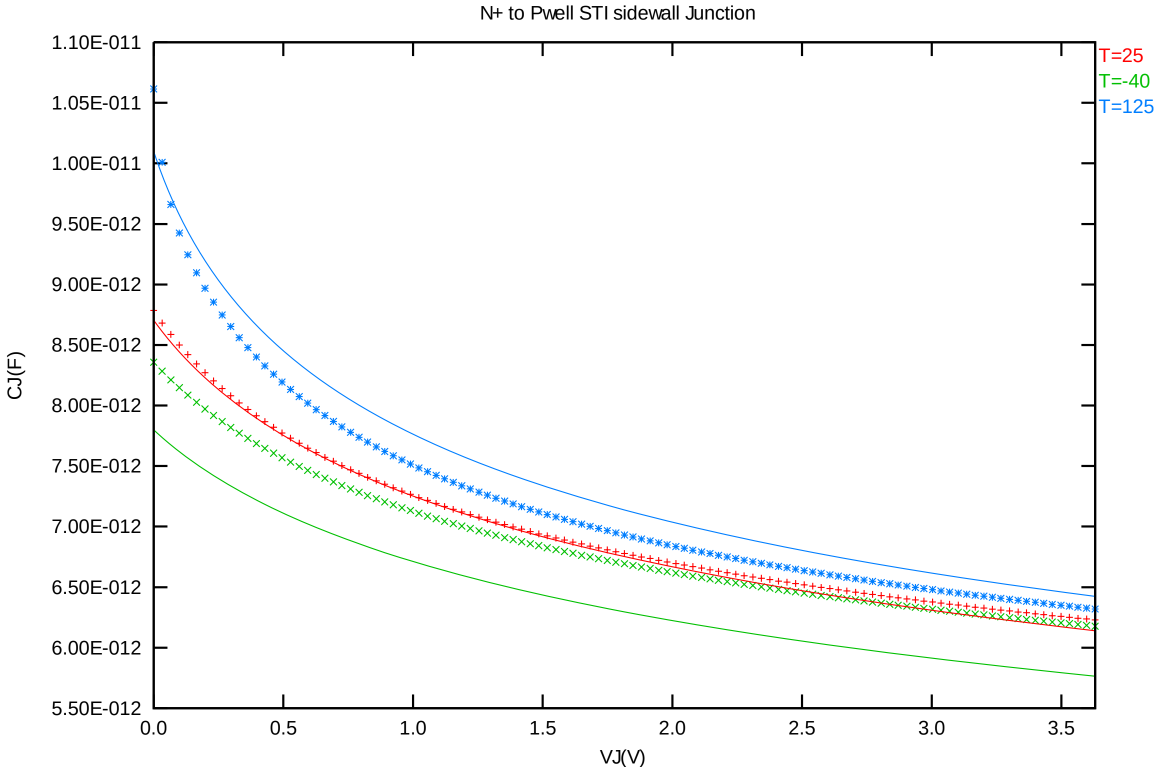

Figure 2-28 STI bounded sidewall junction capacitance at -40 C, 25 C and 125 C for NMOS

3 Worst Case Model

3.1 Parameter Variations

Table 3-1 Corner parameters

	n_nvt25_l110ae	TT	SS	FF	SNFP	FNSP
1	tox	5.40E-09	+1.62E-10	-1.62E-10	+0.00E+00	+0.00E+00
2	wwl	7.61E-21	+1.59E-20	-1.06E-20	+5.37E-21	-6.49E-21
3	xl	0.00E+00	+1.60E-08	-1.60E-08	+8.00E-09	-8.00E-09
4	xw	0.00E+00	-2.00E-08	+1.75E-08	-1.00E-08	+8.75E-09
5	vth0	-1.93E-01	+5.48E-02	-5.52E-02	+2.77E-02	-2.78E-02
6	lvth0	-3.90E-08	+1.65E-08	-1.60E-08	+7.72E-09	-7.99E-09
7	pvth0	1.60E-14	-4.43E-15	+2.86E-15	-1.35E-15	+1.79E-15
8	k3	1.75E+01	+1.80E+00	-2.13E+00	+1.21E+00	-1.30E+00
9	u0	4.23E-02	-1.38E-03	+1.05E-03	-9.13E-04	+8.58E-04
10	vsat	1.46E+05	-3.58E+03	+2.23E+03	-1.91E+03	+2.25E+03
11	pvsat	1.05E-07	-1.28E-08	+7.74E-09	-9.43E-09	+5.70E-09
12	b0	2.20E-06	-2.89E-07	+3.27E-07	-1.63E-07	+1.71E-07
13	rdsw	5.76E+02	+1.25E+02	-9.00E+01	+6.25E+01	-4.50E+01
14	cgso	2.25E-12	-10.0%	+10.0%	+0.0%	+0.0%
15	cgdo	2.25E-12	-10.0%	+10.0%	+0.0%	+0.0%
16	cgsl	2.50E-10	-10.0%	+10.0%	+0.0%	+0.0%
17	cgdl	2.50E-10	-10.0%	+10.0%	+0.0%	+0.0%
18	cj	1.14E-04	+10.0%	-10.0%	+10.0%	-10.0%
19	cjsw	3.09E-10	+10.0%	-10.0%	+10.0%	-10.0%
20	cjswg	9.93E-11	+10.0%	-10.0%	+10.0%	-10.0%

3.2 Wafer Data vs. Corner Plots

Threshold voltage in linear region V_{TSAT}

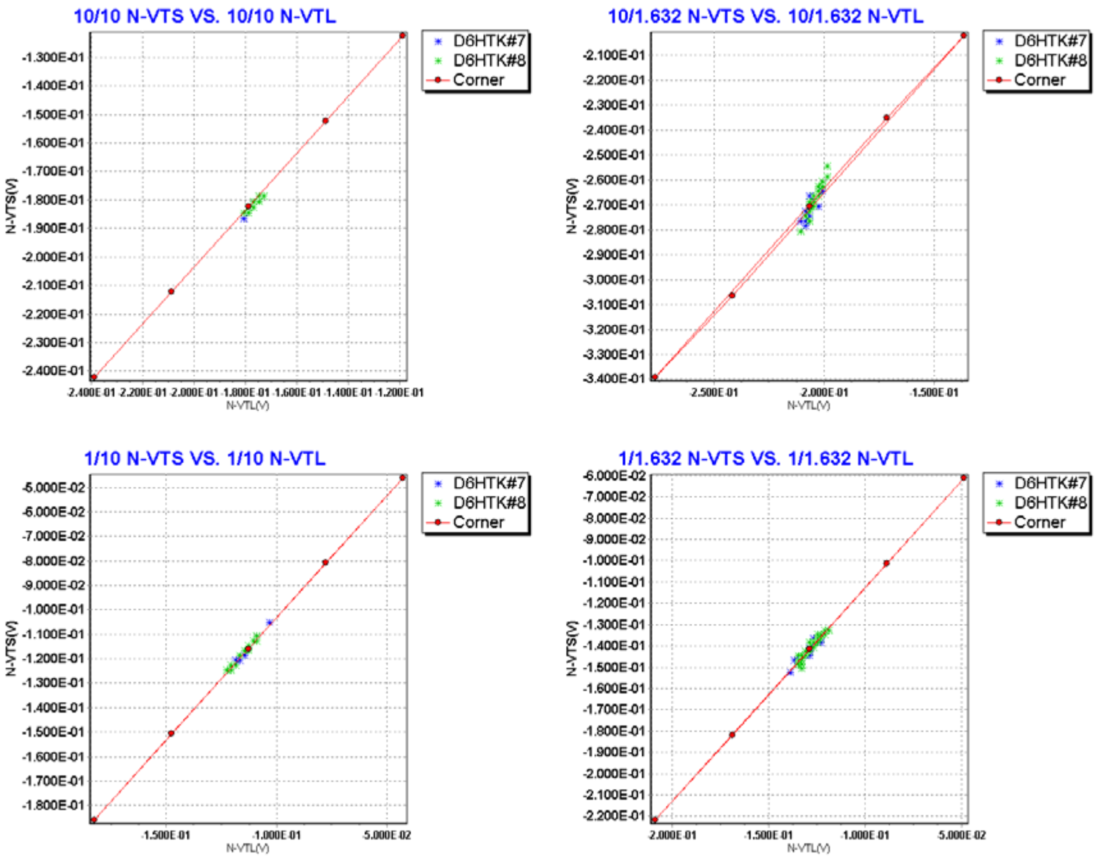

Figure 3-1 Plot of V_{TLIN} vs. V_{TSAT}

Threshold voltage in saturation region I_{DLIN}

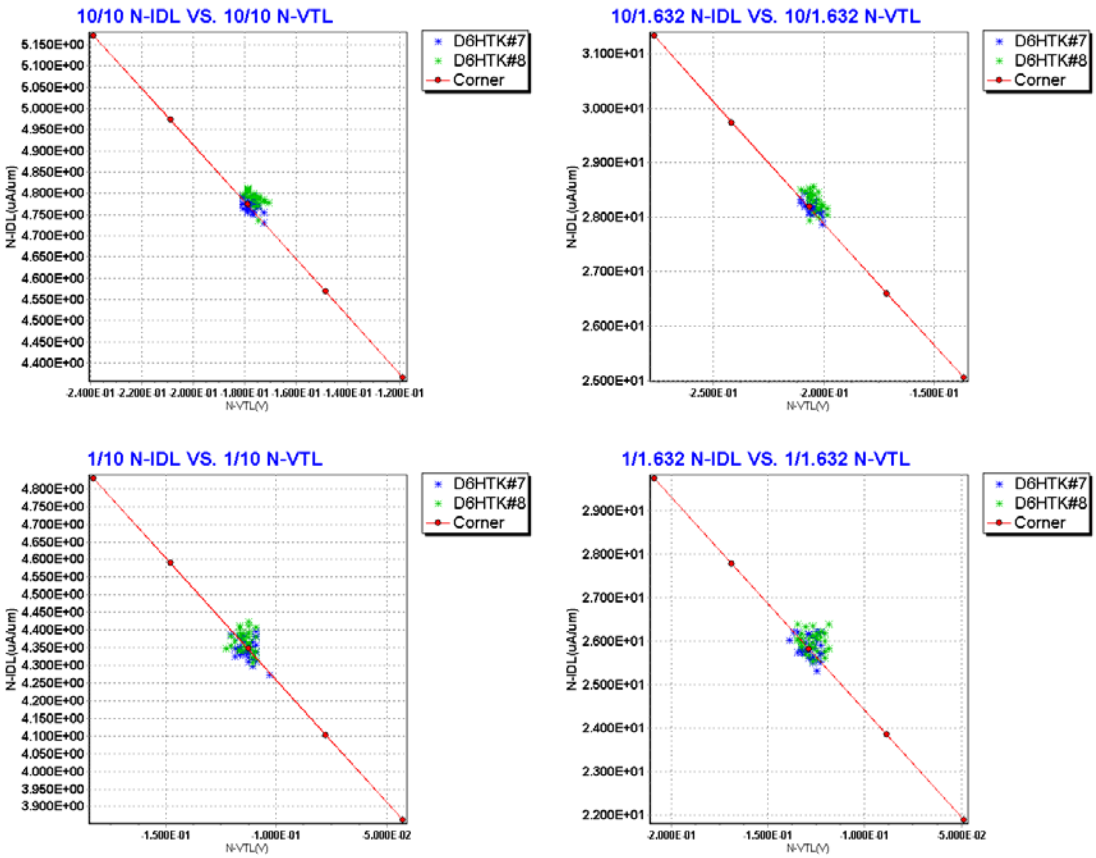

Figure 3-2 Plot of V$_{TLIN}$ vs. I$_{DLIN}$

Saturation current I_{ON}

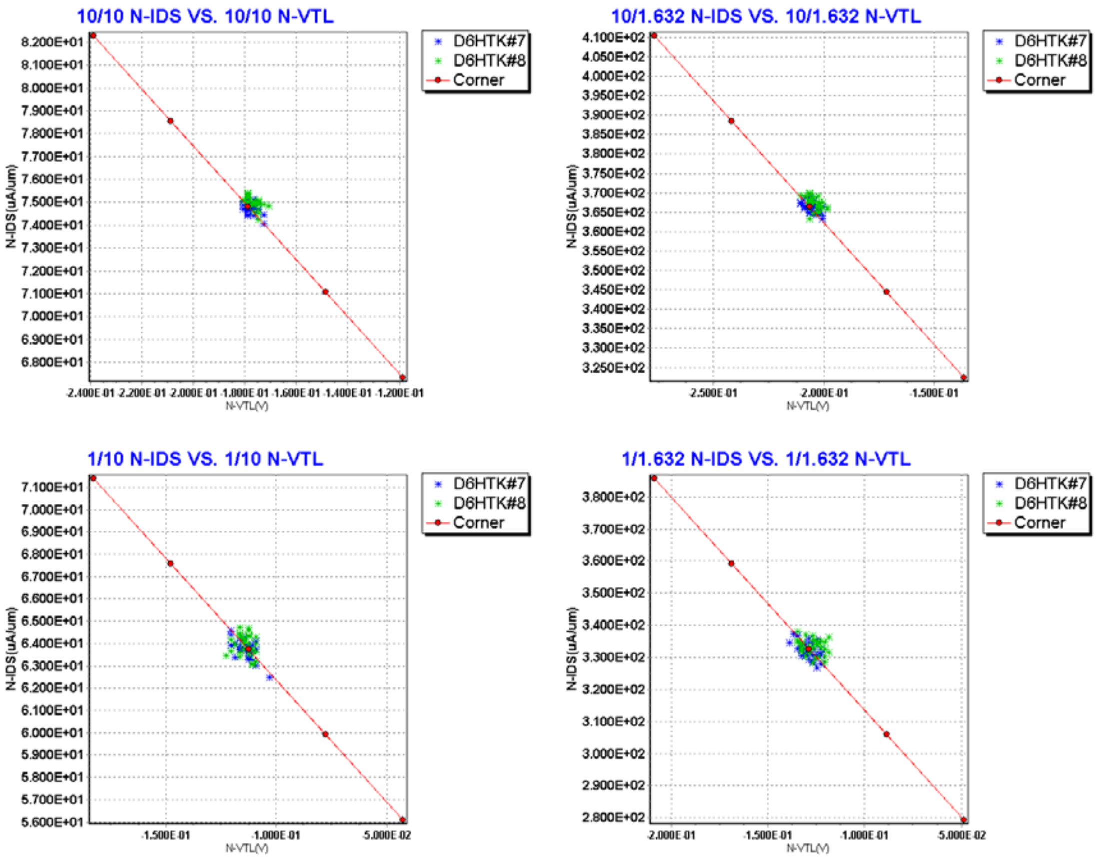

Figure 3-3 Plot of V_{TLIN} vs. I_{ON}

Off state current I_{OFF}

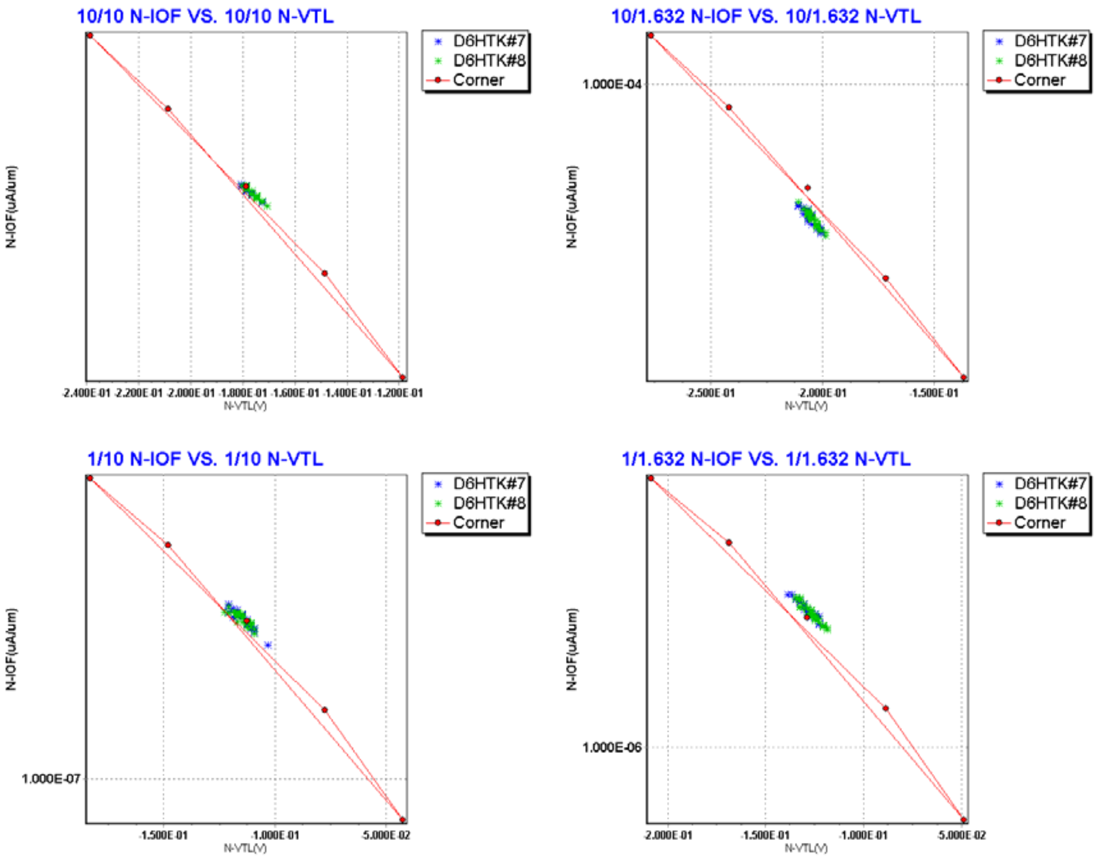

Figure 3-4 Plot of V_{TLIN} vs. I_{OFF}

3.3 Device Performance Reference Tables

** The model simulation results in the following tables have already reflected silicon data that is sizing channel length back for 0nm. It means: The EDR value of size W/L : 10um/1.632um would be the same as the model card simulation result in size of W/L : 10um/1.62um(After 90% shrink without 12nm sizing channel length back).

Table 3-2 NMOS ID current at V_{GS}=2.5 V, V_{DS}=1.25 V and V_{DS}=2.5 V.

NMOS	Temp. (℃)	-40 ℃		25 ℃		125 ℃	
IDSAT (μA)	VDS	1.25V	2.5V	1.25V	2.5V	1.25V	2.5V
TT	9/9	337.72	1034.80	239.95	741.77	162.02	507.65
SS	9/9	289.28	929.00	206.83	667.05	140.88	457.41
FF	9/9	389.06	1140.50	274.82	816.52	184.13	557.91
SNFP	9/9	314.59	980.41	224.33	704.42	152.22	483.70
FNSP	9/9	361.27	1089.20	255.80	779.11	171.95	531.51
TT	9/1.62	1648.90	4222.10	1245.70	3292.70	873.64	2384.30
SS	9/1.62	1389.00	3705.10	1054.50	2896.10	745.90	2104.80
FF	9/1.62	1925.30	4736.80	1448.40	3689.30	1008.20	2663.70
SNFP	9/1.62	1522.20	3961.10	1153.30	3094.20	812.78	2248.10
FNSP	9/1.62	1778.20	4481.20	1340.00	3491.00	935.71	2520.60
TT	1/9	29.50	96.10	21.57	70.44	15.14	49.69
SS	1/9	24.38	84.29	17.95	61.93	12.74	43.81
FF	1/9	35.03	107.92	25.45	78.95	17.70	55.55
SNFP	1/9	27.01	90.06	19.83	66.19	14.01	46.86
FNSP	1/9	32.06	102.15	23.35	74.69	16.31	52.50
TT	1/1.62	154.81	432.67	116.50	332.50	82.90	241.31
SS	1/1.62	122.54	361.56	92.93	279.30	66.90	203.96
FF	1/1.62	189.87	503.64	142.00	385.70	100.04	278.62
SNFP	1/1.62	138.87	396.49	105.01	305.90	75.23	223.19
FNSP	1/1.62	171.31	468.51	128.44	359.10	90.88	259.51

Table 3-3 NMOS ID current density at V_{GS}=2.5 V, V_{DS}=1.25 V and V_{DS}=2.5 V.

NMOS	Temp. (℃)	-40 ℃		25 ℃		125 ℃	
IDSAT (μA/um)	VDS	1.25V	2.5V	1.25V	2.5V	1.25V	2.5V
TT	9/9	37.52	114.98	26.66	82.42	18.00	56.41
SS	9/9	32.14	103.22	22.98	74.12	15.65	50.82
FF	9/9	43.23	126.72	30.54	90.72	20.46	61.99
SNFP	9/9	34.95	108.93	24.93	78.27	16.91	53.74
FNSP	9/9	40.14	121.02	28.42	86.57	19.11	59.06
TT	9/1.62	183.21	469.12	138.41	365.86	97.07	264.92
SS	9/1.62	154.33	411.68	117.17	321.79	82.88	233.87
FF	9/1.62	213.92	526.31	160.93	409.92	112.02	295.97
SNFP	9/1.62	169.13	440.12	128.14	343.80	90.31	249.79
FNSP	9/1.62	197.58	497.91	148.89	387.89	103.97	280.07
TT	1/9	29.50	96.10	21.57	70.44	15.14	49.69
SS	1/9	24.38	84.29	17.95	61.93	12.74	43.81
FF	1/9	35.03	107.92	25.45	78.95	17.70	55.55
SNFP	1/9	27.01	90.06	19.83	66.19	14.01	46.86
FNSP	1/9	32.06	102.15	23.35	74.69	16.31	52.50
TT	1/1.62	154.81	432.67	116.50	332.50	82.90	241.31
SS	1/1.62	122.54	361.56	92.93	279.30	66.90	203.96
FF	1/1.62	189.87	503.64	142.00	385.70	100.04	278.62
SNFP	1/1.62	138.87	396.49	105.01	305.90	75.23	223.19
FNSP	1/1.62	171.31	468.51	128.44	359.10	90.88	259.51

Table 3-4 NMOS V_{TLIN} and V_{TSAT}.

NMOS	Temp. (℃)	-40 ℃		25 ℃		125 ℃	
VTH (V)	VDS	0.1V	2.5V	0.1V	2.5V	0.1V	2.5V
TT	9/9	-0.128	-0.132	-0.178	-0.182	-0.247	-0.252
SS	9/9	-0.068	-0.073	-0.118	-0.122	-0.186	-0.191
FF	9/9	-0.189	-0.192	-0.239	-0.242	-0.308	-0.312
SNFP	9/9	-0.099	-0.103	-0.148	-0.152	-0.216	-0.221
FNSP	9/9	-0.158	-0.162	-0.208	-0.212	-0.277	-0.282
TT	9/1.62	-0.156	-0.219	-0.205	-0.268	-0.275	-0.338
SS	9/1.62	-0.086	-0.151	-0.135	-0.199	-0.203	-0.269
FF	9/1.62	-0.225	-0.288	-0.275	-0.337	-0.346	-0.408
SNFP	9/1.62	-0.121	-0.184	-0.170	-0.232	-0.239	-0.302
FNSP	9/1.62	-0.190	-0.255	-0.240	-0.304	-0.310	-0.375
TT	1/9	-0.063	-0.067	-0.113	-0.116	-0.181	-0.186
SS	1/9	0.006	0.003	-0.043	-0.046	-0.110	-0.115
FF	1/9	-0.133	-0.136	-0.183	-0.186	-0.252	-0.256
SNFP	1/9	-0.028	-0.032	-0.078	-0.081	-0.146	-0.150
FNSP	1/9	-0.098	-0.101	-0.148	-0.151	-0.217	-0.221
TT	1/1.62	-0.079	-0.092	-0.128	-0.141	-0.197	-0.211
SS	1/1.62	0.001	-0.012	-0.048	-0.061	-0.116	-0.129
FF	1/1.62	-0.158	-0.172	-0.208	-0.221	-0.278	-0.292
SNFP	1/1.62	-0.039	-0.052	-0.088	-0.101	-0.157	-0.170
FNSP	1/1.62	-0.118	-0.132	-0.168	-0.182	-0.238	-0.252

Table 3-5 NMOS ID current at V_{GS}=0 and V_{DS}=0.1 V or V_{DS}=2.5 V (off current).

NMOS	Temp. (℃)	-40 ℃		25 ℃		125 ℃	
IOFF (A)	VDS	0.1V	2.5V	0.1V	2.5V	0.1V	2.5V
TT	9/9	3.38E-06	4.33E-06	3.67E-06	5.43E-06	3.64E-06	6.78E-06
SS	9/9	1.42E-06	1.66E-06	2.10E-06	2.71E-06	2.51E-06	4.09E-06
FF	9/9	5.95E-06	8.72E-06	5.51E-06	9.37E-06	4.91E-06	1.04E-05
SNFP	9/9	2.32E-06	2.81E-06	2.86E-06	3.95E-06	3.08E-06	5.37E-06
FNSP	9/9	4.58E-06	6.26E-06	4.53E-06	7.20E-06	4.23E-06	8.39E-06
TT	9/1.62	2.53E-05	6.17E-05	2.58E-05	6.45E-05	2.43E-05	6.85E-05
SS	9/1.62	1.05E-05	2.98E-05	1.44E-05	3.62E-05	1.62E-05	4.33E-05
FF	9/1.62	4.42E-05	1.07E-04	3.92E-05	1.03E-04	3.33E-05	1.01E-04
SNFP	9/1.62	1.74E-05	4.38E-05	2.00E-05	4.90E-05	2.03E-05	5.51E-05
FNSP	9/1.62	3.42E-05	8.31E-05	3.21E-05	8.25E-05	2.85E-05	8.38E-05
TT	1/9	1.41E-07	1.58E-07	2.19E-07	2.71E-07	2.74E-07	4.25E-07
SS	1/9	2.77E-08	3.05E-08	7.92E-08	8.87E-08	1.51E-07	2.01E-07
FF	1/9	3.84E-07	4.77E-07	4.19E-07	5.99E-07	4.24E-07	7.57E-07
SNFP	1/9	6.80E-08	7.52E-08	1.40E-07	1.63E-07	2.10E-07	3.02E-07
FNSP	1/9	2.47E-07	2.89E-07	3.11E-07	4.13E-07	3.44E-07	5.73E-07
TT	1/1.62	1.04E-06	1.39E-06	1.51E-06	2.16E-06	1.78E-06	3.12E-06
SS	1/1.62	1.81E-07	2.53E-07	4.92E-07	6.49E-07	9.03E-07	1.35E-06
FF	1/1.62	2.92E-06	4.21E-06	3.01E-06	4.95E-06	2.88E-06	5.84E-06
SNFP	1/1.62	4.81E-07	6.49E-07	9.33E-07	1.26E-06	1.32E-06	2.13E-06
FNSP	1/1.62	1.86E-06	2.56E-06	2.20E-06	3.37E-06	2.29E-06	4.33E-06

4 Noise Characteristics

4.1 Verification Plots

NMOS, W=10um L=10um

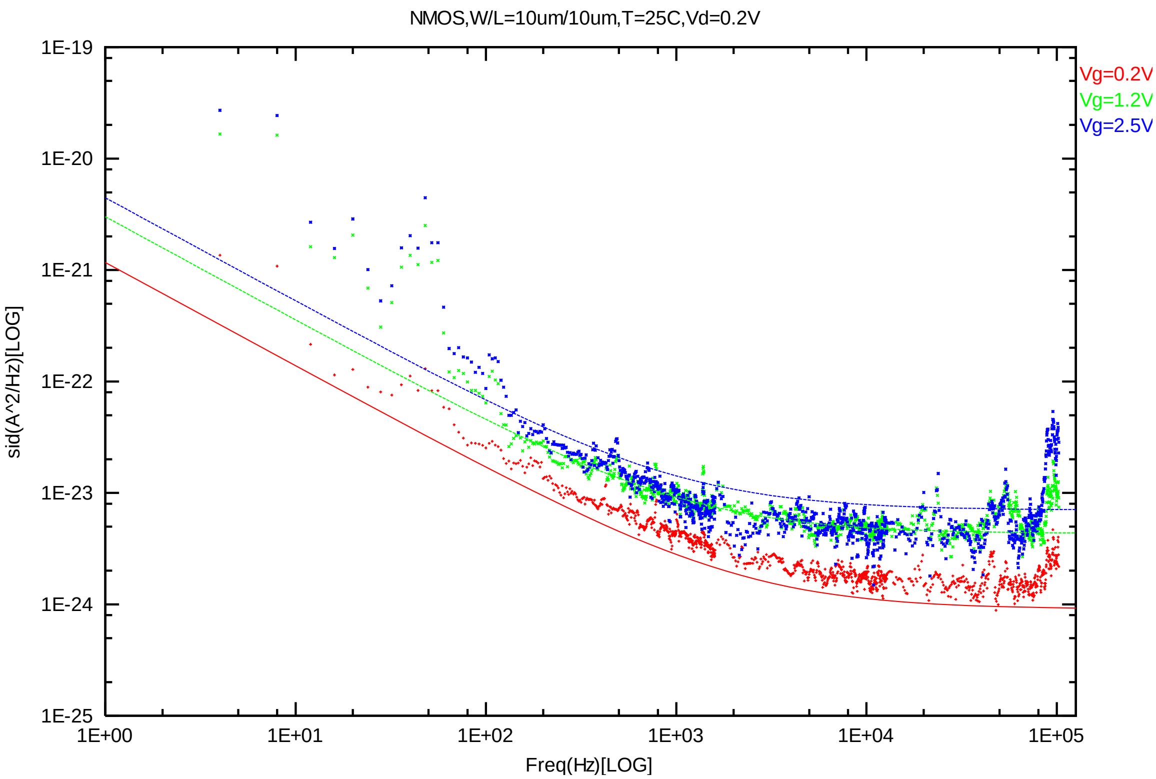

Figure 4-1 NMOS W/L=10um/10um @ Vd=0.2 V

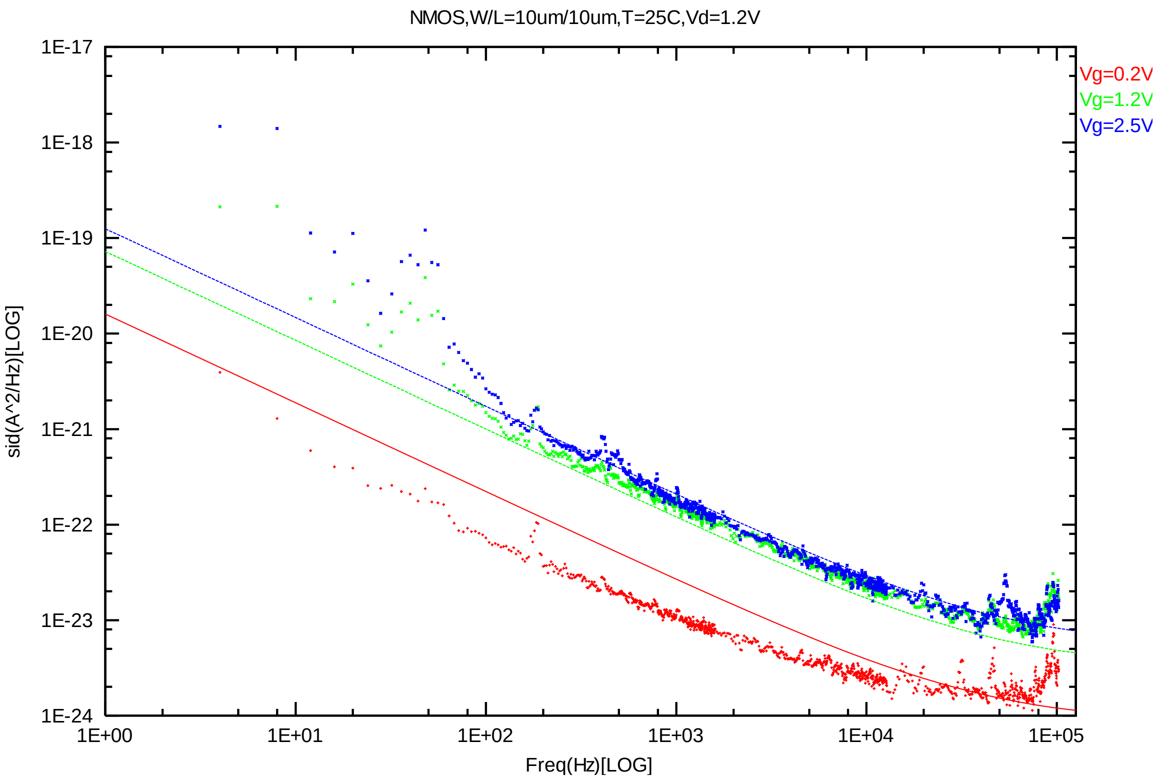

Figure 4-2 NMOS W/L=10um/10um @ Vd=1.2 V

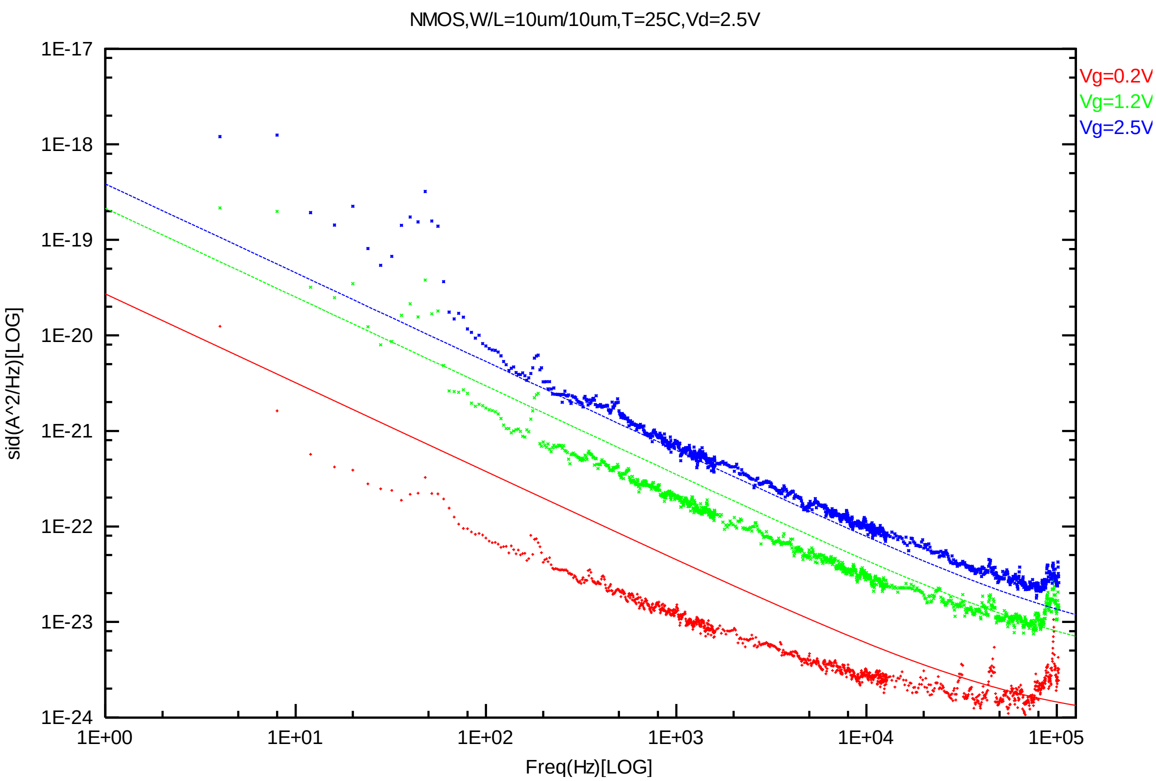

Figure 4-3 NMOS W/L=10um/10um @ Vd=2.5 V

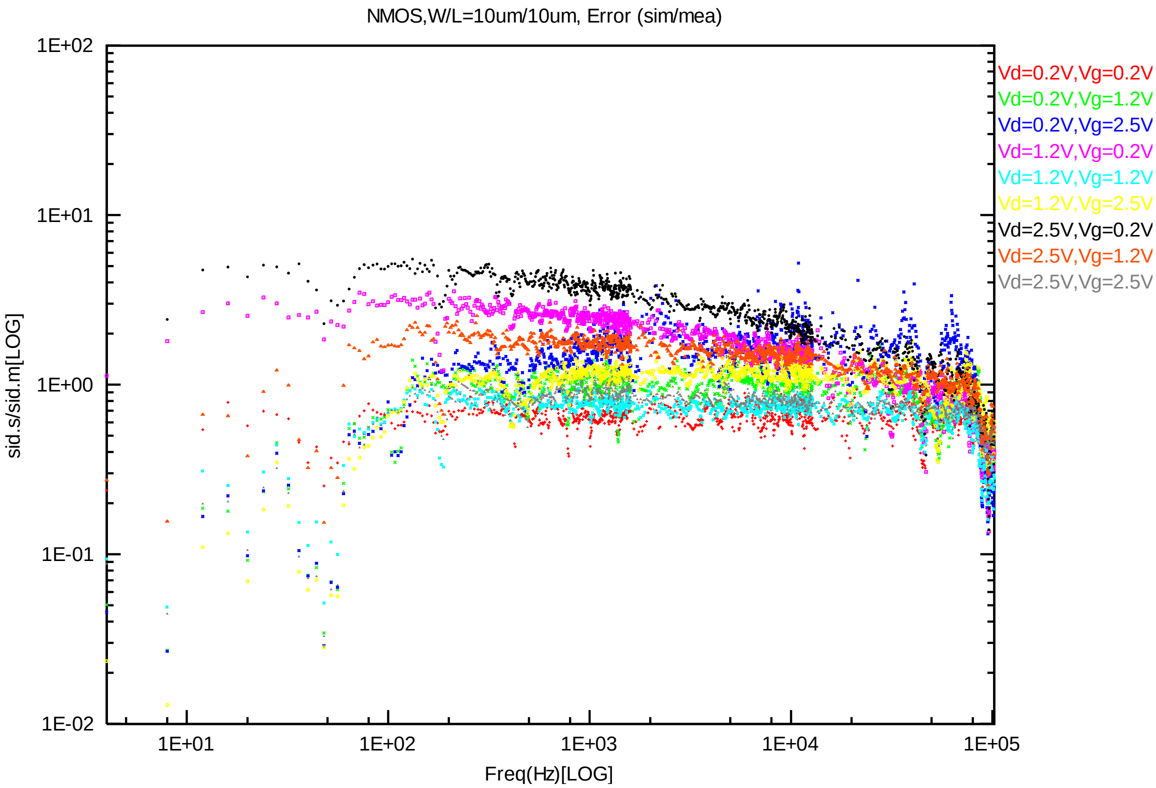

Figure 4-4 NMOS W/L=10um/10um error distribution plot

NMOS, W=10um L=7um

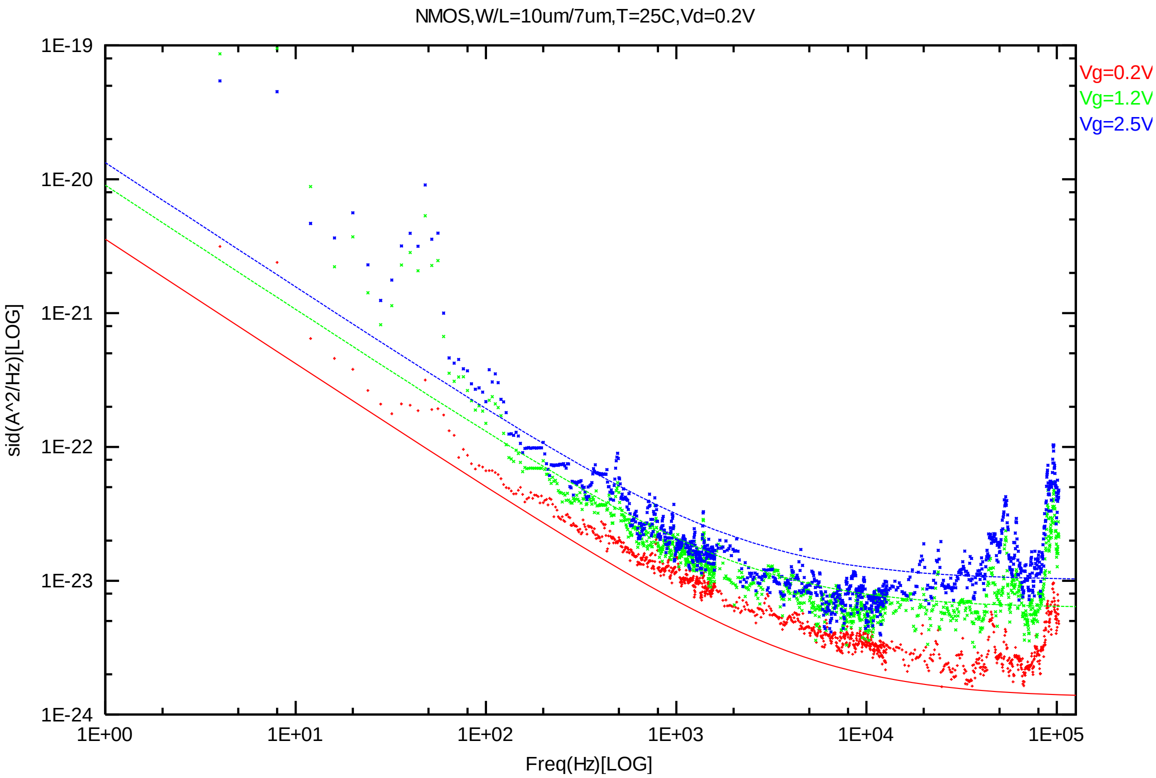

Figure 4-5 NMOS W/L=10um/7um @ Vd=0.2 V

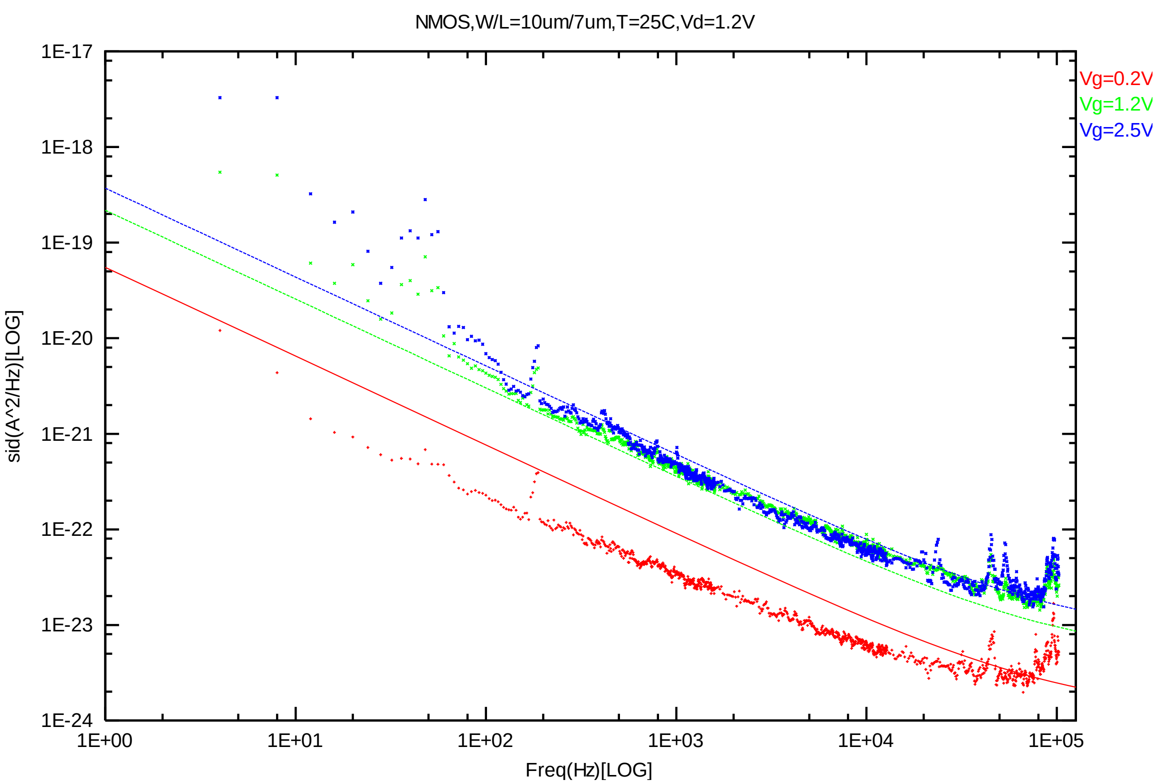

Figure 4-6 NMOS W/L=10um/7um @ Vd=1.2 V

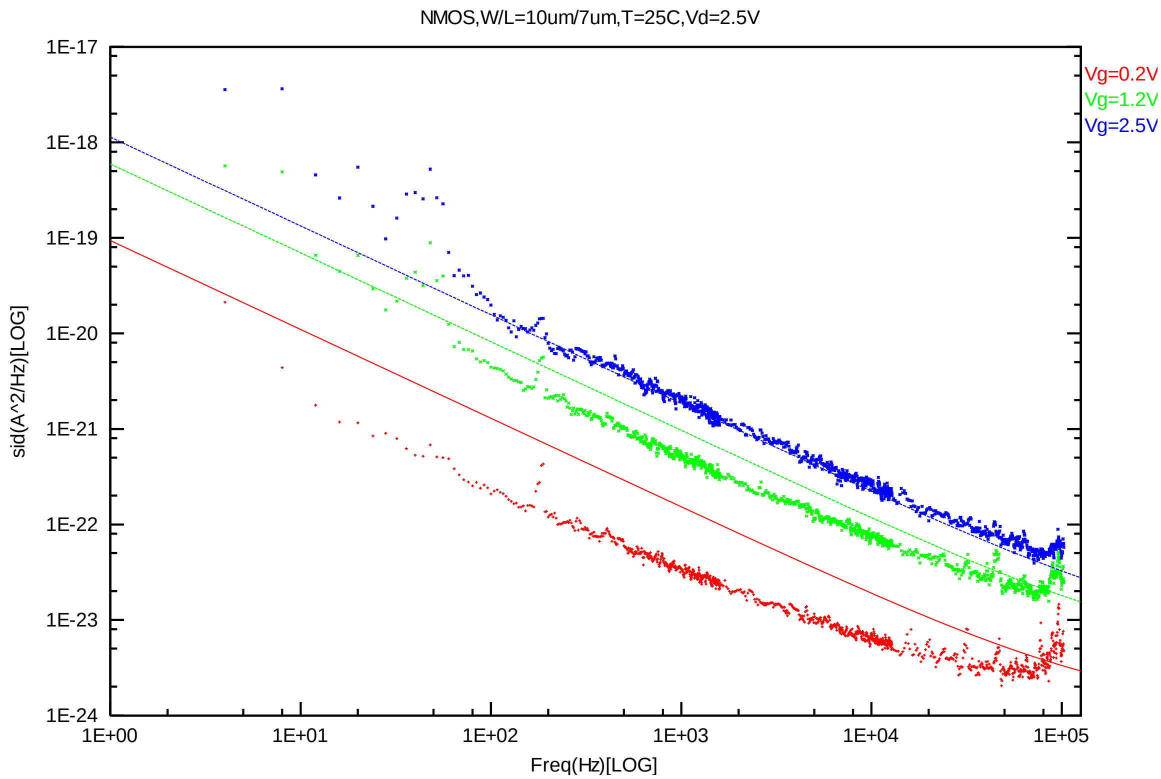

Figure 4-7 NMOS W/L=10um/7um @ Vd=2.5 V

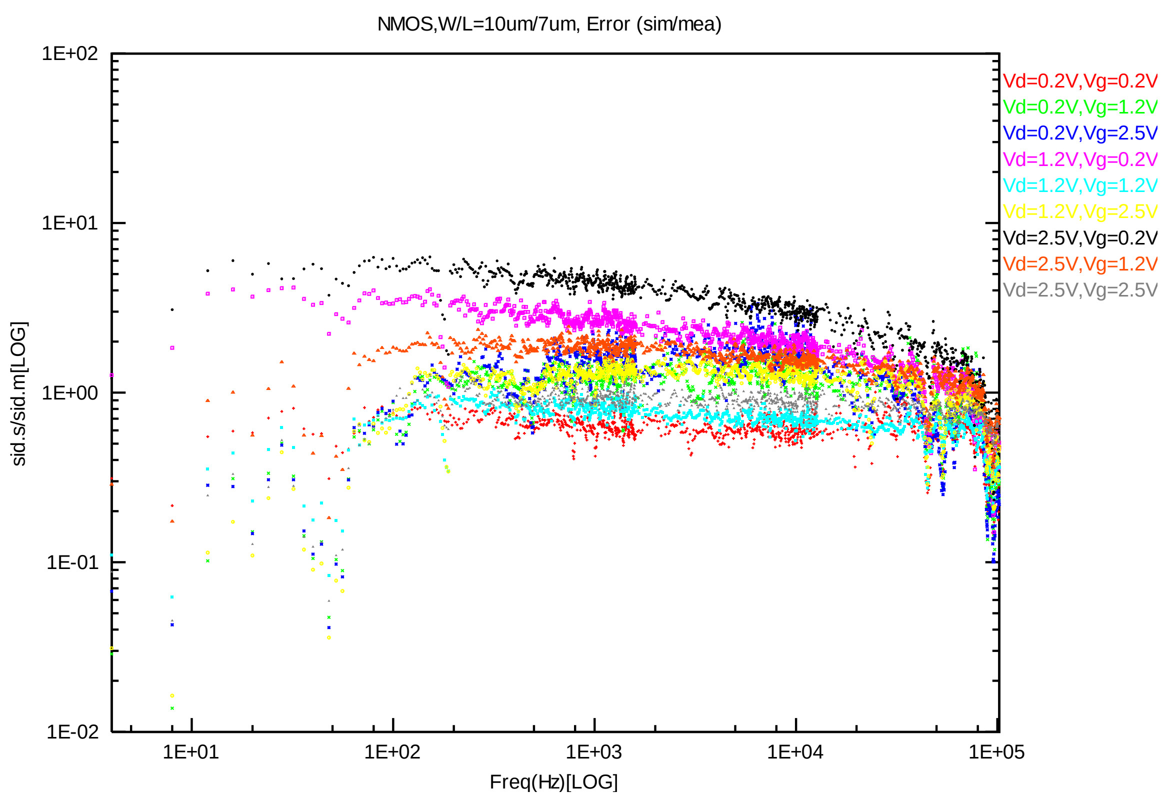

Figure 4-8 NMOS W/L=10um/7um error distribution plot

NMOS, W=10um L=5um

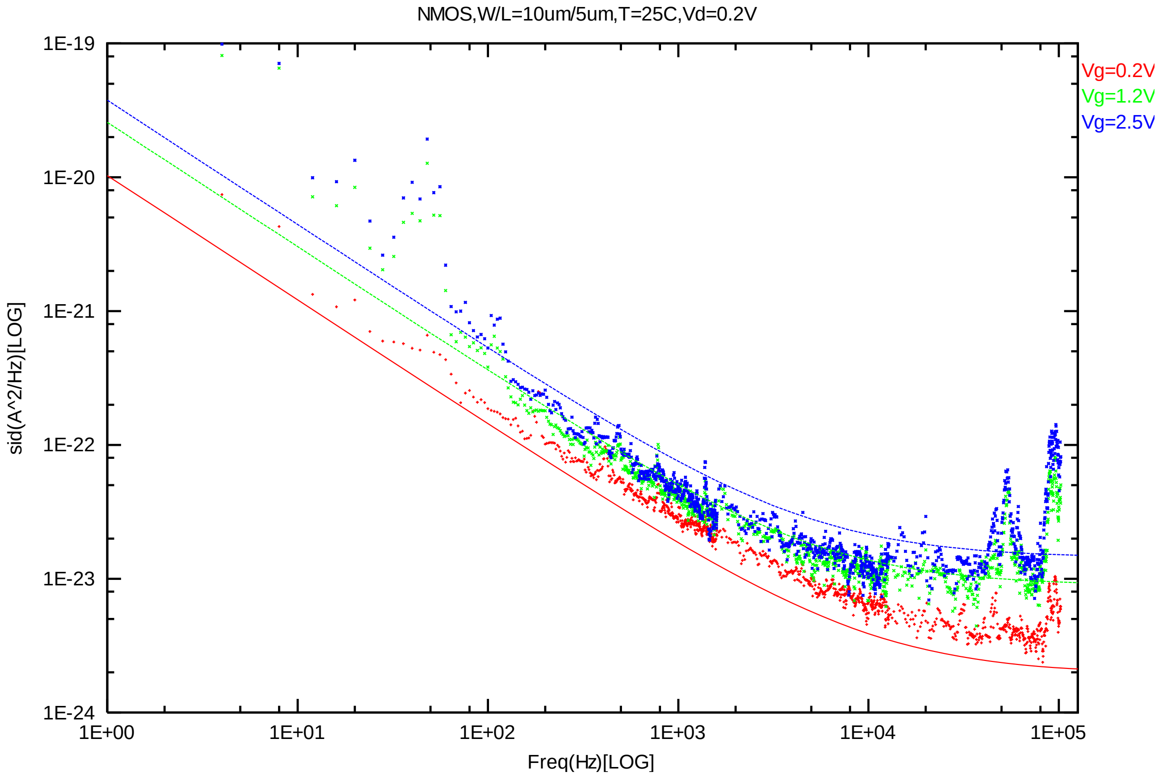

Figure 4-9 NMOS W/L=10um/5um @ Vd=0.2 V

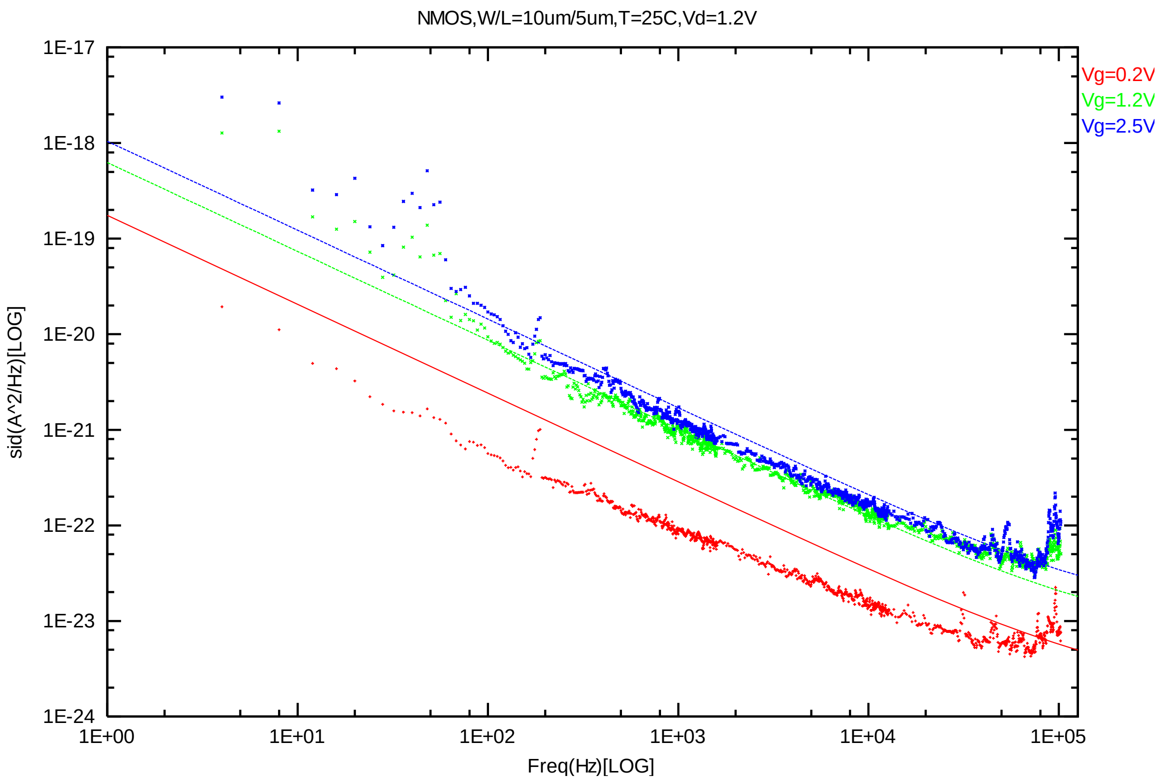

Figure 4-10 NMOS W/L=10um/5um @ Vd=1.2 V

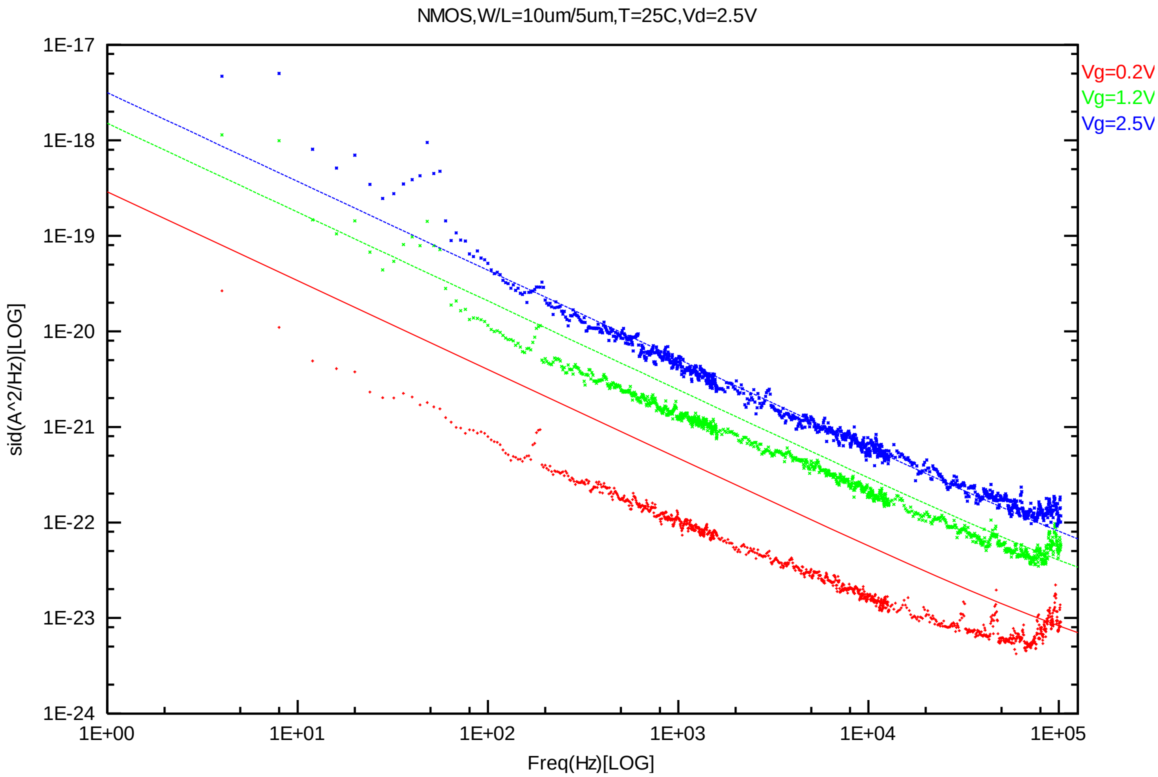

Figure 4-11 NMOS W/L=10um/5um @ Vd=2.5 V

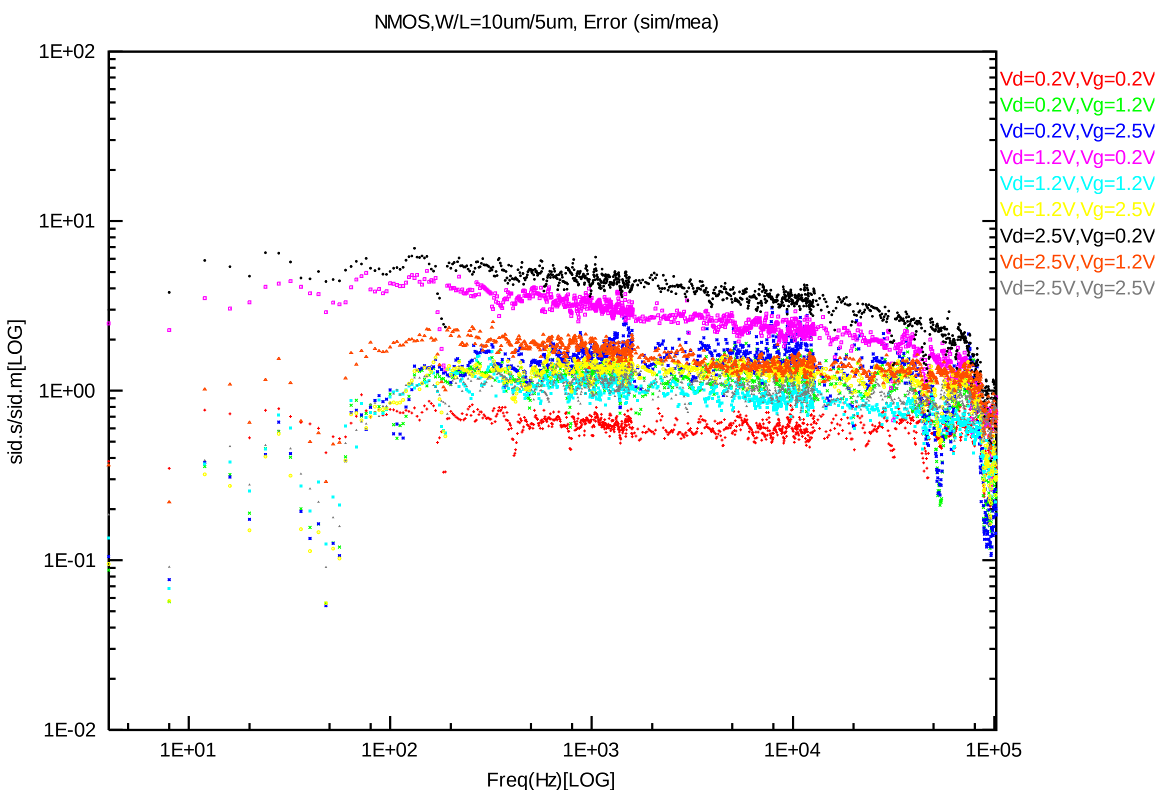

Figure 4-12 NMOS W/L=10um/5um error distribution plot

NMOS, W=10um L=3um

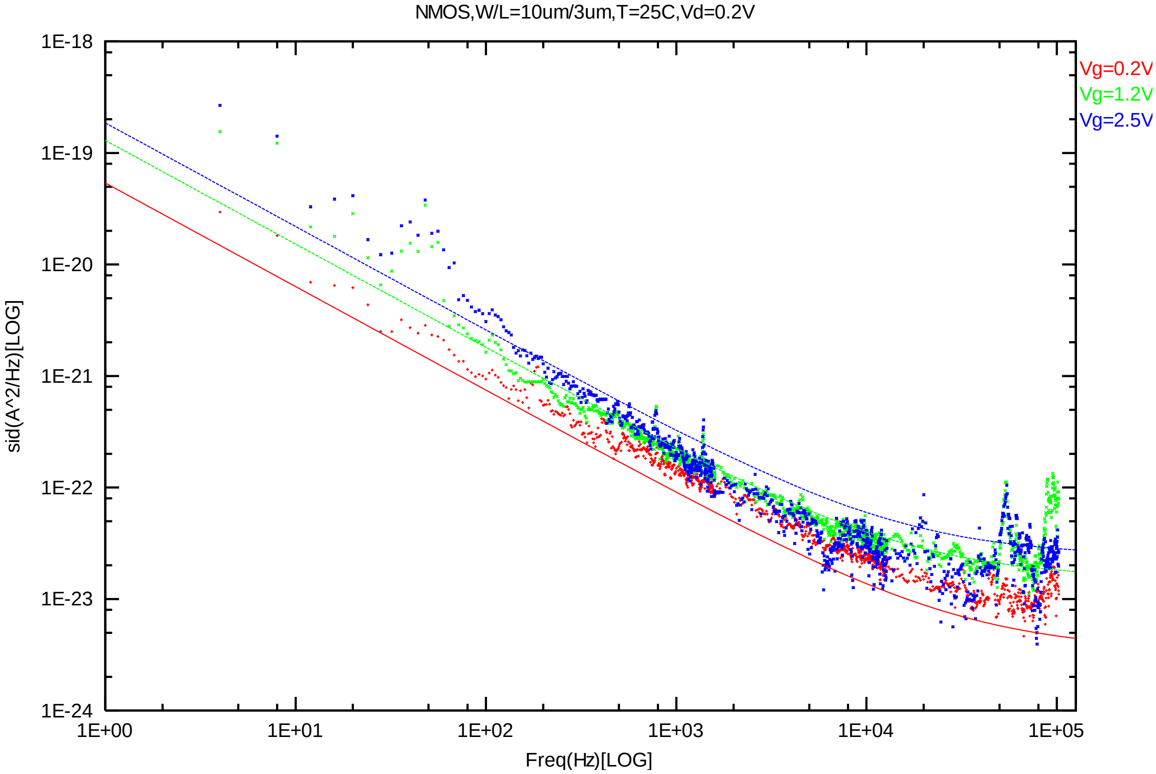

Figure 4-13 NMOS W/L=10um/3um @ Vd=0.2 V

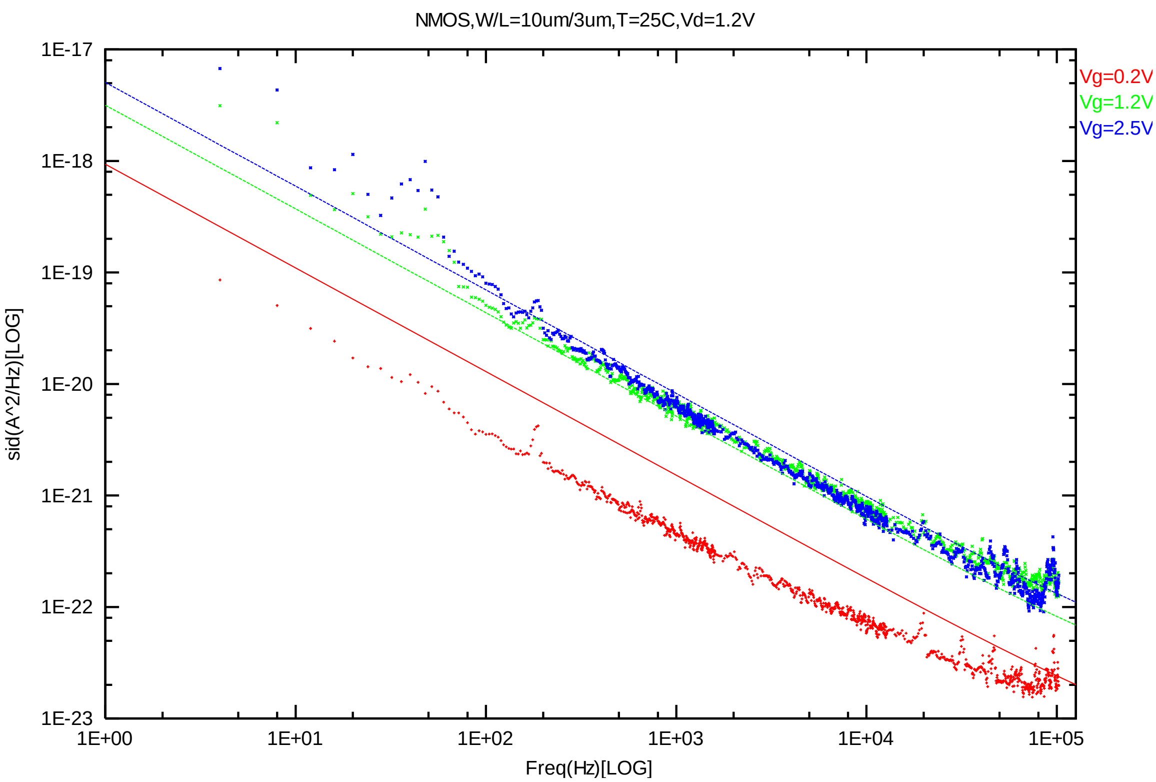

Figure 4-14 NMOS W/L=10um/3um @ Vd=1.2 V

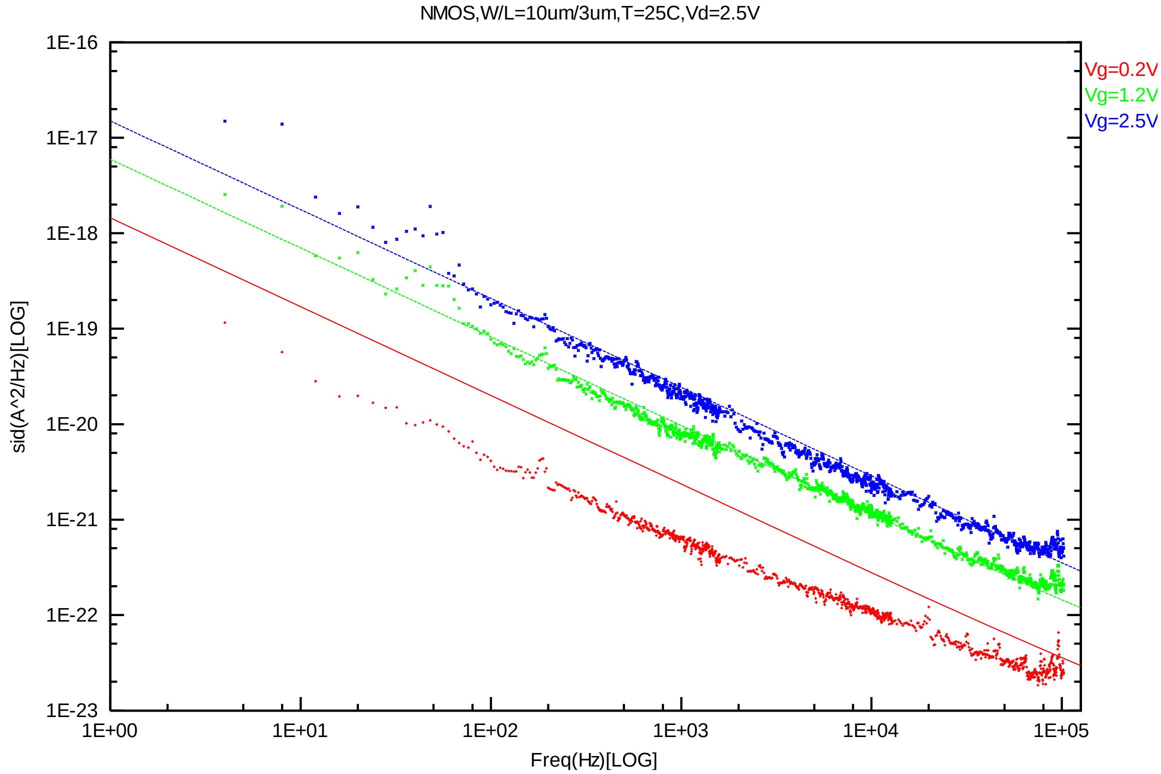

Figure 4-15 NMOS W/L=10um/3um @ Vd=2.5 V

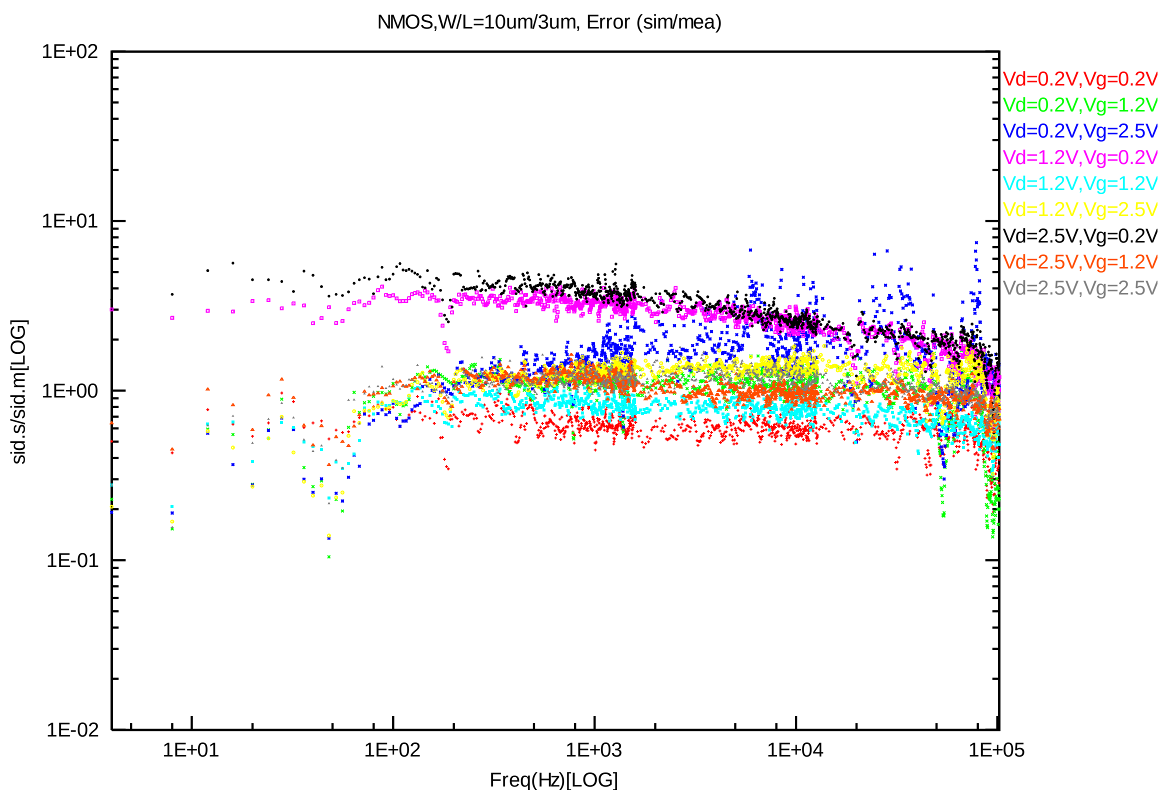

Figure 4-16 NMOS W/L=10um/3um error distribution plot

NMOS, W=10um L=1.8um

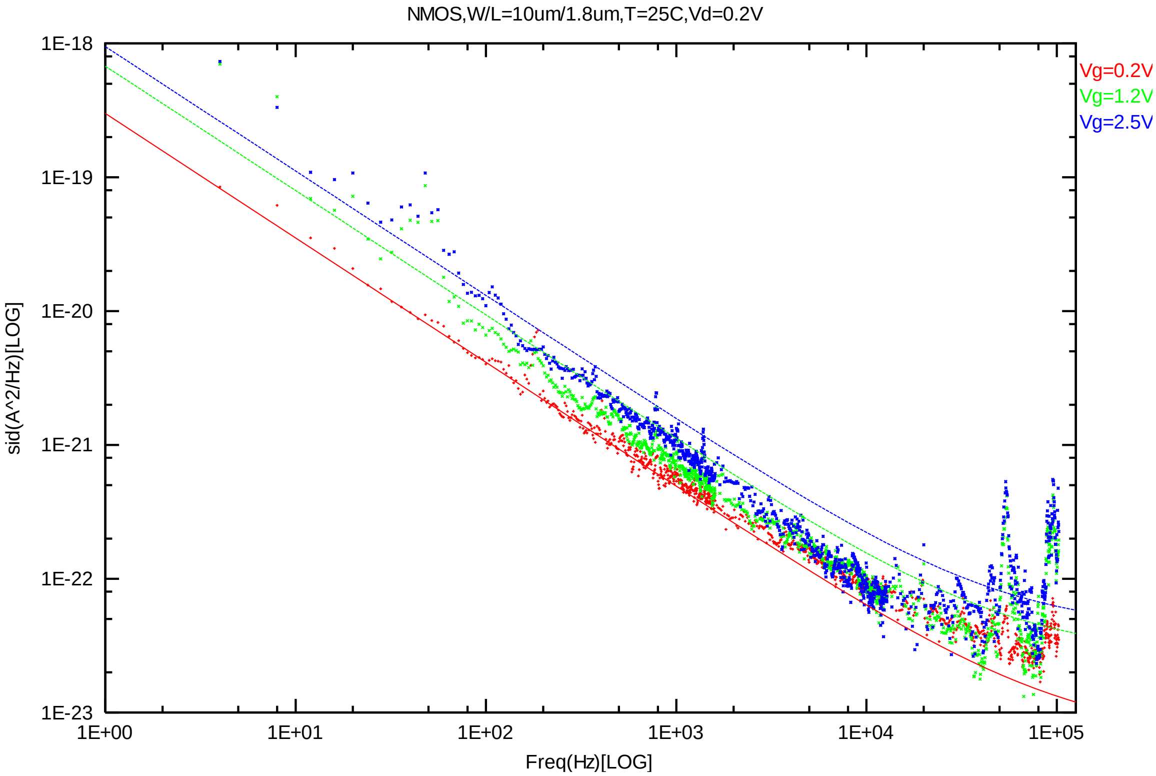

Figure 4-17 NMOS W/L=10um/1.8um @ Vd=0.2 V

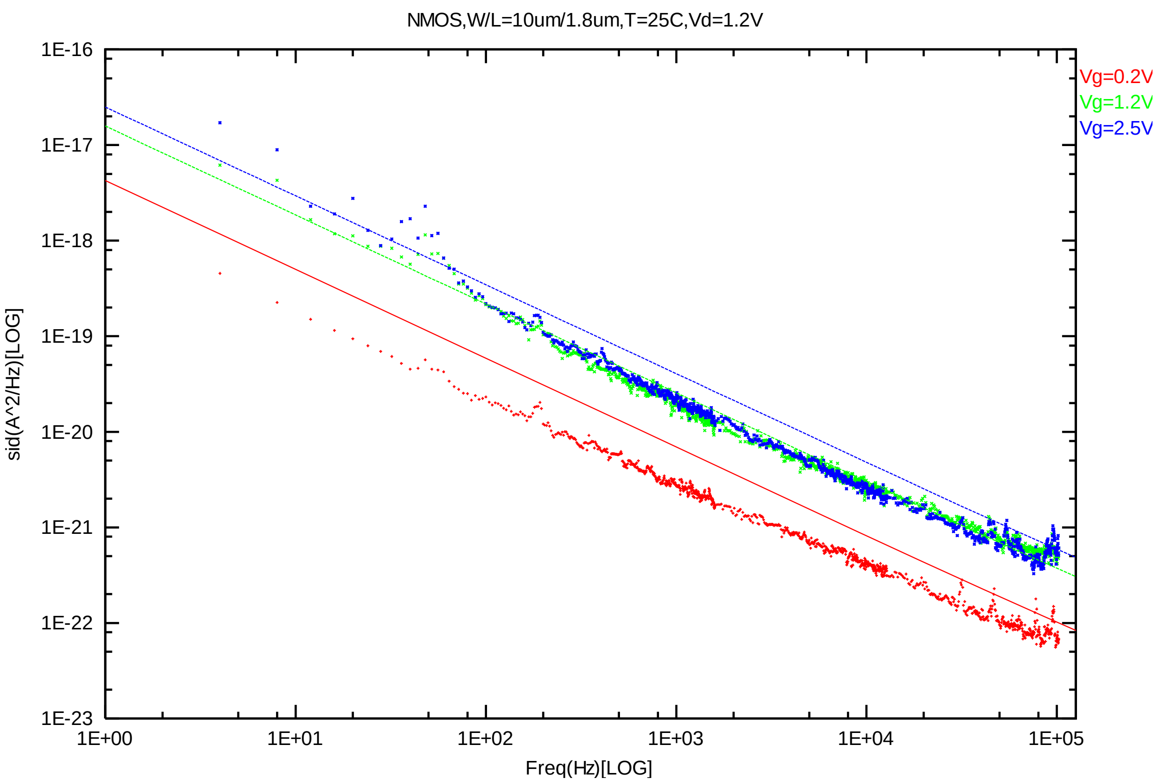

Figure 4-18 NMOS W/L=10um/1.8um @ Vd=1.2 V

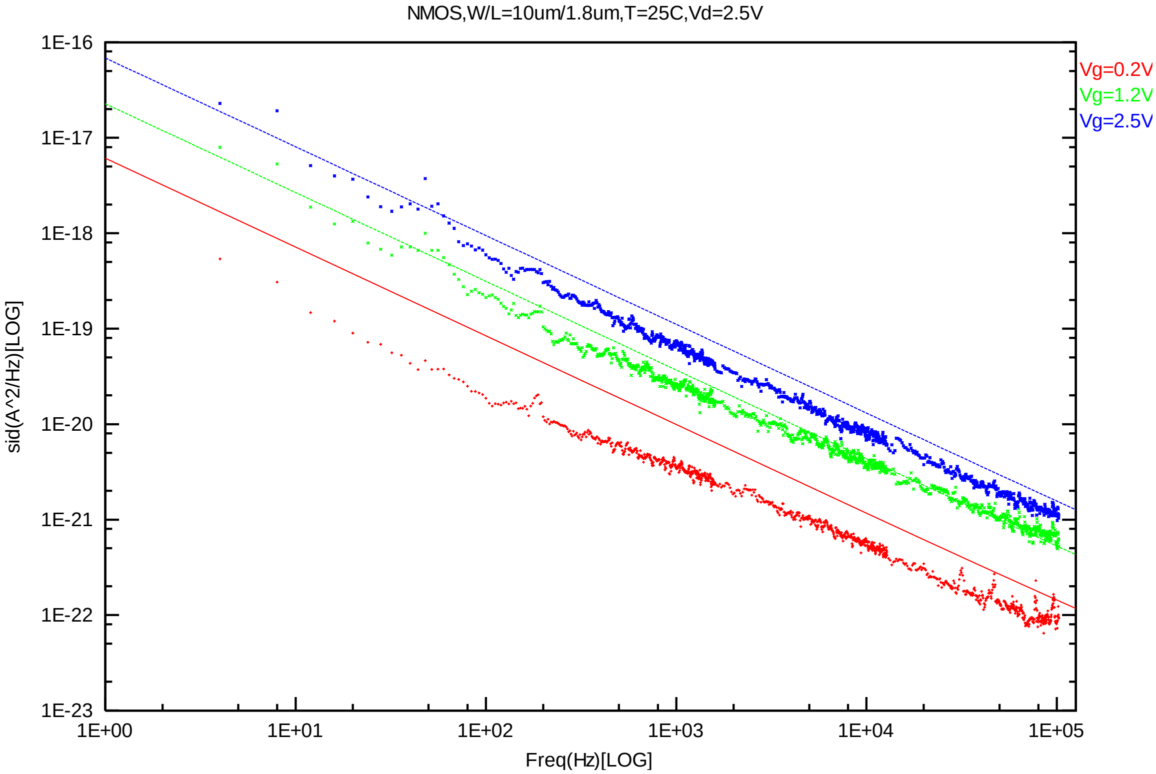

Figure 4-19 NMOS W/L=10um/1.8um @ Vd=2.5 V

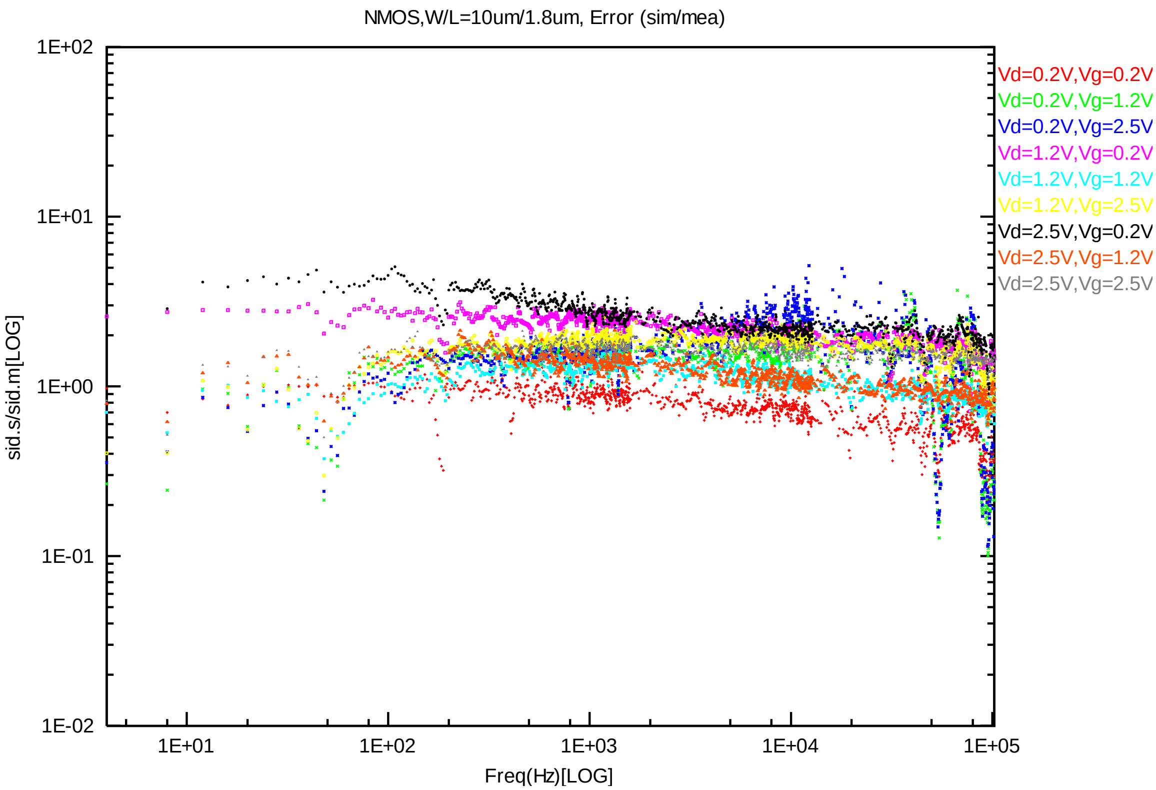

Figure 4-20 NMOS W/L=10um/1.8um error distribution plot

4.2 Trend Plots at Frequency=100Hz

A. NMOS

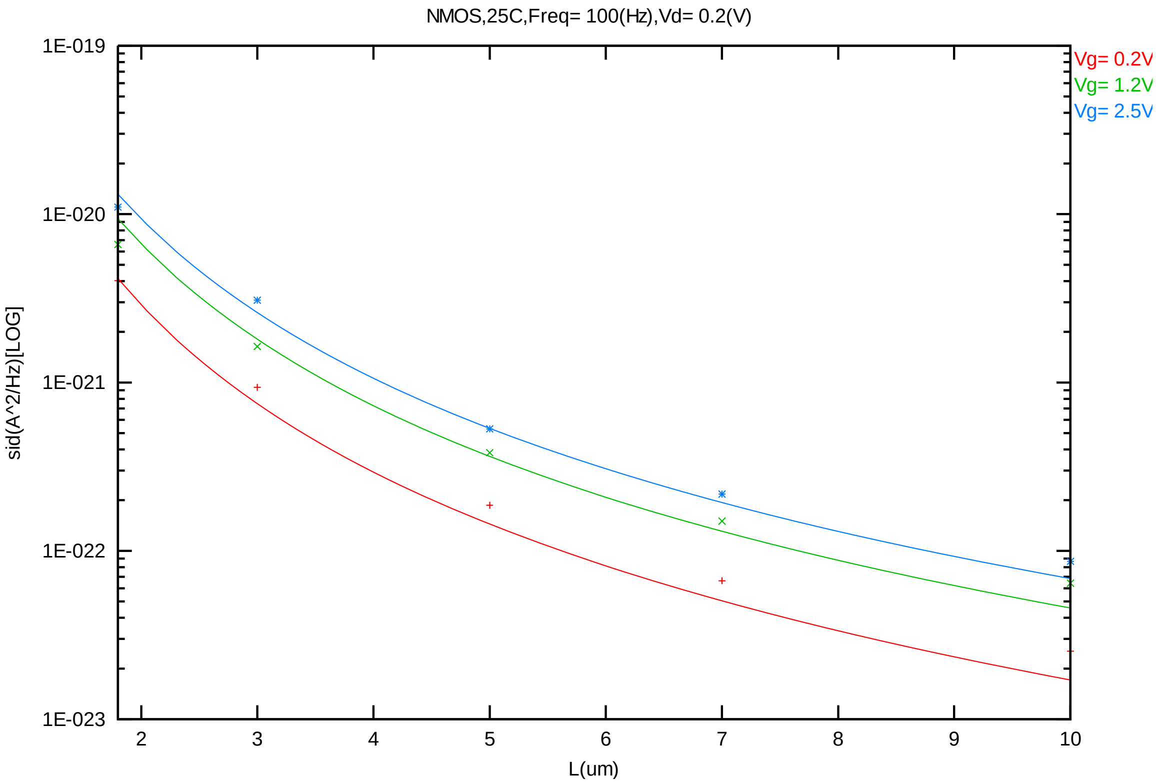

Figure 4-21 Noise vs. Length for NMOS @ Vd=0.2 V

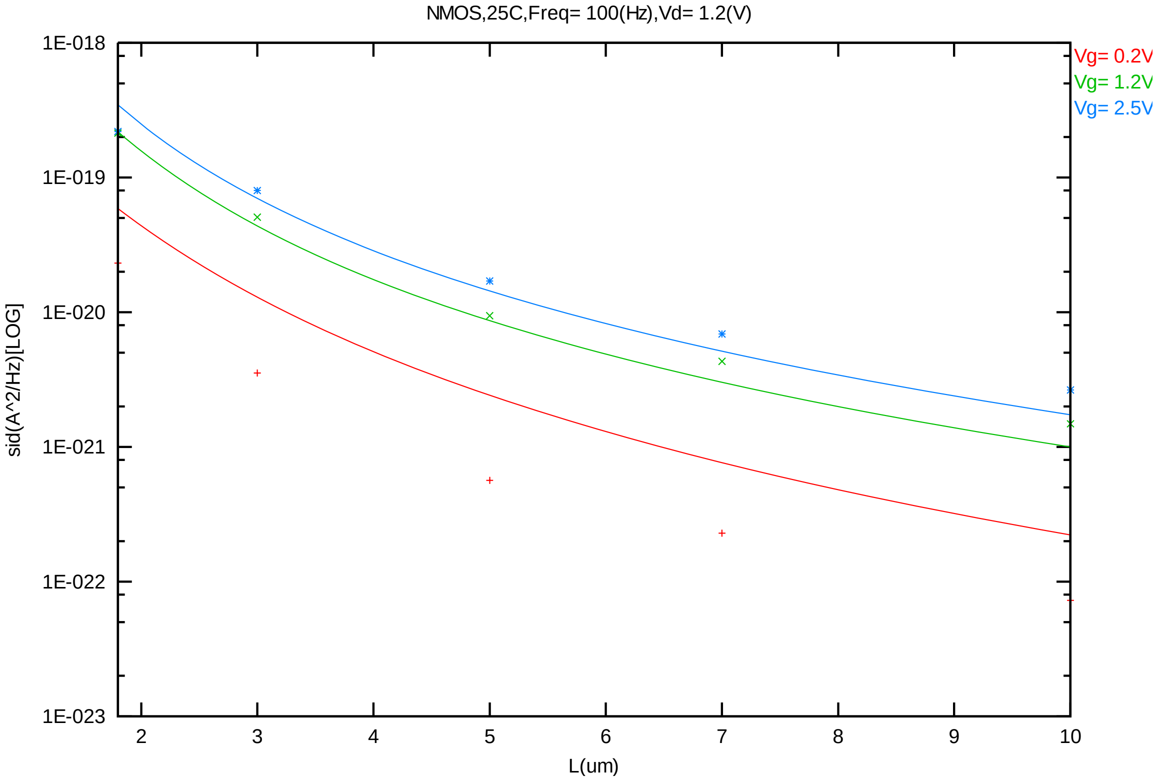

Figure 4-22 Noise vs. Length for NMOS @ Vd=1.2 V

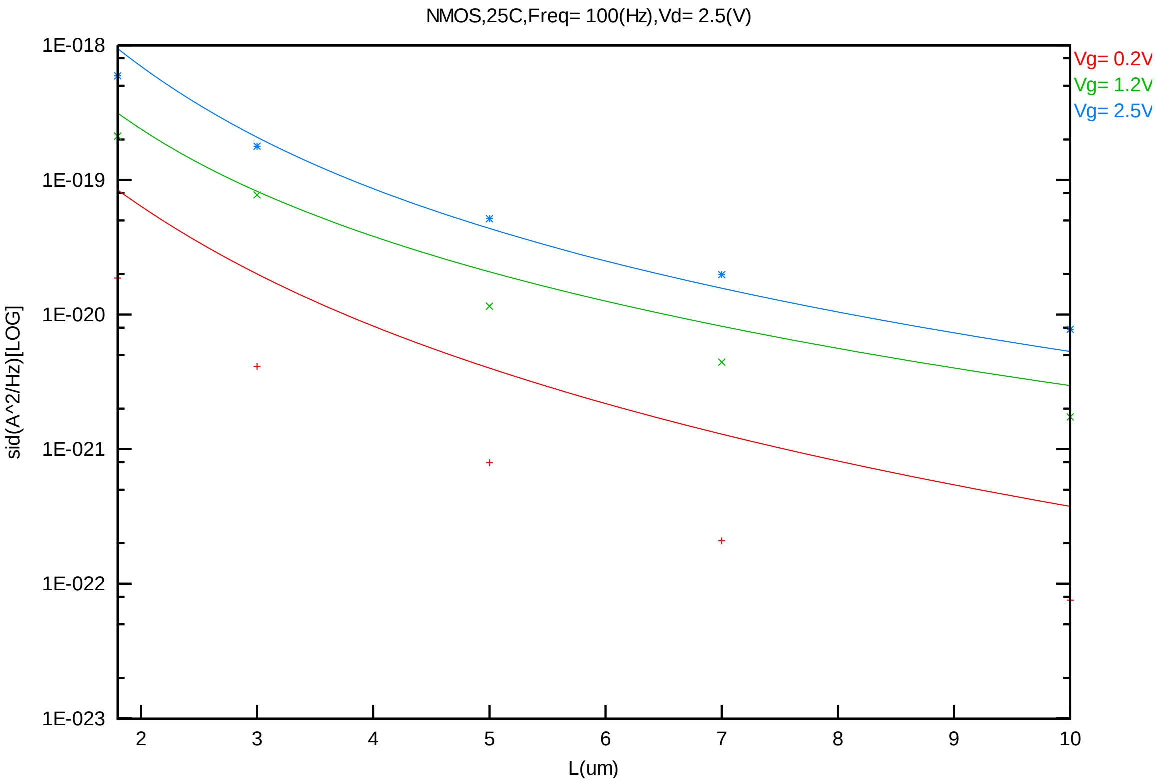

Figure 4-23 Noise vs. Length for NMOS @ Vd=2.5 V

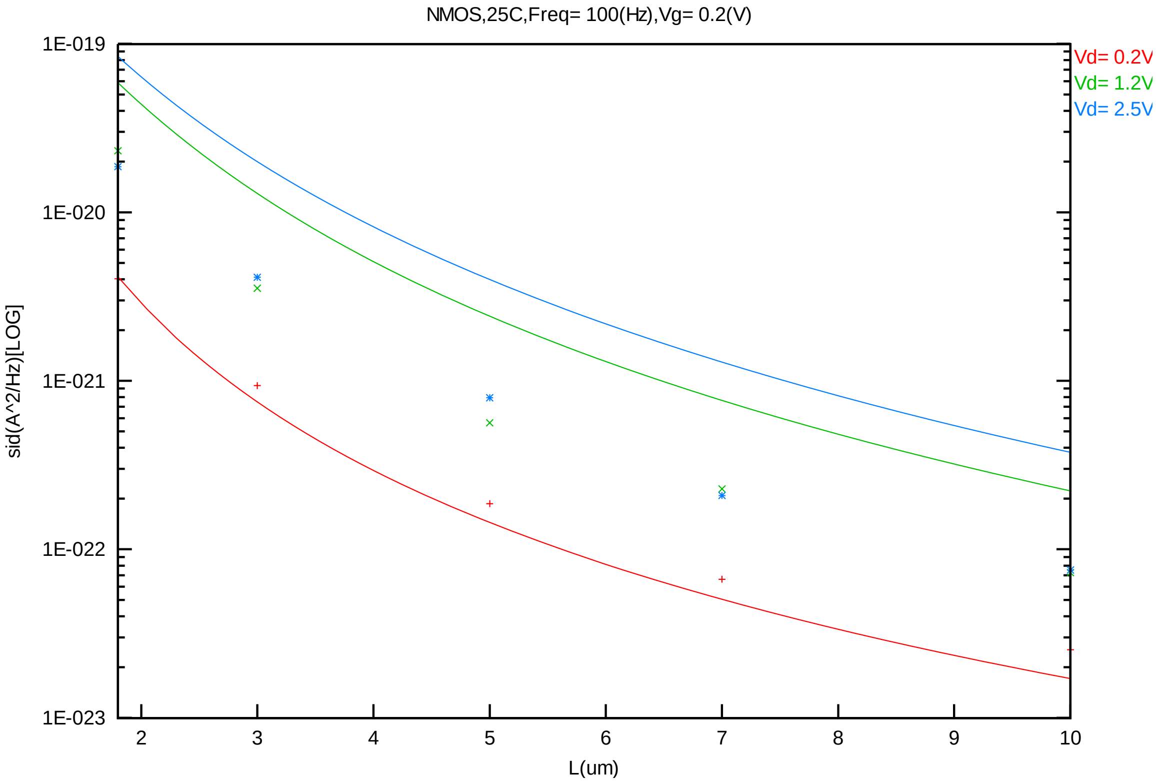

Figure 4-24 Noise vs. Length for NMOS @ Vg=0.2 V

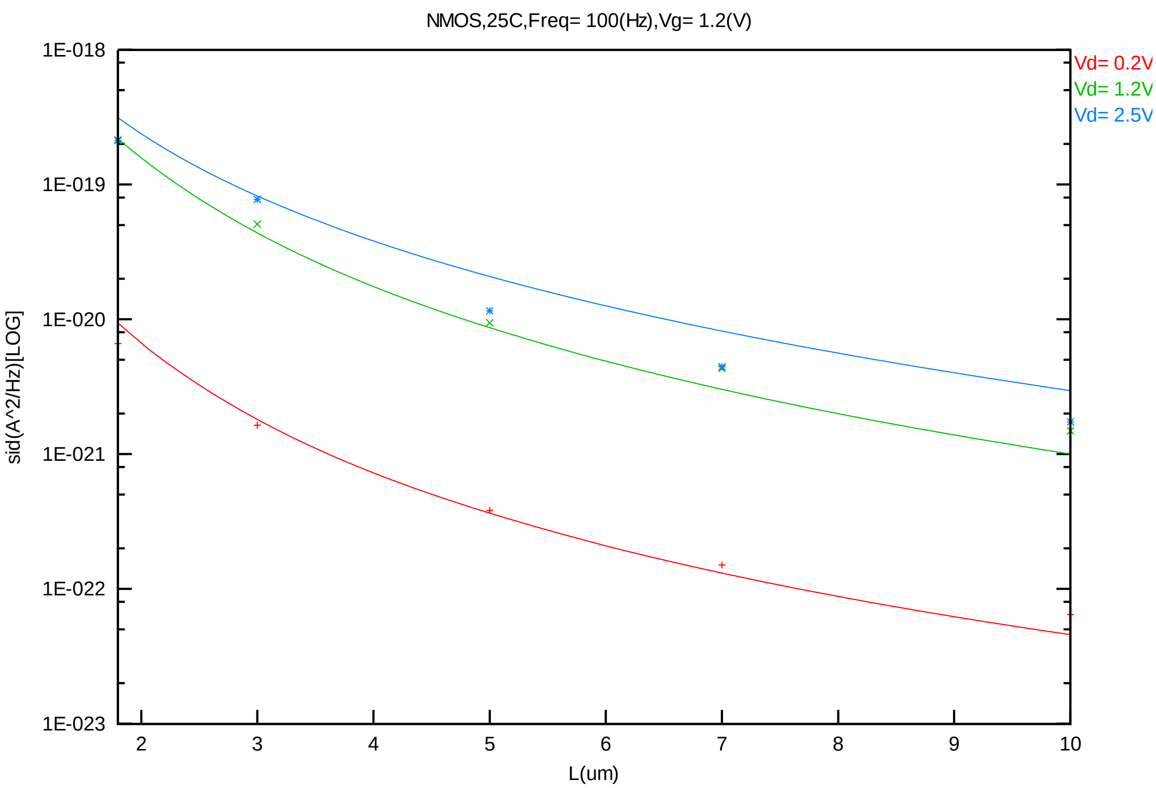

Figure 4-25 Noise vs. Length for NMOS @ Vg=1.2 V

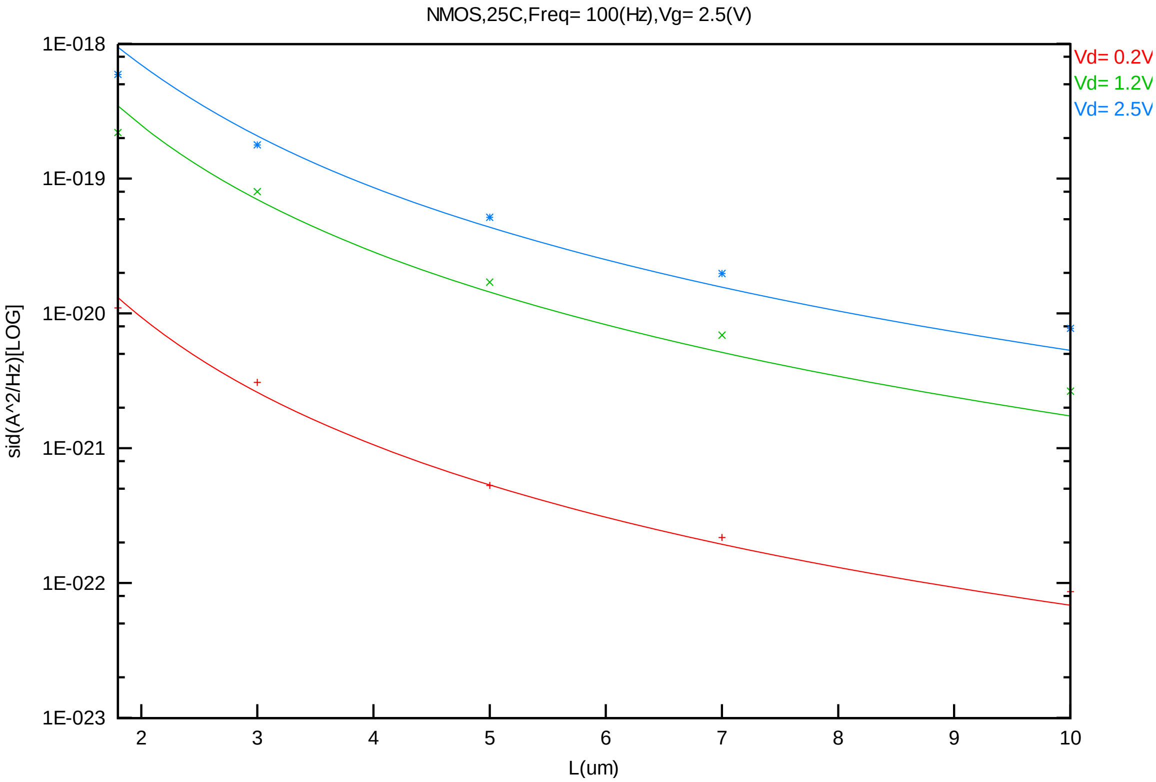

Figure 4-26 Noise vs. Length for NMOS @ Vg=2.5 V

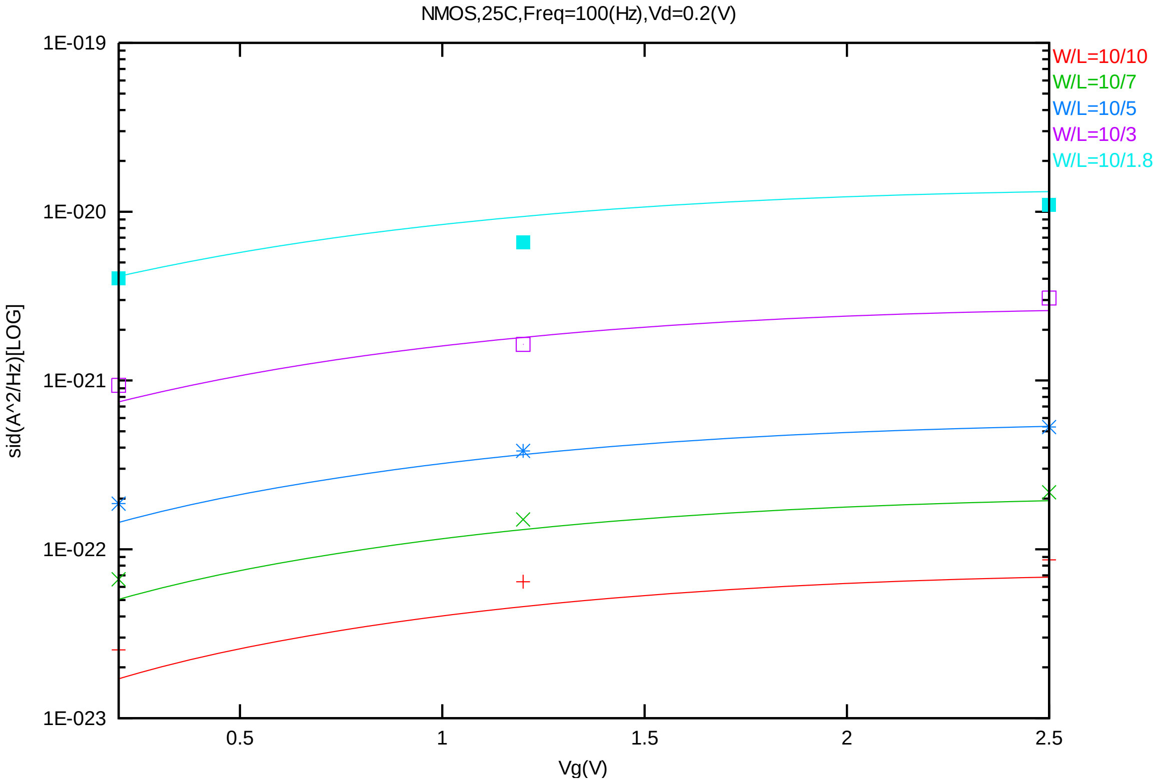

Figure 4-27 Noise vs. Vg for NMOS @ Vd=0.2 V

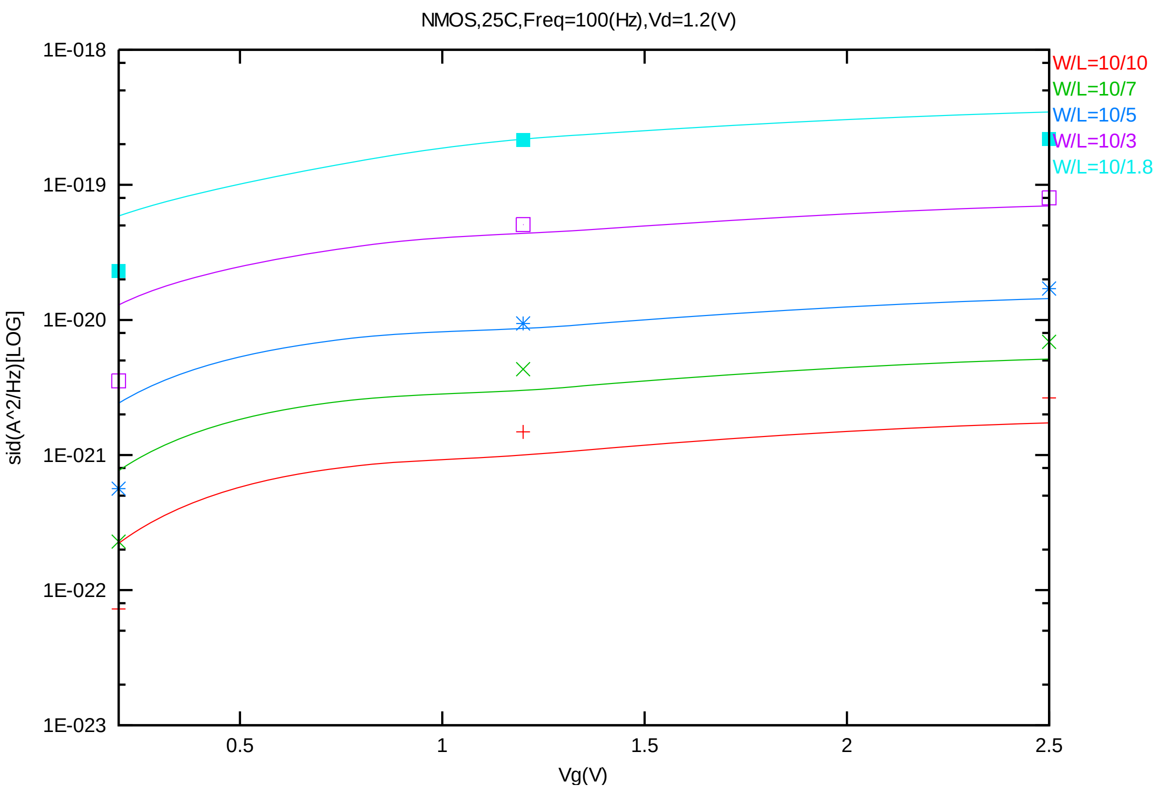

Figure 4-28 Noise vs. Vg for NMOS @ Vd=1.2 V

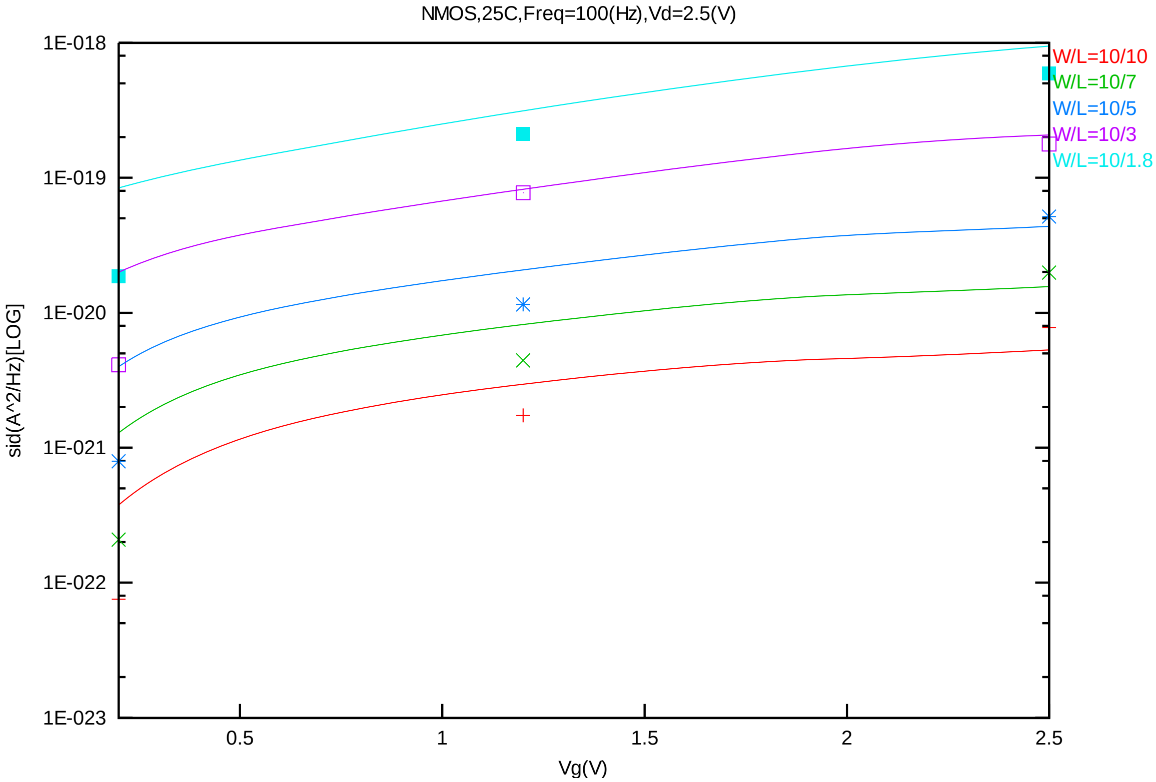

Figure 4-29 Noise vs. Vg for NMOS @ Vd=2.5 V

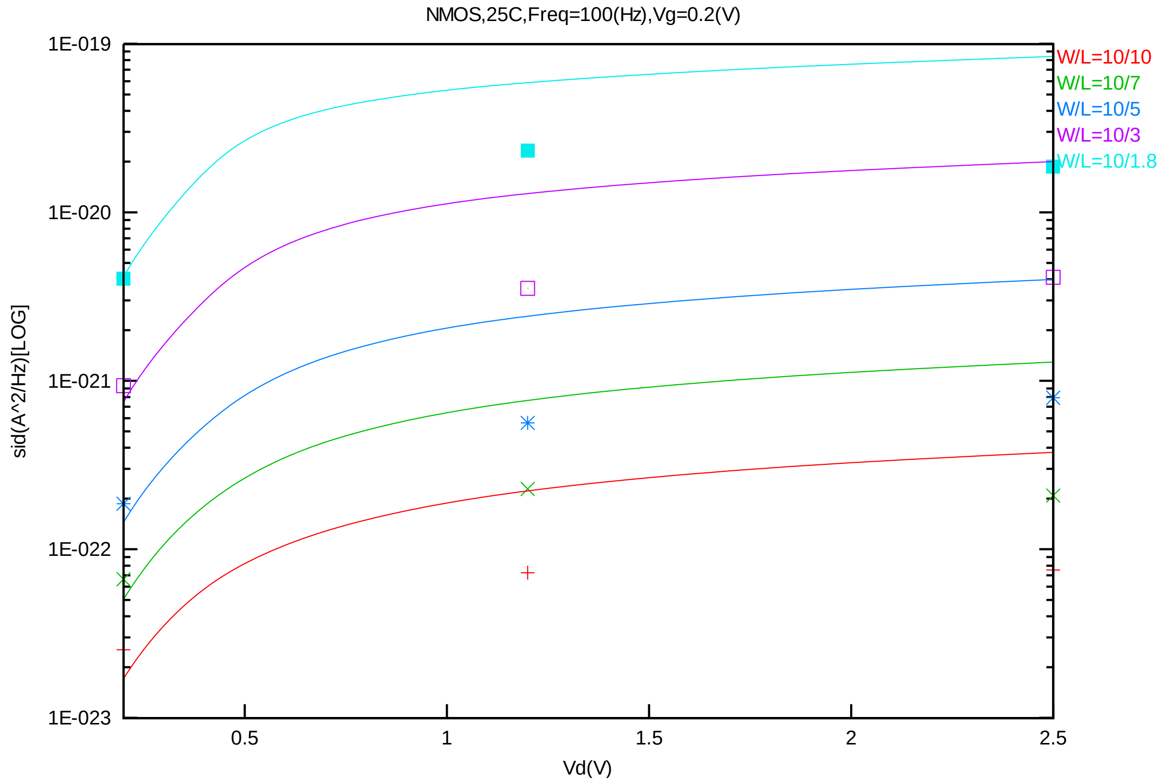

Figure 4-30 Noise vs. Vg for NMOS @ Vg=0.2 V

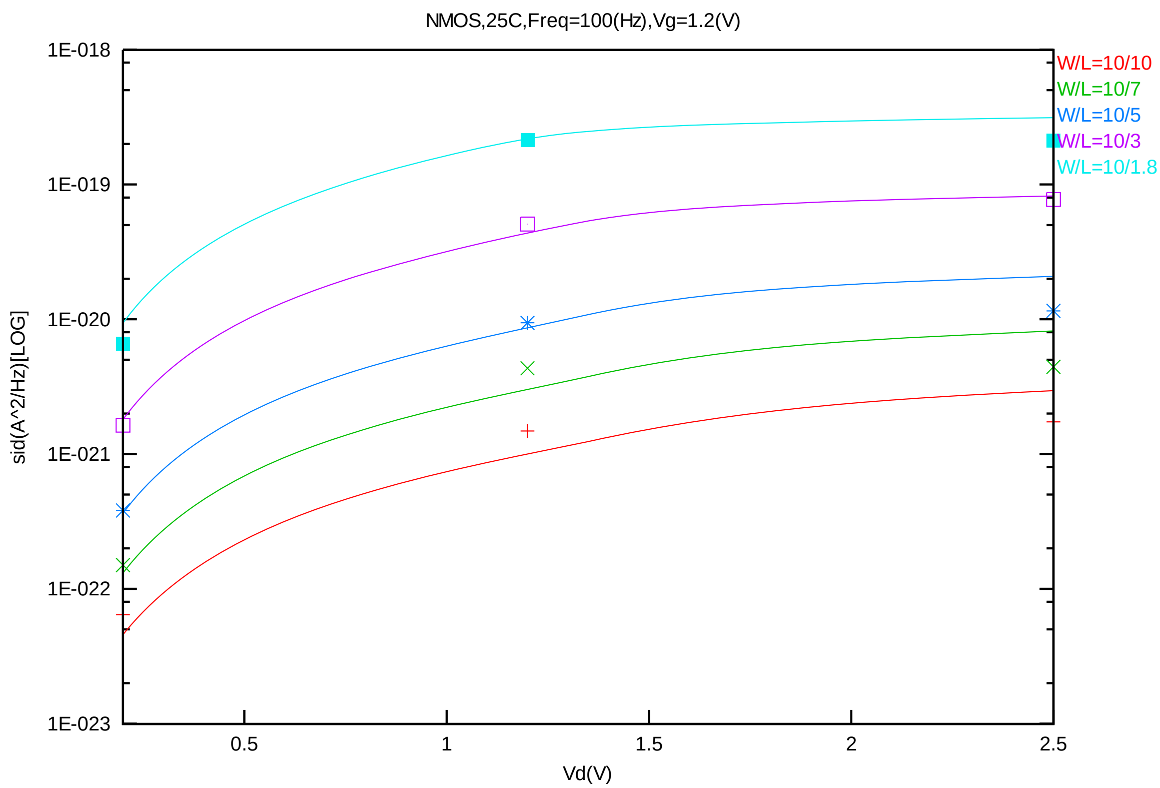

Figure 4-31 Noise vs. Vg for NMOS @ Vg=1.2 V

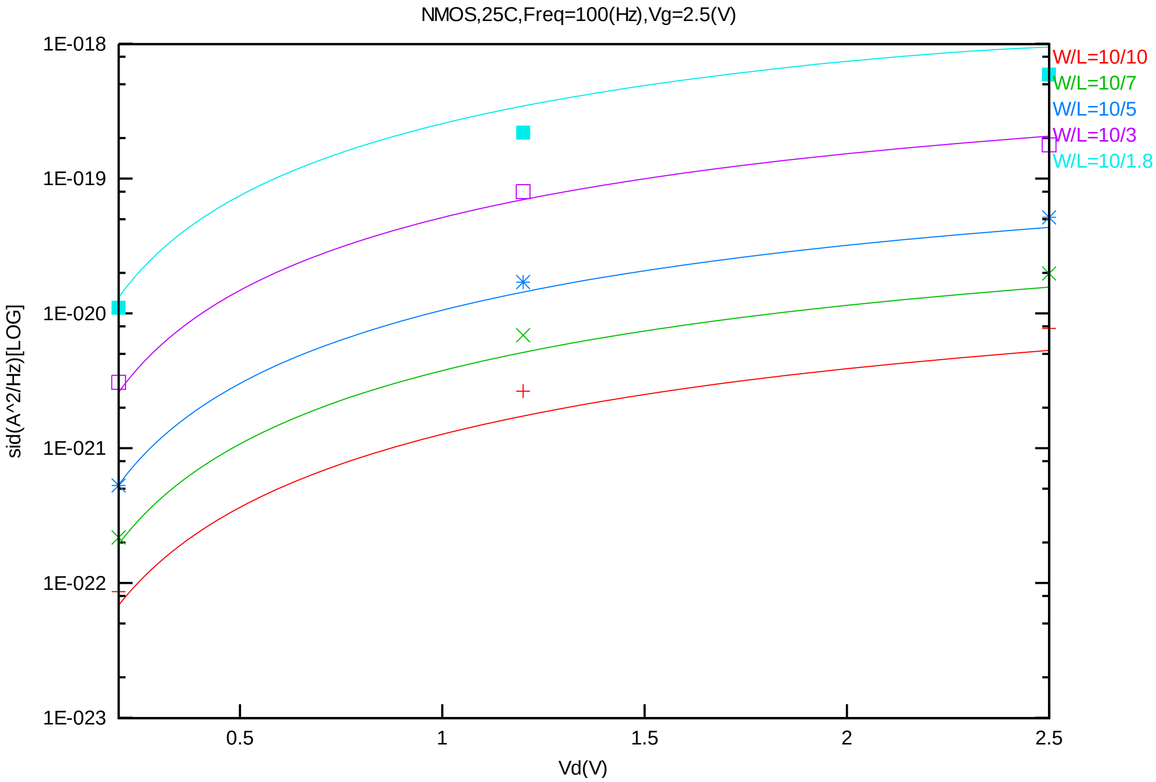

Figure 4-32 Noise vs. Vg for NMOS @ Vg=2.5 V

5 Appendix

5.1 HSPICE Warning Message

Table 5-1 Hspice warning message

NO.	Warning Message	Explanation
1.	-	-

5.2 ELDO Warning Message

Table 5-2 Eldo warning message

NO.	Warning Message	Explanation
1.	-	-

5.3 SPECTRE Warning Message

Table 5-3 Spectre warning message

NO.	Warning Message	Explanation
1.	-	-